THE EPIDEMIOLOGY AND CONTROL OF HAND, FOOT AND MOUTH DISEASE

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Declaration

I hereby declare that this thesis is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in the thesis.

This thesis has also not been submitted for any degree in any university previously.



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Summary

Hand, foot and mouth disease (HFMD) is the result of viral infection of many potential etiological agents, such as human enterovirus A71 (EV-A71), coxsackieviruses (CV) A16, A10 & A6. It occurs throughout the world and is a problem in many areas of East and South East Asia, temperate or tropical. Many jurisdictions in Asia, including mainland China, Hong Kong SAR, Macau SAR, Japan, South Korea, Singapore, Vietnam, and Thailand have stepped up on surveillance efforts in recent years, and routinely reports figures to WHO for monitoring.

Policies against HFMD are usually implemented after outbreaks of EV-A71 which could cause very serious neurological complications that leads to death. These policies could include isolation of sick children, closure of schools, surveillance, which are implemented to different extents in various countries, and are usually set up when there was no luxury of time to properly optimize. Today, there is a large body of literature on the disease from research in the past few decades, which should be investigated to see how policies stood the test of time and how can researchers contribute further to the growing pool of knowledge on HFMD.

Investigating the epidemiology of HFMD is particularly challenging because of multiple reasons: Not only is it caused by multiple etiological agents, the infection rate is typically so high that it is impractical to track all cases. In Singapore for instance, a recent study showed that almost 85% of the population is seropositive to either EV-A71, CV-A16 and CV-A6. Also, the disease severity can vary widely from asymptomatic and non-specific illness, to severe neurological complications and death. The combination of these factors means that information from surveillance is unlikely to capture all the required details to high accuracy. Even mortality rates due to HFMD might not be consistent: For instance, mortality

from encephalitis due to EV-A71 might not be properly documented as such because post-mortem serological tests might not be performed.

In this thesis, we will tap on the evidence base and historical data and re-examine it using modern approaches in biostatistics and Bayesian modeling. The purpose is to consolidate all the accumulated knowledge base and to inform current research agendas towards the gaps in the literature required to tackle this disease. This thesis has a particular focus in generating health statistics for policy and is also part of a greater project meant to investigate the primary contacts of sick children - their parents and the staff of childcare facilities, so that we can identify risk factors and methods that can bring greater effect to HFMD incidence.

Specifically, Chapter 1 provides an overview of HFMD, the options that policymakers have in disease control, and how is Singapore handling HFMD. Chapter 2 moves the discussion internationally and identifies the gaps on HFMD epidemiology and provides a general critique of papers through a rigourous systematic review. Chapter 3 addresses some of the gaps identified in the review through a combination of meta analysis data and Singapore surveillance data through a Bayesian modeling framework. Chapter 4 provides an alternative but important look at disease management to answer the question of how can we design HFMD control policies which parents are more likely to cooperate using a combination of Bayesian modeling and machine learning methods. Finally, Chapter 5 concludes the thesis with some further analysis, recommendations for policymakers and suggestions for future research.

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Chapter 1

Introduction

Enterovirus species A viruses, including enterovirus 71 (EV-A71) and coxsackievirus A16 (CV-A16), have caused large hand, foot and mouth disease (HFMD) epidemics worldwide. Since EV-A71 was first identified in 1969 from an infant suffering from encephalitis in California [1], outbreaks of HFMD, herpangina, aseptic meningitis and encephalitis associated with this virus have been documented. EV-A71 infection is occasionally associated with severe neurological and pulmonary complications leading to childhood mortalities in epidemics around the world [2], while infection from other agents such as CV-A16, CV-A10, CV-A6, CV-A2 are less likely to cause severe illness [3, 4, 5]. Since 1997, EV-A71-related HFMD epidemics in the Asia-Pacific region have been increasingly reported, including China[6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21] , Taiwan[4, 5, 22, 23, 24, 25, 26], Hong Kong [27, 28], India [29, 30], Japan [31, 32], Korea [33, 34], Malaysia [35, 36], Singapore [3, 37, 38], Thailand[39, 40] and Vietnam[41].

We will begin the thesis by providing background information about HFMD with a particular focus on policy and control and discuss the range of policies that could be implemented by the authorities, and the factors which will affect how effective are they likely to be. We are particularly interested in how knowledge of epidemiological characteristics translate to evidence-based decisions. Our analysis on virology will be limited to serology and how to model asymptomatic infection rates. We will then move to the HFMD situation in Singapore, which is a tropical city-state that experiences high symptomatic incidence rates of about 9 in 100 children <5 years [42, 43], one of the highest among the Asia Pacific countries. Finally, we will cover our motivations for our research projects which led to the creation of this thesis.

1.1 The Control of Hand, Foot and Mouth Disease

Current prevention and control measures for HFMD were mainly designed to interrupt virus transmission, although pharmacological intervention is possible today with the development of EV-A71 vaccines in China which had underwent phase 3 trials [44] with 10,245 participants and was shown to provide high efficacy and safety. Some primary measures include:

1. Surveillance systems, which can be sentinel based and voluntary, or mandatory for all healthcare providers and schools [45].
2. Education on good hygiene, such as television programs, children songs or talks conducted by healthcare professionals [45].
3. Regulations on childcare facilities and schools, such as mandatory closures during outbreaks [46].
4. Improving case-management at the clinical level, and equipping doctors with the skills and knowledge of severe manifestations of EV-A71 which can quickly lead to death.
5. Potentially, outbreaks or high incidence can be predicted and influenced to mitigate the effects of subsequent epidemics [47, 10, 48].

How effective each of these measures are will depend on the characteristics of individual countries, such as hygiene levels, the level of healthcare infrastructure, and demography such as population density [45]. As every country is fundamentally different, no one-size-fits-all approach exists and fine-tuning of policies must be conducted for each country. For example, school closure strategy for each country can be different depending on the economic cost, which is in turn affected by the availability of caregivers, the ability of parents to take time off, and the GDP per capita, which is a proxy for the 'value' for each day of work. School closures can be tweaked in many ways: Closure at 2nd case is more likely to stop transmission but there will be a lot of false positive 'outbreaks' (school closed wrongly); Closure at 20% attack rate could be too late, depending on the percentage of cases which can be found. Or perhaps, HFMD might be so infectious that outbreaks will always occur in populations with high seronegative rates, making school closures merely a measure that

postpones the inevitable. To optimise control policies at a national level requires knowledge of epidemiological parameters that, as the literature review in the next chapter attests, may not be available. Some of these parameters include infectious period, which will affect the optimal duration of school closure; reproduction number, which will affect vaccination strategies and herd immunity; severity, which will affect how to conduct surveillance.

Many decisions on health interventions will still have to be made despite the lack of definitive scientific and technical evidence [45]. Policy decisions on infectious diseases often were made quickly, especially if the outbreak led to fatalities. For instance, surveillance programs in Singapore were set up in Oct 2000 after an outbreak which killed 4 children; data collection on severe-EV-A71 started in Taiwan after the 1998 outbreak (Feb 15 to Dec 20) which killed 78 children; similarly for Malaysia after an outbreak of EV-A71 which killed 34 children in 1997 [49]; and in China after major outbreaks in 2007 and 2008 [50]. Many of these interventions persisted till today. In the next section, with the benefit of hindsight, we will critically evaluate some of these measures, the likelihood of the measure being effective, and what information will be required for evidence-based fine-tuning.

1.1.1 Surveillance

Surveillance programs are ongoing collection of health-related data, so that the data can eventually be used to monitor the epidemiology of diseases, guide public health policy, serve as early warning of outbreaks, and also to understand the impact of interventions. For HFMD, surveillance systems usually collect data of symptomatic cases of varying levels of severity, such as mild HFMD, to complicated HFMD, to death, with or without laboratory confirmation. The surveillance systems in Asia are made up of sentinel hospitals and clinics (Malaysia [51], Taiwan [52], etc), laboratories (Japan [48], Singapore [53], China [50]) and could also involve schools and childcare facilities [46].

Data on HFMD are frequently put up in the respective government websites, but the reporting standard is not the same. The Taiwan National Infectious Disease Statistics System provides information on 'Severe Enterovirus' infections [54], while the Department of Health in Hong Kong provides data on notified HFMD per GP Visit [55]. The National

Institute of Infectious Diseases, Japan, provides weekly data of notified HFMD per sentinel clinic [56], while the Ministry of Health, Singapore, provides the actual number of notified HFMD [43]. The data that is provided is a reflection of the way surveillance is collected: Singapore collects data from all physicians and it is mandatory to do so, and thus is more likely to have higher notification rate, and therefore makes sense to provide the number of cases. For sentinel networks which is known to be incomplete, reporting incidence as a ratio of the number of sentinel clinics will make more sense. As an example, during Taiwan's outbreak in 1998 which resulted in over 100k cases [52], the sentinel system consisted of only 850 physicians out of a total of 9427 around the island, so it is likely that 100k cases is still likely to be underreported, perhaps by a factor of 11 times, assuming that there is no systematic differences between those who reported and those who did not.

The choice of surveillance system will be dependent on the characteristics of the disease. For instance, a disease with high severity should probably call for a more stringent surveillance system to identify new cases quickly. For HFMD which is mostly a low severity disease (refer to Chapter 3 for more information), a comprehensive surveillance net might potentially be excessive. Another factor to consider is the incidence rate, and whether is it reasonable to find all cases and to expect doctors to report every single case that they see. It is certainly more reasonable to expect doctors to report all cases of dengue, rather than all cases of influenza, for example. Lastly, we should consider whether can we achieve the purpose of surveillance by observing a subset of the total number of cases. For example, will observing influenza cases only from public clinics and hospitals (and not from the private sector) be sufficient enough to construct a early warning system? All these factors should be analysed before deciding on what type of surveillance system do we need.

1.1.2 Education of the Public

HFMD is a disease spread through the oral-fecal route [57] and thus personal hygiene is naturally a risk factor for disease transmission. Education of personal hygiene could come in the form of instructions to caregivers in childcare facilities, government awareness campaigns or advice given by doctors and other medical practitioners. These education programs could

serve to prevent caregiver-to-child transmissions, especially childcare centre teachers who might handle multiple children at once. A paper from China showed that children under caregivers who washes their hands (the caregiver's hands) before feeding are 25 times less likely to get HFMD [16]. Other research include a case-control study from Korea [34], which identified the role of water and public toilets in HFMD transmission. These information are all collated through the systematic review presented in Chapter 2 of the thesis.

Another aspect which is important for policy makers is how likely will caregivers follow the set regulations. For instance, mandatory isolation policies directed at sick children will require the childcare centres to be diligent in screening for sick children before allowing the children to mix. It also requires parents to be understanding and know why their child is being isolated, otherwise they can still continue to bring them to public playgrounds and spread HFMD to the community. This is the primary reason for our work in Chapter 4, which is part of a bigger project targeted at both parents and teachers of preschools to understand their perceptions and knowledge of the disease.

1.1.3 Regulations on Childcare Facilities and Schools

Being a gathering place for large groups of susceptible children daily, preschools are implicated in the spread of HFMD. Although it is not clear whether most HFMD cases spread in these places (discussed heavily in Chapter 2), their ostensible role as venues for disease transmission means they also may play a vital role in preventing it.

The success of school closures in stopping disease transmission depends largely on the window of opportunity for virus spread within the preschool prior to the closure. Consider how the effectiveness of closures can vary with different attack rate triggers and how asymptomatic infection rate can complicate the optimisation of this policy. We modeled the rate of asymptomatic infection using serological data in Chapter 3 of the thesis and obtained estimates up to 70% for young children. This means that closures triggered based on 25% attack rate will experience an average of 85% actual infection rate, removing any tangible benefit to the action. It is a delicate balancing act between closing too many schools and closing too little, too late. Also on the topic of asymptomatic infections, another factor to

consider is whether such cases are transmissible. There is evidence that shedding of virus in faeces may last for weeks after symptoms have ceased, so that even with closure or isolation, there is a risk of subsequent outbreaks [58]. A complete school closure policy should also have an optimized duration of closure based on incubation and infectious period of the disease to ensure that the isolation is long enough to break transmission but not cause excessive burden to the parents, which will be discussed in greater detail in Chapter 2.

School closures and isolation policies can have profound impacts on families. Discussed in greater detail in Chapter 4, the effect of temporary school closures in Singapore is not clear: on the one hand, many families have domestic help, live near their extended families, or can take advantage of government schemes to allow workers to take leave to look after their children; on the other hand, the female participation rate in the workforce is high, and few families have stay at home parents, forcing them to take leave or even leave their jobs to care for their children [46]. Deciding under what circumstances schools should be closed must account for both health effects and social effects; both of which require a greater evidence base than currently exists.

1.1.4 Outbreak Prediction and Mitigation

Policies of this type are meant to mitigate diseases 'ex-ante' - stopping the outbreak before it begins by manipulating causal factors, which can be challenging to identify. Meteorological parameters have been frequently mentioned as a predictive factor [48, 59], since HFMD is a 'summer disease' (see Chapter 2). The biological plausibility could be explained by more rapid virus decline during dry seasons compared to wet seasons [48]. Warmer temperatures/humidity have been also been mentioned as a potential factor [60] for high incidence, although it is also equally possible that colder weather is associated with lower incidence, which could be the result of decreased social contact during winters. Currently, we still do not understand why HFMD is epidemic during summer, while other droplet-borne diseases such as influenza are epidemic in the winter. Manipulating ambient temperatures to decrease HFMD incidence, with the current level of evidence, seems sketchy at the moment, and perhaps a more controlled experiment is required to investigate the phenomenon.

Some other interesting approaches include the potential use of UV lamps to sterilize classrooms, supported by a case-control study in Beijing [61]. Spatial-modeling also showed similar results, that sunshine could be protective against HFMD [10]. These findings are inconsistent with the fact that HFMD epidemics occur in high incidence in the tropics and in the summers of temperate countries, and not the winter. Overall, using meteorological variables for disease prediction (and eventually, mitigation) seems unlikely until these differences could be reconciled with stronger evidence.

1.1.5 Vaccines

A number of vaccines for EV-A71 have been developed in China in the recent years. Double-blind, randomized, placebo-controlled phase 3 trials have been completed by Beijing Vigoo Biological [44] in 2012, Institute of Medical Biology, Chinese Academy of Medical Sciences [62] in 2014 and Sinovac Biotech [63] in 2014, with vaccine efficacy are above 90% for all three trials. These trials are also large with over 10,000 children aged between 6 to 71 months.

These vaccines might be a potential way to mitigate the risks of HFMD outbreaks. Considerations on whether EV-A71 vaccines are a viable control measure will require robust evidence in order to assess their potential value alongside or in place of the isolation and school closure policies that are now in place. Some of the required information will include the disease burden due to HFMD, covered in Chapter 3, and an economic study which investigates the cost-effectiveness of implementation, which lies outside the scope of the thesis.

1.2 HFMD in Singapore

HFMD surveillance in Singapore begun when HFMD was made a notifiable disease under the Infectious Diseases Act on 1 October 2000 during an EV-A71-associated outbreak between September and October 2000 that caused four pediatric deaths and over 3000 cases in two months [37, 64]. During this outbreak, Singapore implemented mandatory closure of all preschools to break transmissions, and the preschool closure policy have persisted until today.

Even though HFMD case-fatality rates are low in Singapore, HFMD remains as a concern

because it is a significant contributor to childhood morbidity. Between 2000 to 2015, annual incidence rates varied from 125 to 702 per 100,000, with large outbreaks occurring almost every year, reaching 37,000 cases in 2012, but dropped to 28,000 in 2015 (as a comparison, there are 41,000 live births in Singapore in 2017). Children under age 4 are disproportionately affected, rising from an incidence of 1,640 per 100,000 in 2001 to 9,029 per 100,000 in 2010 [42, 43]. Yet, even 9% peak incidence is not high enough to explain the evidence from serological studies. In a recent paper published by Ang et al [53], serological data collected from hospitalized patients aged 1-17 between 2008 to 2010 revealed that in the 700 subjects, 83.9% of them were seropositive to at least one of the three viruses: EV-A71, CV-A6 and CV-A16, showing that the infection rate for HFMD-causing agents is so high that close to the entire population gets infected at some point. Further evidence from two serological studies from Singapore in 1996/97 and 2008/10 suggests that between 25% and 50% of Singaporean children are infected by the time they reach formal schooling [64, 65].

Given the high incidence of HFMD, along with the past experience of death of children due to EV-A71, it is not surprising for HFMD to be taken seriously in Singapore. The surveillance system requires all healthcare practitioners and all childcare facilities to report HFMD cases to the ministry, who then consolidate the information and puts it on their website [66]. Pediatric clinics, general practitioners and the media are also involved with the education of the public on HFMD. Guidelines are available for infectious disease control in childcare centres [67]. Also, there are a number of school closure/isolation policies in force:

1. Preschools to notify the Ministry of Health whenever at least 2 cases of HFMD occur in the same 10-day period;
2. Home isolation of children diagnosed with HFMD for up to 10 days;
3. Publishing on a government website the names of schools with more than 10 cases or an attack rate greater than 13% over a period of 16 days; and
4. Closing preschools with more than 16 cases or an attack rate greater than 23% over 24 days [46].

The measures employed against HFMD appears to be effective — the mortality rate of

HFMD had not reached the levels observed in 2000 in the past 17 years but there was no comparable data before and after the implementation of the closure to determine whether there is causality behind the temporality of the measure.

Although symptoms of HFMD are generally mild, disease control efforts are potentially burdensome to caregivers socially and economically. There is qualitative evidence to show that parents have the tendency to voluntarily distance their healthy children during outbreaks leading to children missing school by a month or more [46]. Siegel, Cook and La (2017) also showed that both parents and teachers perceive HFMD as a severe disease and experience a sense of helplessness when hygiene and social-isolation efforts were not effective, and thus willing to accept high social costs if isolation/closure policies are proven effective.

Although studies have identified risk factors and transmission pattern of EV-A71 infections in the Asia-Pacific region such as Taiwan, China, Malaysia and Japan, many open questions remain. In the face of limited information, policy makers will use heavier strategies to curb the immediate threat. For instance, the response of the Singapore government to the H1N1 pandemic included isolation of confirmed cases, mandatory quarantining, school closures, temperature screening in hospitals and public places - policies that were effective against SARS but did not prevent a local epidemic of pandemic influenza [68]. Through this thesis, we revisit all the key assumptions behind the policies and conduct a detailed investigation to provide recommendations to how should this disease be handled from this point onward.

1.3 Our Approach

As identified in the previous section, there are many unclear areas of HFMD despite being endemic in Asia for many decades. This thesis is written with a focus on making discoveries which can drive policy discussions and targets three specific problems in three Chapters.

In Chapter 2, we identify what are the known and unknown unknowns through a rigourous systematic review which covers eight specific research questions on HFMD in Asia. The eight specific research questions are as follows:

1. What time of the year do HFMD outbreaks occur, and with what seasonal factors are

outbreaks associated?

2. How long have EV-A71 and CV-A16 been circulating in Asia?
3. What age groups are at higher risk of infection?
4. What risk factors are associated with infection and severe outcomes?
5. Where do infections predominantly occur (home or school)?
6. What is the incubation period?
7. What proportion of infections are symptomatic?
8. What is the basic reproduction number for HFMD by virus?

The findings from this research project provides a current state assessment which will guide policy decisions and researchers on what are the potential areas for further research for HFMD.

In Chapter 3, we address some of the unknowns identified in the systematic review with a modeling paper that is made of four different models, combined to calculate the disease burden (in DALY) and severity (proxied by the probability of entering into more severe states) of HFMD, which forms the basis of comparison against other diseases. One of the consequences in this paper is the estimation of asymptomatic rates which could prove to the key factor for reconciling the estimates of the basic reproduction number in the literature.

In Chapter 4, we shift the focus towards policy and identify parents' preferences and their corresponding willingness-to-pay for different intervention/disease outcomes, and their adherence to recommended practices such as withdrawing children from school if sick. The direct impact of the findings of this paper is to allow policymaking not purely from the economic standpoint, but also from the receiver's viewpoint. Policies set with consideration from both points of view will lead to greater success through greater adherence.

Chapter 2

The Epidemiology of Hand, Foot and Mouth Disease in Asia: a Systematic Review and Analysis

2.1 Abstract

Context: Hand, foot and mouth disease (HFMD) is a widespread pediatric disease caused primarily by human Enterovirus 71 (EV71) and Coxsackievirus A16 (CA16).

Data Sources: PubMed, Web of Science and Google Scholar were searched up to December 2014.

Study Selection: Two reviewers independently assessed studies for epidemiological and serological information about prevalence and incidence of HFMD against predetermined inclusion/exclusion criteria.

Data extraction: Two reviewers extracted answers for eight specific research questions on HFMD epidemiology. The results are checked by three others.

Results: HFMD is found to be seasonal in temperate Asia with a summer peak, and in sub-tropical Asia, with spring and fall peaks, but not in tropical Asia; evidence of a climatic role was identified for temperate Japan. Risk factors for HFMD include hygiene, age, gender and social contacts, but most studies were underpowered to adjust rigorously for confounding variables. Both community-level and school-level transmission have been implicated but their relative importance for HFMD is inconclusive. Epidemiological parameters are poorly understood: No supporting quantitative evidence was found for the incubation period of EV71; the symptomatic rate of EV71/CA16 infection was 10% to 71% in four studies; while

the basic reproduction number was between 1.1 to 5.5 in three studies. The uncertainty in these estimates inhibits their use for further analysis.

Limitations: Diversity of study designs complicates attempts to identify features of HFMD epidemiology.

Conclusions: Knowledge on HFMD remains insufficient to guide interventions such as the incorporation of an EV71 vaccine in pediatric vaccination schedules. Research is urgently needed to fill these gaps.

2.2 Introduction

Hand foot and mouth disease (HFMD) has become an endemic childhood disease in East and South East Asia. Its main etiological agents are Human Enterovirus 71 (EV-A71) and Coxsackievirus 16 (CV-A16). Although usually mild with symptoms limited to >38°C fever, malaise, rashes on the volar regions of the hands and feet, herpangina and difficulty eating and drinking - more rarely, infection can lead to complications of the nervous or cardiopulmonary systems. Such cases can result in long term sequelae such as cognitive and motor disorders [69, 70] or death, usually from pulmonary edema or brainstem encephalitis [2]. Although complications are rare, the number of children being infected in high incidence countries such as China (about 2.7M cases in 2014 [2]) means the death toll can be substantial (384 deaths in China in 2014 [2]). The EV-A71 virus seems to be responsible for more severe outcomes, while CV-A16 and other Coxsackieviruses, such as CV-A2, CV-A6 and CV-A10, usually present milder symptoms that resolve within a few weeks [3, 4, 5].

There are nearly 25 years of literature from Asia that describes the epidemiology of HFMD, drawing on pediatric cohorts, national surveillance systems, outbreak investigations and clinical data, and from disparate countries that span stages of economic development and with climates that range from tropical to temperate. This diversity complicates attempts to identify general features of HFMD epidemiology and conceals gaps in the body of knowledge of this important pediatric disease.

The objective of this paper is to provide a robust systematic review of the epidemiology

of HFMD that informs public health policy making about HFMD epidemics. The review covers three major areas: (i) history and seasonality of HFMD, and the efforts in predictive modeling; (ii) risk factors for infection, to guide control; and (iii) global epidemiological parameters, such as the incubation period and basic reproduction number, which may determine the effectiveness of control policies.

2.3 Methods

2.3.1 Search Strategy and Selection Criteria

Using a combination of search terms, including 'Hand foot and mouth disease', 'Hand foot and mouth', 'HFMD', 'Enterovirus', 'Enterovirus 71', 'EV71', Coxsackie A16', 'CA16', 'CVA16', we searched 'All Fields' in PubMed, Thomson Reuters Web of Science and Google Scholar to identify 1305, 1255 and 100 articles respectively (Figure 2.1).

Eligibility criteria were articles that:

1. Were published in peer reviewed journals from January 1957 to Dec 2014;
2. Were studies with epidemiological and/or serological information (quantitative/qualitative) about incidence, prevalence of HFMD; and/or
3. Contained information about factors associated with prevalence and incidence and/or
4. Employed statistical models to derive the above.

Articles not in English, not related to HFMD, or HFMD articles that did not cover epidemiological or clinical factors were excluded.

Two independent readers examined each of the 407 abstracts to determine if specific research questions were answered. The eight specific research questions are as follows:

1. What time of the year do HFMD outbreaks occur, and with what seasonal factors are outbreaks associated?
2. How long have EV-A71 and CV-A16 been circulating in Asia?

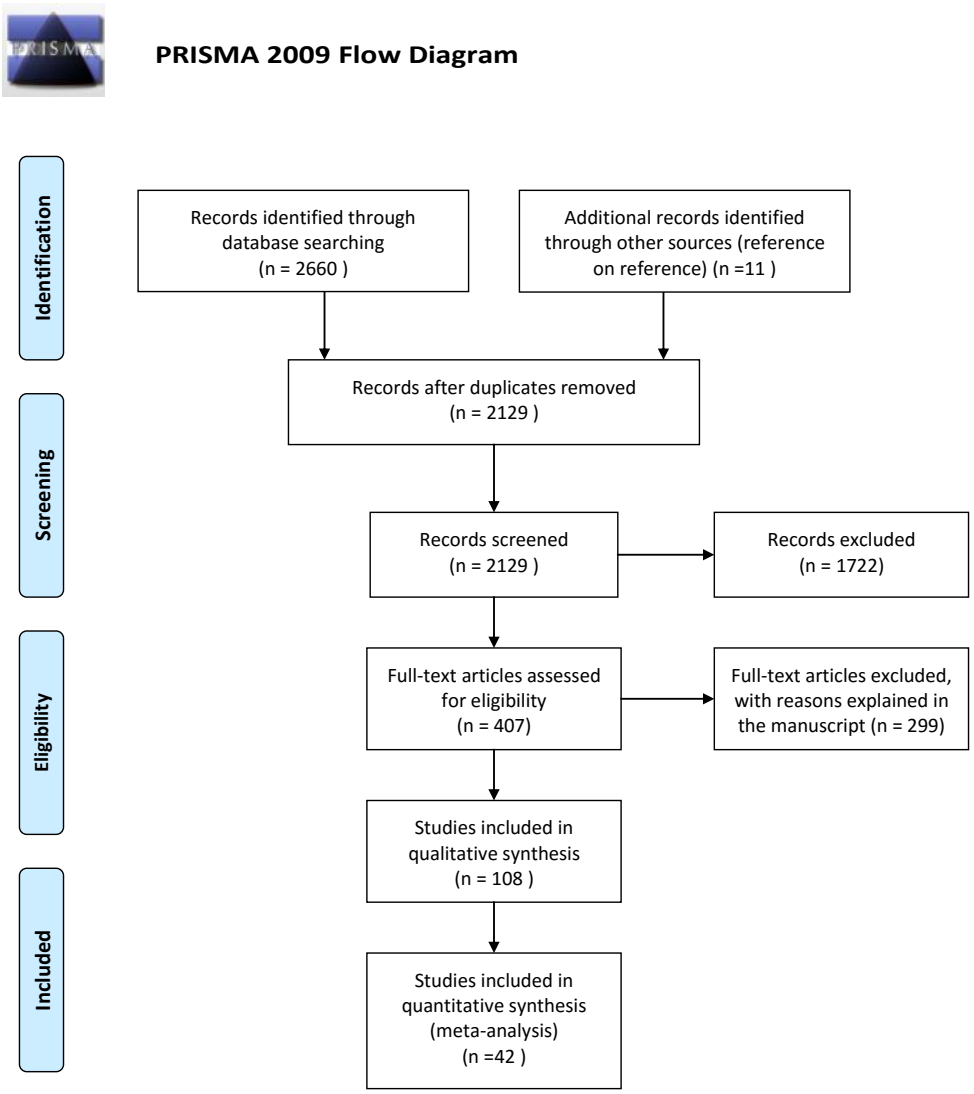


Figure 2.1: PRISMA 2009 Flow Diagram

3. What age groups are at higher risk of infection?
4. What risk factors are associated with infection and severe outcomes?
5. Where do infections predominantly occur (home or school)?
6. What is the incubation period?
7. What proportion of infections are symptomatic?
8. What is the basic reproduction number for HFMD by virus?

An article was retained as long as both readers indicated that it answered at least one specific research question and was discarded if both readers agreed that no questions were answered. A third independent reader arbitrated when there was a disparity between the original readers.

The two original readers each read the full text of half of the articles to identify answers to the questions. A second pair of independent readers read the articles again. Finally, the first author compiled all answers to the specific questions and compared the extracted answers to the original text. Relevant references from these papers were included in the analysis, in particular to identify non-English and early references. In total, information from 242 papers was compiled and 108 papers were used in data synthesis.

Hourly weather data were downloaded from the Weather Underground and aggregated at a weekly scale. Incidence data from Tokyo, Hong Kong, Taiwan and Singapore were extracted from routine surveillance data published by government agencies (the National Institute of Infectious Diseases, Japan [56], the Department of Health, Hong Kong [55], the Taiwan National Infectious Disease Statistics System [54], and the Ministry of Health, Singapore).

To assess how general the relationship between climate and transmissibility of HFMD was, we took incidence data from Tokyo, Hong Kong, Taiwan, and Singapore (Table 2.1), i.e. spanning temperate, sub-tropical and tropical latitudes, and used autoregressive models to investigate the effect of meteorological variables after correcting for contagion via autoregression.

A lag 2 model can be specified as follows:

$$HFMD_t = c + A_1 HFMD_{t-1} + A_2 HFMD_{t-2} + B_0 WEA_t + B_1 WEA_{t-1} + B_2 WEA_{t-2} + \epsilon_t$$

The A coefficients are the coefficients for the AR terms, while the B coefficients represent how a change in weather is correlated with changes in HFMD incidence. The number of lag terms are determined by the Akaike information criterion values of the regression models. This same model was used for 4 countries - Japan (lag 2), Hong Kong (lag 3), Taiwan (lag 3) and Singapore (lag 2), for three meteorological parameters - temperature, absolute humidity and relative humidity. As this model carries autocorrelated terms, we used Generalized Least Squares (GLS) for model fitting. Coefficients from the fitted models are presented in Figure 2.3 (bottom) and in detailed tables (Tables 2.8, 2.9, 2.10).

Another model which delivered similar results was the Structural Vector Autoregressive (SVAR) model which the 1-lag version can be specified as follows:

$$HFMD_t = \gamma_{10} - b_{12}WEA_t + \gamma_{11}HFMD_{t-1} + \gamma_{12}WEA_{t-1} + \epsilon_{1t}$$

$$WEA_t = \gamma_{20} - b_{21}HFMD_t + \gamma_{21}HFMD_{t-1} + \gamma_{22}WEA_{t-1} + \epsilon_{2t}$$

which can alternatively be written in matrix form:

$$\begin{bmatrix} 1 & b_{12} \\ b_{21} & 1 \end{bmatrix} \begin{bmatrix} HFMD_t \\ WEA_t \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \gamma_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} \begin{bmatrix} HFMD_{t-1} \\ WEA_{t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}$$

where ϵ are exogenous normally distributed error terms with means 0, and different standard deviations. The endogeneity of $HFMD_t$ and $temp_t$ is determined by the values of b_{12} and b_{21}.

We can assume that HFMD incidence will not affect the weather, and thus set b_{21} and $\gamma_{21} = 0$:

$$\begin{bmatrix} 1 & b_{12} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} HFMD_t \\ WEA_t \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \gamma_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ 0 & \gamma_{22} \end{bmatrix} \begin{bmatrix} HFMD_{t-1} \\ WEA_{t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}$$

Rewriting the equation as $\boldsymbol{B y_t = \gamma_0 + \Gamma_1 y_{t-1} + \epsilon_t}$, we can see that multiplying $\boldsymbol{B^{-1}}$ to the equation will convert the model to $\boldsymbol{y_t = B^{-1}\gamma_0 + B^{-1}\Gamma_1 y_{t-1} + B^{-1}\epsilon_t}$ which is just a normal VAR model which can be calculated by the R package 'vars' [71]. The results from the SVAR model are shown in Figure 2.4.

Non-tabular data were extracted from figures using Plot Digitizer [72]. Data on weather and incidence were analyzed using a time series model. Symptomatic proportions were pooled by aggregating denominators and numerators. Other analyses used standard statistical methods and were conducted using R [73].

2.4 Results

2.4.1 Timing and Seasonality of HFMD Outbreaks

Outbreaks of HFMD do not occur uniformly throughout the year across Asia. In Fukuoka, Japan, for example, weekly numbers of HFMD cases have been found to increase with average temperature and humidity, especially among younger children [74]. By digitizing incidence data from publications on Japan [74, 31, 48, 32] and North China [6, 75, 7, 76, 8, 77] (Figure 2.2), we observe that May through July are the months with highest incidence in temperate regions of Asia. However, this relationship is less clear for tropical and subtropical Asia. The extracted data on Southwest China [6, 9], South China [6, 10, 78, 79], Hong Kong [27, 28] and Taiwan [4, 80, 81] shows that outbreaks typically happen in late spring and fall. No distinct pattern is obvious for tropical regions as seen from data in Thailand [82, 39, 40], Vietnam [41, 83], Malaysia [51] and Singapore [3, 37, 84, 85], where outbreaks occur sporadically throughout the year, although models have been developed for Singapore (around 1° North) that show a positive statistical relationship between maximum daily temperature above 32°C with HFMD incidence in the subsequent 1-2 weeks [84].

After controlling for contagion via autoregression terms, the effect of meteorological factors was weak: a small positive increase in transmissibility with rising absolute humidity/temperature during the current week in Tokyo and Singapore. There was no evidence for temperature and humidity in having the same effect in Hong Kong or Taiwan, although rising relative humidity seems to decrease transmissibility in Singapore.

The earliest recorded cases of HFMD in Asia are from Japan (1967) [86], Singapore (1970) [87], Taiwan (1980) [88] and Shanghai, China (1981) [89]. Since then, outbreaks have been reported in many parts of Asia, including mainland China [7, 76, 9, 10, 89, 11, 90, 91, 92, 93,

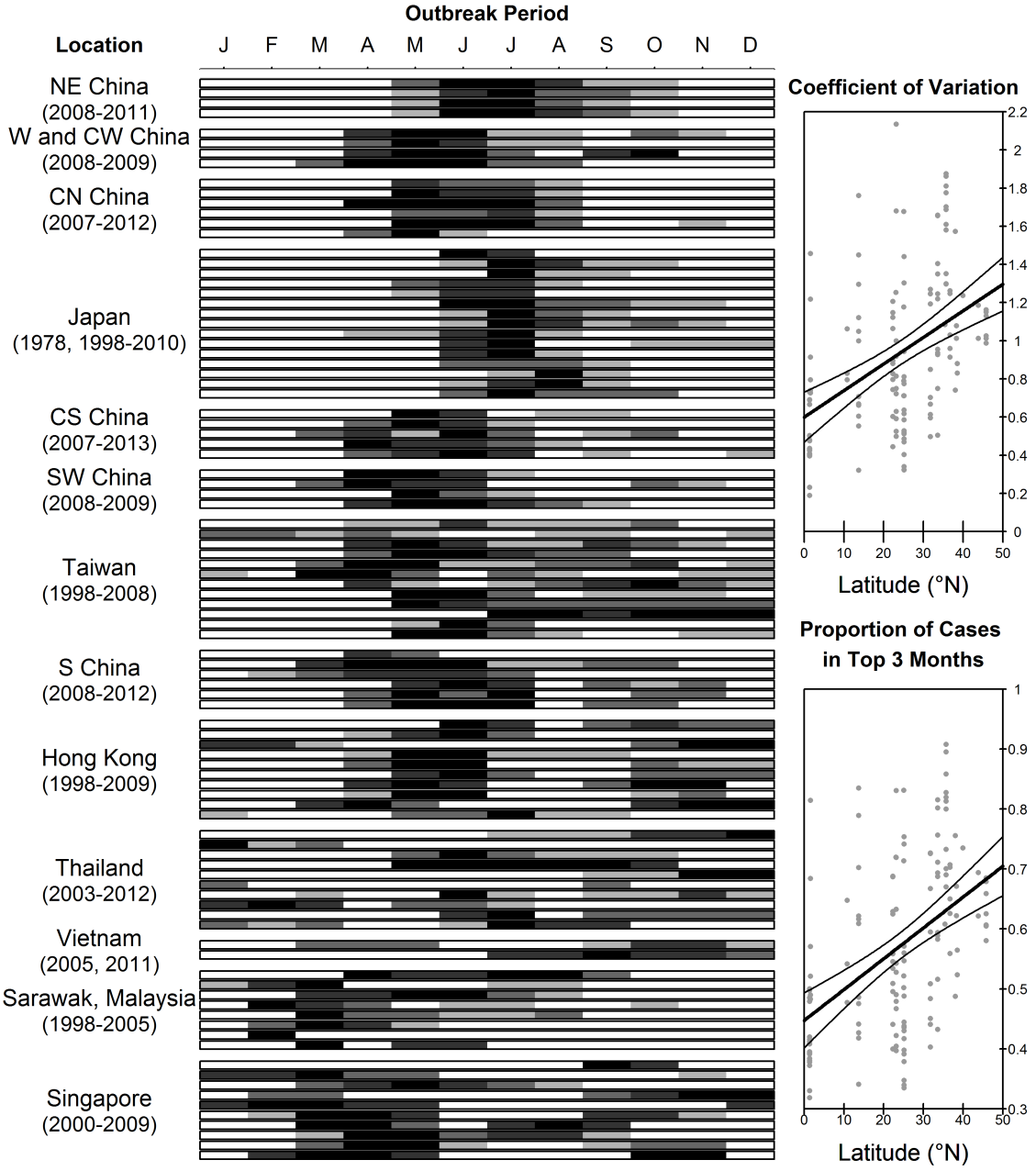


Figure 2.2: Temporal Patterns of HFMD Outbreaks in Asia, by Latitude. Left: Plot Digitizer is used to convert charts into numbers. White boxes are the months where HFMD cases fall below the year's median. The remaining cells are then shaded into 4 darker shades by octiles. The regions of China were based on Wang et al's classification[**?**] (C standing for central). The regions are arranged by latitude. South China, Hong Kong and Taiwan have subtropical climates. Areas further north are temperate, while the South East Asian regions are tropical. Right: The coefficient of variation is the ratio of the standard deviation to its mean, and the proportion of cases in top 3 months is the proportion of cases of the 3 months with highest incidence to the annual incidence. Points represent one year per region. The lines are obtained from ordinary least squares regression with Latitude as the independent variable, and show how clearly defined epidemics become the further north from the equator.

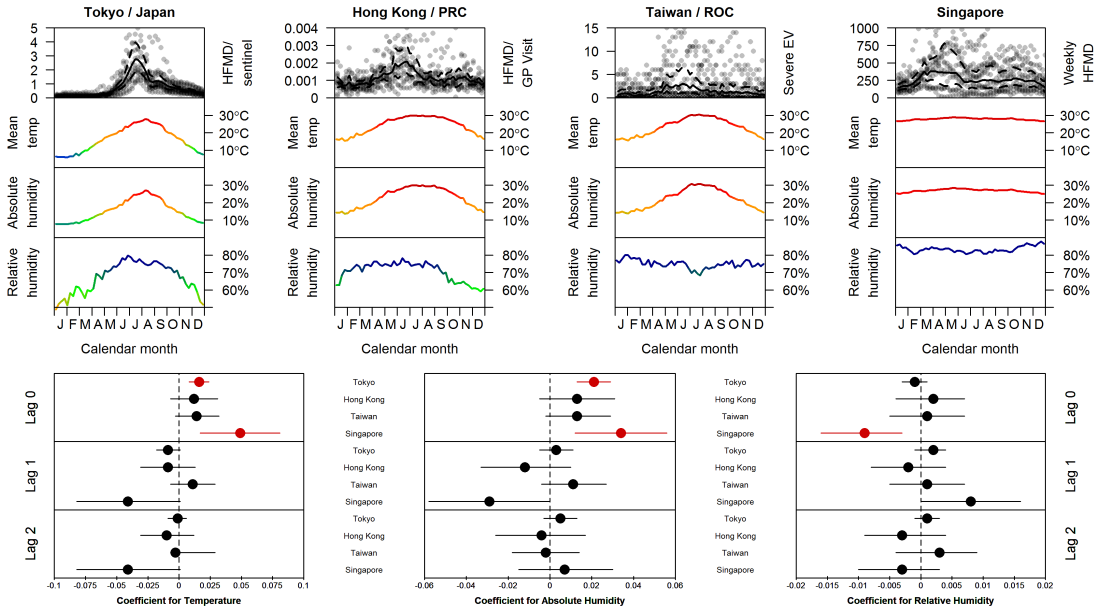


Figure 2.3: Temporal Incidence of HFMD or Enterovirus and Climatic Factors for Four Asian Cities Spanning Temperate, Subtropical and Tropical Latitudes. Top: incidence, temperature, absolute and relative humidity. Top panels indicate incidence (data in gray, mean and 95% interval in black) for the time period Jan 2001-Mar 2012 (Tokyo), Jan 2001-Dec 2009 (Hong Kong, Peoples' Republic of China), Jan 2001-Dec 2011 (Taiwan, Republic of China), Jan 2001-Jan 2012 (Singapore). Middle and bottom panels show mean daily temperature, absolute and relative humidity at Tokyo Narita, Hong Kong International, Taipei Taoyuan International and Singapore Changi airports, downloaded from the Weather Underground. Bottom: Coefficients of meteorological variables at 0 to 2 week time lags in autoregressive models of z-scored HFMD case counts to facilitate comparison between locations. Each city is analyzed separately using a model in which HFMD incidence in week t is (auto)regressed on incidence in weeks t-1 up to t-3 (using the Akaike information criterion to select the order of the autoregression component) and, independently, on each meteorological variable. Weather parameters are not regressed together in a single model due to collinearity. The effect of weather on HFMD incidence can be seen by coefficient mean (points) and 95% confidence intervals (lines), colored red if statistically significant at the 5% level.

Table 2.1: Data Source for Climate-Incidence Analysis

Country	Data	Source
Singapore	Notified Cases (low underreporting, as notification is mandatory and laws are strict)	Ministry of Health, Singapore
Japan	Notified Cases, Cases per sentinel (not mandatory reporting)	National Institute of Infectious Diseases, Japan (Released weekly)
Hong Kong	Notified Cases per 1000 GP consultations (Not mandatory reporting)	Department of Health (Website and digitized from historical documents)
Taiwan	Enterovirus with complications	Taiwan National Infectious Disease Statistics System

61, 94, 95, 96, 97, 98, 99, 100, 12, 13, 14, 15], Korea [101, 33, 102], Japan[31, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116], Taiwan [5, 115, 52, 117, 22, 118], Hong Kong [27, 28, 119], India [29, 120, 121, 122, 30, 123], Thailand [82, 40, 124], Vietnam [41], Malaysia [51, 115, 49, 35, 125, 126, 127], Singapore [3, 38] and Brunei [128], as summarized in Figure 2.5. These reported outbreaks are unlikely to reflect the true first outbreaks of HFMD, as serological studies provide evidence that by the time surveillance systems were established, EV-A71 and CV-A16 were already endemic in many of these countries. Early serological tests conducted in Japan in 1970 show evidence of EV-A71 and CV-A16 circulation[129]. Serum taken in the late 1990s in Singapore, before the start of surveillance in 2000, shows that around 50% children and 44% cord blood, indicating maternal infection, had already seroconverted to EV-A71 [64]. Blood samples from Taiwan from 1989 to 1997 show 3% to 11% EV-A71 incidence per year, and up to 68% of children [130] had serological evidence of EV-A71 infection before the large HFMD outbreak of 1997. Similarly, although China has reported millions of HFMD cases since the beginning of the HFMD surveillance program in 2008, evidence from Anhui [99] shows high seroprevalence of up to 74.6% in older children before the 2008 outbreaks. Retrospective seroepidemiologic tests from blood serum

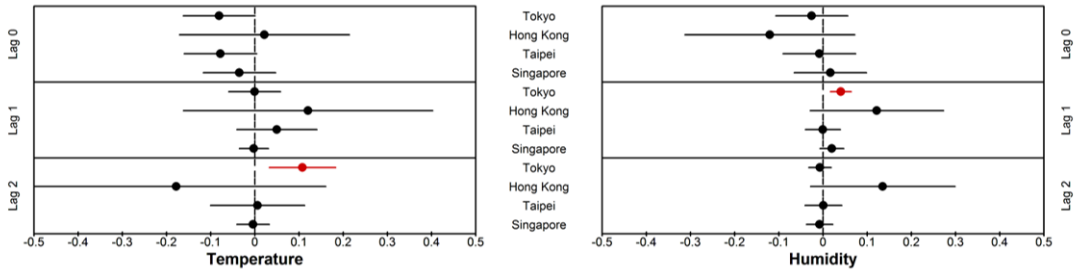


Figure 2.4: Coefficients of Meteorological Variables on Disease Incidence in Structural VAR model. Coefficients of meteorological variables at 0 to 2 week time lags in SVAR models of z-scored HFMD case counts to facilitate comparison between locations. Each city is analyzed separately using a model in which HFMD incidence in week t is (auto)regressed on incidence in weeks t-1 up to t-3 and independently on each meteorological variable. Weather parameters are not regressed together in a single model due to collinearity. The effect of weather on HFMD incidence can be seen by coefficient mean (points) and 95% confidence intervals (lines), colored red if statistically significant at the 5% level. Lag 0 parameter is b_{12}, while the lag 1 to 4 parameters are γ_{12} and the corresponding coefficients for WEA_{t-2} to WEA_{t-4}.

collected in 2005 [131] also show that China had positive rates of 32.0% to EV-A71 and 43.4% to CV-A16, indicating that outbreaks happened earlier but were simply not reported in the literature.

2.4.2 Risk Factors

Risk factors for infection are depicted in Figure 2.6 and summarized below.

Hygiene: Evidence from Qiaosi, China [16] indicates the importance of hygiene for protection against HFMD infection. Children who always wash their hands before meals are about 50 times less likely to contract HFMD, while those whose caregivers wash their hands before feeding are about 25 times less likely. Additional protective habits include washing of hands after play, washing of hands more than 4 times per day, using soap, and not sucking fingers. A study in Korea [34] revealed that drinking unboiled water (3.34 OR [1.59-6.99]), a change in water quality such as color, taste, smell, presence of precipitation or floating materials (6.93 OR [2.17-22.15]), using communal toilets/toilets outside the house (2.77 OR [1.14-6.74]), and eating outside the home (37.0 OR [5.1-269.5]) were risk factors for HFMD.

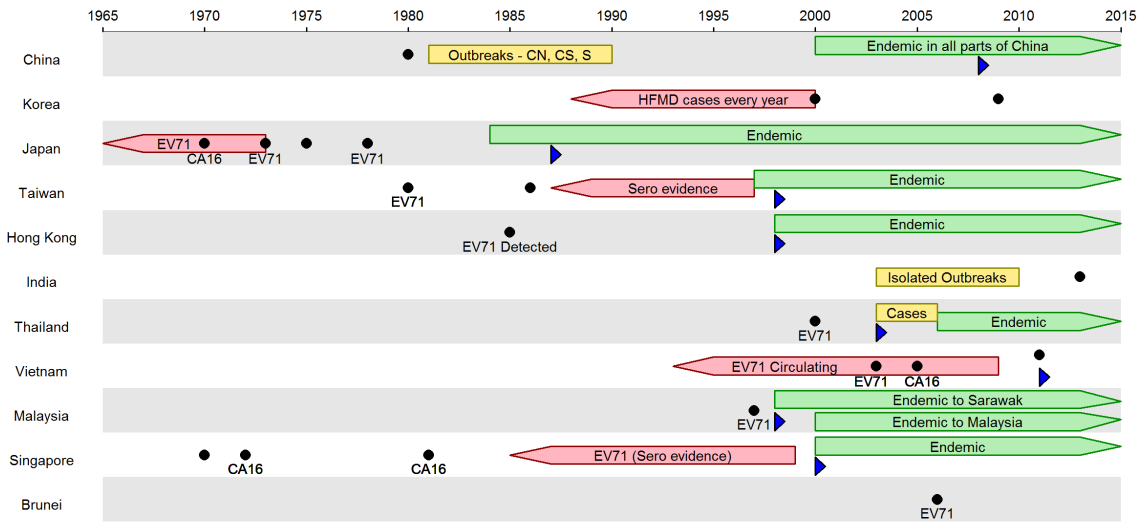


Figure 2.5: Historical Establishment of HFMD in Asia. Dots represent a reported outbreak in that year, with the main causal agent written below. Boxes with right arrow indicate endemicity of HFMD, evidenced by repeated reporting of HFMD. Boxes with left arrow indicate seroepidemiological evidence that the pathogen is already in existence, even if no significant outbreaks were documented previously. Triangles indicate the point where data on HFMD started to be collected systematically, for example, through government surveillance. The length of the left arrows are arbitrary as there is no way to know how long has HFMD been circulating prior to the tests.

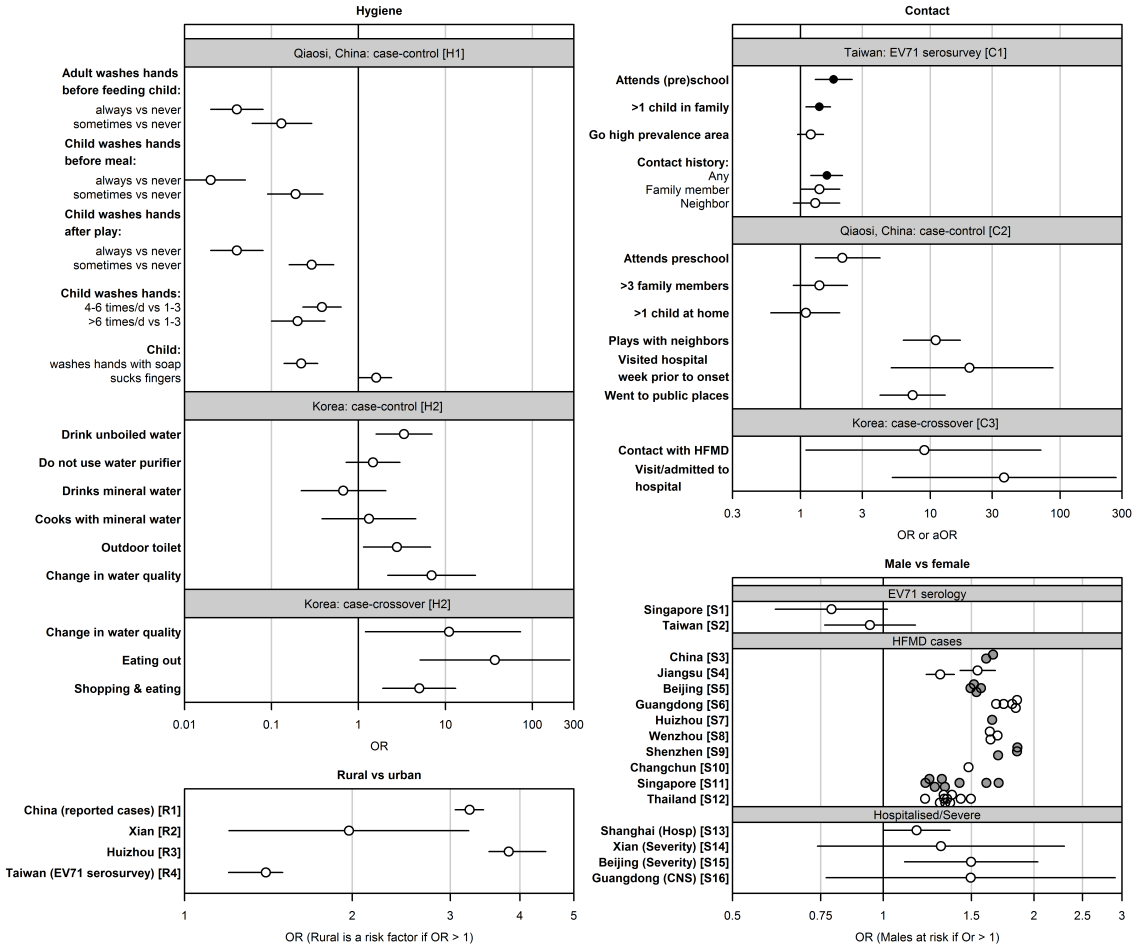


Figure 2.6: Excerpts from Studies on Risk Factors. Ruan et al's case-control study in Qiaosi [16], China, was conducted by taking 273 diagnosed HFMD/Herpangina as cases (age 6 years or younger), and 273 stratified random sample as controls. Park et al's case-control study [34] uses hospital cases of enteroviral Aseptic Meningitis (n = 205) and HFMD (n=116), and non-enteroviral disease controls (n = 170). Their case-crossover design uses only cases. 1-7 days prior to admission was set as the hazard period, and 22-28 days prior to admission was set as the non-hazard period. Both studies gather data via questionnaires. White points indicate non-adjusted odds ratios (ORs), black, adjusted (aOR). For the effect of sex (bottom right), confidence intervals are omitted from complete case notifications, and alternatingly colored gray and white for visual distinction.
Data sources:
Hygiene: H1 [16], H2 [34]
Rural/Urban: R1 [14], R2 [132], R3 [133], R4 [134]
Contact: C1 [134], C2 [16], C3 [34]
Sex EV71 Serology: S1 [65], S2 [134]
Sex HFMD Cases: S3 [6], S4 [14], S5 [92], S6 [10, 11], S7 [133], S8 [17], S9 [79], S10 [75], S11 [3, 37], S12 [40, 124]
Sex Severe: S13 [18], S14 [132], S15 [8], S16 [15]

Rural vs urban areas: All papers [14, 134, 132, 133] that compared urban with rural areas agreed that the latter conferred a higher risk for HFMD. However, this might be confounded by socioeconomic status and hygiene practices.

Sex: While most papers show that being male is a risk factor for both mild [3, 6, 75, 10, 79, 40, 37, 11, 14, 92, 124, 133, 17] and severe[8, 15, 132, 18] HFMD (OR ranges between 1.2 to 2), surprisingly, serological evidence does not support this finding: a study from Singapore [65] shows marginal evidence that females are more likely to have seroconverted to EV-A71 (OR 0.79 [0.61-1.01]) while a Taiwanese [134] study shows no statistically significant differences (OR 0.94 [0.76-1.16]). Taken together, these suggest that infection rates are comparable, but that boys are more likely to develop symptoms, more involved in propagation of outbreaks or more likely to be brought for medical care than girls.

Other: A case-control study in Xi'an, China [132], found that breastfeeding may lower the risk of developing severe HFMD (aOR 0.57 [0.33-0.98]), even though breastfeeding does not apparently lower the chance of being infected by EV-A71 (OR 1.1 [0.93-1.3]) [134]. It further found that patients with a history of Epstein - Barr virus are at greater risk of contracting severe, rather than mild, HFMD (aOR 2.6 [1.5-4.4]). A spatial-temporal model of Guangdong [10] showed that sunshine could be protective against HFMD. This is agreed by a matched case-control study of preschoolers in Beijing [61] which showed that UV radiation in classrooms is associated with lower HFMD attack rate (p-value 0.027), and recommended installing UV lamps to sterilize unoccupied classrooms. These findings are however, inconsistent with the seasonal nature of HFMD, where outbreaks in temperate countries tend to occur in summer, when sunlight and UV exposure is strongest.

2.4.3 Age Distribution of HFMD Cases

The age distribution of HFMD cases in Asia, compiled from a variety of sources including surveillance and cohort data, is summarized in 2.7. Data from China[6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21] and Taiwan[4, 5, 22, 23, 24, 25, 26] are particularly abundant. Other sources include Hong Kong[27, 28], India[29, 30], Japan[31, 32], Korea[33, 34], Malaysia[35, 36], Singapore[3, 37, 38], Thailand[39, 40] and Vietnam[41].

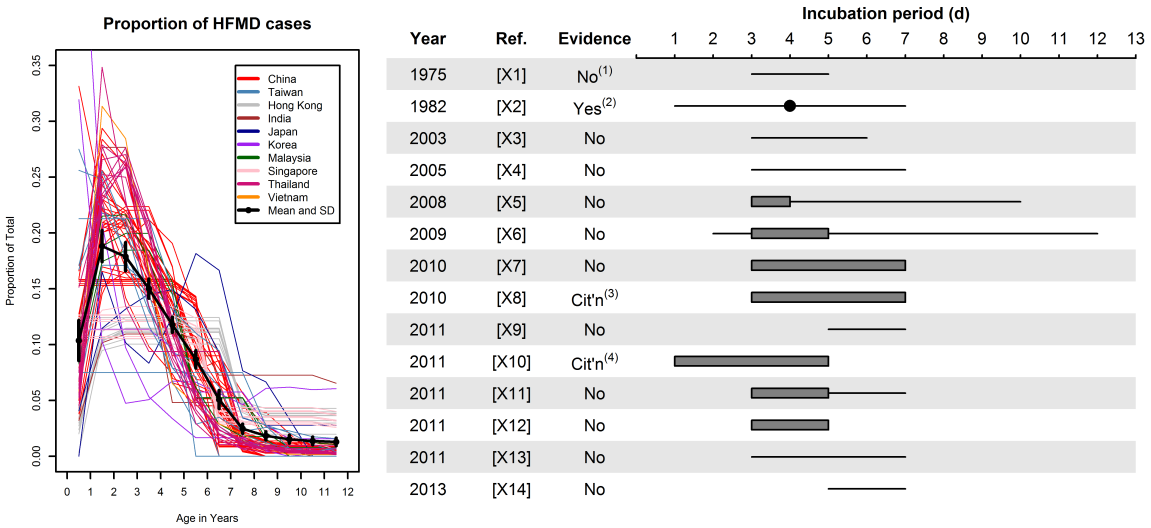


Figure 2.7: HFMD Cases by Age and Estimates of Incubation Period. Left: Each line indicates a unique data set (total 79 lines). Distributions within age ranges were assumed to be constant. The black dots are average proportion for that age (with 95% CI). Right: Reported incubation periods for EV-A71, by year of publication and provision of evidence to support claimed period. Lines indicate that the incubation period "is X days". Gray bars indicate that the incubation period is "usually" or "typically" X days. Gray bars with lines shows an extended interval "can be up to X days". The point indicates a median. Notes: (1) provides information within the paper which is inconsistent this estimate. (2) uses generation interval distribution as a proxy for incubation period. (3) Cites a US CDC factsheet on aseptic meningitis, which in turn provides no supporting evidence. (4) Cites Goh et al[38]. Data source: X1 [135], X2 [38], X3 [136], X4 [29], X5 [88], X6 [26], X7 [137], X8 [34], X9 [94], X10 [74], X11 [138], X12 [12], X13 [36], X14 [133]

There is a wide variation in the proportion of cases of HFMD within the narrow 0 to 6 year age-band. The greatest proportion of cases occur at ages 1 (18.8% [17.4-20.2%]) and 2 (17.9% [16.6-19.2%]). By the age of formal schooling, from 6 in most Asian countries, the proportion is substantially lower (8.7% [7.9-9.5%]). Overall, 82.6% (82.2-82.9%) of all cases occur before age 6. The lower proportion during the first year of life could be due to lack of contact with other children or to presence of maternal antibodies [64].

2.4.4 Community versus School as Medium for Infection

The literature is ambiguous about the importance of locations for transmission. Four studies showed that contact with a case, particularly a household member, is as or more significant a risk factor than preschool attendance [82, 38, 134, 23]. An early study in Singapore observed 60 families with secondary cases and found the secondary attack rate amongst children below 12 years old to be 77% [38]. Similarly, in a large seroepidemiological study of EV-A71 in Taiwanese children [134], multivariate analysis showed attendance at a preschool imparted a similar magnitude of risk as contact with a case (adjusted ORs of 1.6 [1.2-2.1] and 1.8 [1.3-2.5], respectively), as well as a strong concordance (84%) between seropositivity in younger and older siblings.

Also, a number of studies showed that a higher percentage of diagnoses occurred among children who did not attend a nursery or preschool [92, 14]. Liu et al [12] note that about half of symptomatic cases in Nanchang, China, are among children under 3, the age at which preschooling starts in China.

Conversely, some studies suggest that preschool attendance is a key risk factor [3, 139]. For example, a seroepidemiological study in 1996-7 in Singapore showed that seropositivity to EV-A71 increases rapidly from age 2 to age 59 [56], when attendance at childcare or preschool is the norm. Also, a case-control study in Japan [139] showed that preschool attendance was associated with increased risk of severe disease.

Other studies suggest that both locations are important. In Shanghai, China, [18] there was a marked shift from 2007 to 2008 in the proportion of cases among children in preschools (from 59% to 37%) with a concurrent shift from local to migrant children, suggesting that

the importance of routes of transmission can vary over time within the same locale. A case-control study from Zhejiang [16] showed that while attending preschool is a risk factor (2.1 OR), other factors such as contact with neighbours (11 OR), going to hospital (20 OR), and going to parties (31 OR) impart greater risk. Yet, a Korean case-control study [34], found no significant relationship between infection and school attendance or household size.

Overall, the evidence points to both home and school environments contributing to transmission, but the relative importance of these venues remains murky.

2.4.5 Incubation Period of EV-A71

Several papers describe the incubation period (Figure 2.7), though it is striking that the majority do not provide a source to justify the claimed period. These unsupported claims vary substantially from paper to paper, from the incubation period 'is' 3 - 6 days [136], or 3 - 7 days [29], 'is usually 3 - 4 days, but can be ... 10 days or more' [88], or 'is usually 3 to 5 days (range, 2 to 12 days)' [26], is 'typically' 3 - 7 days [137] or 3 - 5 days [12], ranges from 5 to 7 days [94, 133] or 3 to 7 days [36], and the 'usual period' is 3 to 5 days 'with longest period of 7 days' [138]. Only a few provide evidence to justify the claim: one reports [34] that the incubation period is usually 3 to 7 days, citing a US CDC factsheet on aseptic meningitis. Another cites [74] an early study from Singapore [38] that presented the median and range for the serial interval (3 days [1 - 7]), not the incubation period. Another early study [135] states that the incubation period is 'said to be' 3 to 5 days, but notes that this is inconsistent with the serial interval observed in the study. It appears that there is no empirical support whatsoever for any distribution of incubation periods.

2.4.6 Symptomatic Proportion

Although several studies report that the asymptomatic rate of EV-A71 infection is high, few studies report data (Table 2.2). Two studies, from Taiwan and Shanghai, tested sera for evidence of EV-A71 infection and asked patients or their families to recall past HFMD infection, deriving estimates of 29% [140] and 10% [20] of symptomatic infection respectively. Some HFMD cases may have been caused by other enteroviruses, biasing these estimates

upwards, while some may have been diagnosed as another viral illness or forgotten, biasing them downwards.

Table 2.2: Estimates of Symptomatic Proportion from the Literature.

Location	Ref	Year	Inclusion Criteria	Sympt	Inf	Symp/Inf (95% CI)
Taiwan	[140]	1998	Stratified sampling. Infection determined using serology vs EV-A71 and recall of previous infection history	140	484	28.9 (24.9 - 33.0)
Shanghai	[20]	2010-11	Routine blood samples and recall of previous infection history	12	122	9.8 (4.6 - 15.1)
Taiwan	[23]	2001-02	Household members of hospital symptomatic cases were swabbed for EV-A71	119	176	67.6 (60.7 - 74.5)
Taiwan	[141]	2006-09	Prospective cohort of neonates with repeat serology (EV-A71) and swabs taken upon subsequent illness	20	28	71.4 (54.7 - 88.2)
Total				291	810	35.9 (32.6 - 39.2)

Two additional studies in Taiwan found much higher symptomatic infection rates. The first [23] recruited symptomatic cases suspected of having EV-A71 infection, and took throat and rectal swabs or stool samples, of cases and their household members with repeat telephone interviews. Excluding the 94 symptomatic index cases, 68% of confirmed infections in the household were symptomatic (88% of infected children and 47% of infected adults). A second study [141] prospectively followed a cohort of neonates over 3 years, taking repeat

sera, requesting that parents report suspected HFMD and giving reminders during HFMD epidemics. This study found that 71% of serologically confirmed infections were symptomatic, though the sample size is only 28.

The discrepancy between these two pairs of papers is substantial, undoubtedly due to differences in methodology. An overall estimate, combining the four studies, is 36% (33%-39%) but given the large discrepancy between studies this estimate does not appear reliable. The latter pair of studies are prospective, thereby circumventing recall bias, and thus appear to provide a more accurate description of the epidemiology of enterovirus infection.

2.4.7 Basic Reproduction Number for HFMD by Virus

Only three papers have sought to estimate the reproduction number for HFMD or the viruses that cause it. One [6] estimates what they call the 'local effective reproduction number' in China - meaning using the average number of secondary cases from a randomly selected index to estimate the cases that would be caused in a fully susceptible population (note, this is substantially different from the effective reproduction number [142] in a partially susceptible population) - using a sophisticated Poisson regression model that incorporated infection from the environment, the prefecture and neighboring prefectures. This model did not, however, account for the accumulation of herd immunity, and required arbitrary assignment of the infectious period, so the estimated local effective reproduction number of 1.1 to 1.2 during peak periods may be biased.

A second paper [138] used a method from Choi and Pak [143] to estimate the basic reproduction number to be 5.5 (IQR 4.2 to 6.5) for EV-A71 and 2.5 (IQR 2.0 to 3.7) for CV-A16. These estimates are likely inaccurate because the method assumes (i) a known generation time distribution, labelled incubation period in the paper; (ii) a completely immunonaÃŕve population, though applied to groups of individuals for whom past exposure was highly plausible; and (iii) an early exponential growth period, despite being applied to complete outbreak data.

The third paper [144] attempted to estimate the reproduction number using a SEIQRS (susceptible, exposed, infectious, quarantined, recovered) simulation model, and obtained

an estimate of 1.1 for the years 2009 to 2012 in China. However, the model used 10% of China's population as the initial susceptible population but did not conduct a sensitivity analysis on this vital parameter.

2.5 Discussion

Despite the substantial number of papers on HFMD, this systematic review shows that many fundamental questions about EV-A71 and CV-A16 persist. Both viruses occur year round in tropical Asia, but are epidemic in the summer in North-East Asia. A role for temperature or humidity therefore seems plausible [48, 10, 60, 145, 146], although given the relative lack of seasonality in equatorial Asia, it is not clear whether prediction of outbreaks is possible there. In Japan, summer temperatures peak after HFMD incidence does, suggesting correlation but not temporality, and that it may not be possible to provide early warning of impending epidemics. This also differs from other human enteric viruses including Poliovirus 1 (also an Enterovirus), Hepatitis A, and Adenovirus that have been shown to survive longer on colder surfaces [59].

Urashima et al (2003) [48] claimed that enteroviruses experience a more rapid virus decline during dry seasons than during wet seasons, which could explain the seasonality. This result is supported by Wang et al (2011) [147], where they showed that precipitation patterns has the most similar structure as HFMD incidence, more so than other meteorological variables, albeit with only 11 months of data.

While any causal relationship between climate and HFMD is unknown, speculations include a lower HFMD incidence due to decreased social contact during temperate zones' winter [74, 148]. In contrast, increased social contacts during winter have been speculated to facilitate spread of other droplet borne diseases, such as influenza [149], which are epidemic in winter. Given the unknowns surrounding this issue, further research is clearly required to ascertain whether meteorological factors or seasonal social contact patterns is an adequate explanation for the seasonality of HFMD.

The next step to analyzing the dynamics of HFMD seasonality is likely to involve social

and environmental factors - another under-researched area for this pediatric disease. For instance, the literature is unclear on the relative importance of school versus community transmission, with evidence to support both, yet knowledge of where HFMD most often is transmitted is important as school closure policies are employed to control outbreaks in some countries. Further, the environment of schools in Asia may vary widely, and attributes such as hygiene practices should be characterized and quantified to allow more definitive results and conclusions in future studies.

Even without being able to determine the relative importance of school versus community transmission, the effectiveness of school closure to prevent large scale HFMD outbreaks is questionable, as the interruption to social networks cannot be enforced while children are out of school. Additionally, while we know little about the infectiousness of asymptomatic cases of HFMD, the proportion of infections that are asymptomatic is substantial, and so even quite modest school closure attack rate thresholds, such as Singapore's 25% [150], corresponds to a possible majority of students being infected prior to the trigger for closure being met. Further, EV-A71 can be found in fecal samples for up to 54 days after infection [151], and thus continue to be shed after a school is closed, disinfected, and reopened.

Studies on risk factors were rare, and we identified only 3 papers that describe risk factors for hygiene and contact patterns, making a meta-analysis of risk factors unfeasible. These typically were only powered to provide unadjusted effect sizes, and so provide evidence of correlation, not causation. One interesting finding was the apparent protective effect of a caregiver 'always washing' their hands. This suggests that adult to child transmission might be important, even if adults are mostly asymptomatic with EV-A71 and CV-A16, but may reflect confounding with general hygiene. Future work may elicit hygiene factors at the preschool level and relate these to attack rates. A recently developed EV-A71 vaccine has undergone phase 3 trials in China [44, 62, 63]. To determine the cost-effectiveness of incorporating the vaccine in paediatric vaccination schedules, or of other interventions such as school closure or isolation, would require epidemiological models that account for the protective effects of herd immunity. However, this review indicates that vital parameters for such models remain unknown. The asymptomatic rate and relative infectiousness of asymptomatic cases are both poorly known, while estimates of the incubation period,

although commonly cited as 3 to 5 days, appear to be based solely on expert opinion. Most importantly, estimates of the basic reproduction number range widely from 1.1 to 5.5. This uncertainty prohibits utilitarian estimation of the necessary vaccine coverage to prevent epidemics of EV-A71.

To reconcile the differences between the disparate estimates, the age distributions of the samples need to be considered. As shown in this review, symptomatic HFMD incidence rate differ greatly even between ages 0 to 6, and thus studies conducted predominantly on pre-schoolers may derive higher estimates of R_0 compared to studies in older children. Accordingly, future studies on HFMD should use narrower age bands and also state the distribution clearly to allow adjustments or standardization.

Two final omissions from the literature are quantitative estimates of the impact of infection on complications, child and caregiver absenteeism, and costings of complications, and qualitative evidence on the impact of infection and enforced isolation on families and schools. Given the promising direct effects of the EV-A71 vaccine, and the huge public health impacts of HFMD in East and South East Asia, research is urgently needed to fill these gaps.

The research questions in this systematic review were generally answered only by a limited number of papers, with substantial differences in their study design, and thus most data were not synthesized through meta-analysis. More research to assess risk factors and measure key epidemiological parameters is needed. We were also unable to trace the earliest cases of HFMD in Asia as our scope only covers published material on outbreaks, which leads us back to 1967 in Japan. Finally, we limited the scope of our study to exclude virological characteristics or molecular epidemiology, which have been well reviewed elsewhere [137, 152, 153, 154, 155], and clinical manifestations of EV-A71 and CV-A16 [137, 80, 152, 155, 156]. A recent review of the case-fatality rate has recently been published [157], as has a review of the epidemiology in Taiwan [158].

2.6 Acknowledgments

We thank Alex S Leow, Vincent J Pang for screening some of the papers and Zhao Xiahong for processing the meteorological data.

2.7 Footnote

This chapter is written based on the original Open Access article [159] distributed in accordance with the Creative Commons Attribution Non Commercial No Derivative (CC BY-NC-ND 4.0) license, which does not permit others to distribute work which is adapted or built upon the original paper.

2.8 Additional Materials

2.8.1 Protocol

1. **Objectives of Review**

   To summarise existing literature on the epidemiology of Human Enterovirus species A infections, especially Coxsackievirus A16 and Human Enterovirus 71, the two viruses most associated with hand, foot and mouth disease, in Asia.

2. **Secondary Objectives**

   - What times of the year do HFMD outbreaks occur in countries to which it is common?
   - What are the seasonal factors that influence HFMD in those countries?
   - Is there evidence of how long the two main viruses has been circulating in those countries?
   - What risk factors have been associated with (i) infection or (ii) severe outcomes?
   - What is the incubation period for HFMD by virus?

- What proportion of infections are symptomatic?
- What is the basic reproduction number for HFMD by strain?
- Where do infections predominantly occur (home vs childcare)?

3. **Methods**

   - Search Strategy:

     Sources: PubMed, SSCI Web of Knowledge, Google Scholar

   - Search terms

     Hand Foot and Mouth Disease / Epidemiology / Hand Foot and Mouth (X) / Risk / HFMD /Species A / Enterovirus / Incubation Enterovirus 71 / sero / seroepi / seropre / serocon / seropos / seroin / EV71 Incidence / Coxsackie / A16 / CA16 / CVA16

     For Google Scholar, the first 100 papers will be retained.

4. **Inclusion Criteria**

   Years of interest: 1957 (first outbreak of HFMD reported, Toronto, Canada - see Goh et al, 1982, Bull WHO, 60 (6) [38]) - present Studies with epidemiological and/or serological information (quantitative/qualitative) about incidence, prevalence

5. **Exclusion Criteria**

   Studies with focus outside of Asia, no provision of English or Chinese full texts

6. **Selection Procedure**

   Studies will be selected for retrieval after titles and abstracts have been appraised by an initial search. All identified abstracts will be reviewed by two independent reviewers. Discrepancies will be resolved by a third reviewer.

7. **Data Management**

   Publication information (author, journal, year, abstract text) will be stored in a spreadsheet along with detailed checklist of criteria met/unmet.

Location	Lag (weeks)	Autoregression term (95%CI)	p	Temperature (°C) coefficient (95%CI)	p
Japan	0	-	-	0.016 (0.008, 0.024)	<0.001
	1	1.517 (1.452, 1.582)	<0.001	-0.009 (-0.018, 0.001)	0.07
	2	-0.601 (-0.665, -0.536)	<0.001	-0.001 (-0.009, 0.006)	0.723
Hong Kong	0	-	-	0.012 (-0.007, 0.031)	0.204
	1	0.756 (0.678, 0.835)	<0.001	-0.009 (-0.031, 0.013)	0.415
	2	-0.079 (-0.178, 0.02)	0.116	-0.01 (-0.031, 0.012)	0.388
	3	0.195 (0.117, 0.274)	<0.001	0.017 (-0.001, 0.036)	0.071
Taiwan	0	-	-	0.014 (-0.003, 0.032)	0.108
	1	-0.181 (-0.26, -0.102)	<0.001	0.011 (-0.007, 0.029)	0.222
	2	0.062 (-0.019, 0.142)	0.132	-0.003 (-0.02, 0.015)	0.779
	3	0.192 (0.113, 0.271)	<0.001	0 (-0.018, 0.017)	0.968
Singapore	0	-	-	0.049 (0.017, 0.081)	0.003
	1	1.42 (1.349, 1.491)	<0.001	-0.041 (-0.082, 0.001)	0.054
	2	-0.458 (-0.53, -0.387)	<0.001	0.01 (-0.023, 0.042)	0.564

Figure 2.8: Coefficients for Autoregressive Model using Temperature as Predictor. The outcome variable is the number of reported HFMD cases per sentinel per week (Japan), the number of reported HFMD cases per GP per week (Hong Kong), the number of reported severe enterovirus cases per week (Taiwan), and the number of reported HFMD cases per week (Singapore). To facilitate comparability, the incidence measures were standardized to have mean 0 and variance 1.

8. **Synthesis**

   We will progress through questions (see **Secondary Objectives)**, identifying which papers address these and indicating such in the spreadsheet. Each question will then be allocated to one or more team members who can identify papers from the spreadsheet and synthesise those relevant.

2.8.2 AR models for HFMD Incidence against Meteorological Variables

Location	Lag (weeks)	Autoregression term (95%CI)	p	RH (%) coefficient (95%CI)	p
Japan	0	-	-	-0.001 (-0.003, 0.001)	0.363
	1	1.567 (1.503, 1.631)	<0.001	0.002 (-0.001, 0.004)	0.186
	2	-0.634 (-0.699, -0.569)	<0.001	0.001 (-0.001, 0.003)	0.231
Hong Kong	0	-	-	0.002 (-0.004, 0.007)	0.578
	1	0.942 (0.864, 1.021)	<0.001	-0.002 (-0.008, 0.004)	0.483
	2	-0.242 (-0.349, -0.135)	<0.001	-0.003 (-0.009, 0.004)	0.413
	3	0.215 (0.136, 0.294)	<0.001	0.001 (-0.005, 0.006)	0.753
Taiwan	0	-	-	0.001 (-0.005, 0.007)	0.72
	1	-0.178 (-0.257, -0.1)	<0.001	0.001 (-0.005, 0.007)	0.755
	2	0.062 (-0.018, 0.142)	0.13	0.003 (-0.004, 0.009)	0.426
	3	0.19 (0.111, 0.269)	<0.001	-0.001 (-0.006, 0.005)	0.849
Singapore	0	-	-	-0.009 (-0.016, -0.003)	0.004
	1	1.43 (1.36, 1.501)	<0.001	0.008 (0, 0.016)	0.055
	2	-0.469 (-0.539, -0.398)	<0.001	-0.003 (-0.01, 0.003)	0.349

Figure 2.9: Coefficients for Autoregressive Model using Relative Humidity as Predictor. The outcome variable is the same as the temperature model

Location	Lag (weeks)	Autoregression term (95%CI)	p	AH (%) coefficient (95%CI)	p
Japan	0	-	-	0.021 (0.013, 0.029)	<0.001
	1	0.847 (0.766, 0.928)	<0.001	0.003 (-0.005, 0.011)	0.484
	2	-0.141 (-0.221, -0.061)	0.001	0.005 (-0.003, 0.013)	0.228
Hong Kong	0	-	-	0.013 (-0.005, 0.031)	0.153
	1	0.781 (0.703, 0.86)	<0.001	-0.012 (-0.033, 0.01)	0.285
	2	-0.102 (-0.202, -0.001)	0.047	-0.004 (-0.026, 0.017)	0.703
	3	0.195 (0.117, 0.274)	<0.001	0.013 (-0.005, 0.031)	0.156
Taiwan	0	-	-	0.013 (-0.002, 0.029)	0.096
	1	-0.18 (-0.259, -0.101)	<0.001	0.011 (-0.004, 0.027)	0.158
	2	0.064 (-0.016, 0.144)	0.117	-0.002 (-0.018, 0.014)	0.814
	3	0.194 (0.115, 0.273)	<0.001	0.001 (-0.015, 0.017)	0.893
Singapore	0	-	-	0.034 (0.012, 0.056)	0.003
	1	1.42 (1.349, 1.492)	<0.001	-0.029 (-0.058, 0)	0.047
	2	-0.459 (-0.53, -0.388)	<0.001	0.007 (-0.015, 0.03)	0.536

Figure 2.10: Coefficients for Autoregressive Model using Absolute Humidity as Predictor. The outcome variable is the same as the temperature model

Chapter 3

Severity and Burden of Hand, Foot and Mouth Disease in Asia: A Modeling Study

3.1 Abstract

Background: Hand, foot and mouth disease (HFMD) affects millions of children across Asia annually, leading to an increase in implemented control policies such as surveillance, isolation and social-distancing in affected jurisdictions. However, limited knowledge of disease burden and severity causes difficulty in policy optimization as the associated economic cost cannot be easily estimated. We use a data synthesis approach to provide a comprehensive picture of HFMD disease burden, estimating infection risk, symptomatic rates, the risk of complications and death, and overall disability adjusted life-year (DALY) losses, along with associated uncertainties.

Methods: Complementary data from a variety of sources were synthesized with mathematical models to obtain estimates of severity of HFMD. This includes serological and other data extracted through a systematic review of HFMD epidemiology previously published by the authors, and laboratory investigations and sentinel reports from Singapore's surveillance system.

Results: HFMD is estimated to cause 96,900 (95% CI 40,600 - 259,000) age weighted DALYs per annum in eight high burden countries in East and South East Asia, with the majority of DALYs attributed to Years of Life Lost. The symptomatic case-hospitalization rate of HFMD is 6% (2.8% - 14.9%), of which 18.7% (6.7% - 31.5%) are expected to develop complications. 5% (2.9% - 7.4%) of such cases are fatal, bringing the overall case fatality ratio to be 52.3

(24.4 - 92.7) per 100,000 symptomatic infections. In contrast, the EV-A71 case fatality ratio is estimated to be at least 229.7 (75.4 - 672.1) per 100,000 symptomatic cases. Asymptomatic rate for EV-A71 is 71.4% (68.3% - 74.3%) for ages 1 to 4, the years of greatest incidence.

Conclusion: Despite the high incidence rate of HFMD, total DALY due to HFMD is limited in comparison to other endemic diseases in the region, such as Dengue and URTI. With the majority of DALY caused by YLL, it is possible to mitigate most with increased EV-A71 vaccine coverage.

3.2 Introduction

Hand, Foot and Mouth Disease (HFMD) is a pediatric disease which, although usually self-limiting, can result in complications associated with the central nervous system (CNS) or death [69, 70]. The viruses that cause HFMD are particularly widespread in Asia, where they are responsible for millions of infections each year [50]. An outbreak in Malaysia between April to June 1997 in which 29 children died[160] was a harbinger of fatal outbreaks that have continued to recent years (170 deaths in Vietnam in 2011 [83] and 98 deaths in Cambodia in 2012 [161]). These fatal outbreaks have influenced the perceived severity of HFMD and led to the demand for control measures [46], such as the recent Phase III vaccine trials in China [44], despite the limited information on the disease burden. Many affected countries, such as China, Japan and Singapore, adopted routine control measures akin to those from pandemic preparedness plans - including surveillance, mandatory reporting, isolation, school closures, and social-distancing - but optimal use of such interventions requires the disease burden of HFMD to be quantified so that the public health response can be calibrated accordingly.

The main etiological agents of HFMD are the Human Enterovirus A71 (EV-A71) and Coxsackieviruses 10 and 16, with EV-A71 being associated with more severe outcomes. As with other usually mild diseases where infections vastly exceed severe cases, efforts to track all HFMD infections to determine indices for different levels of severity are impractical due to the large number of cases involved [162]. With the exception of a few jurisdictions, much of the data on HFMD are based on subpopulations, such as the catchment of a health system, a

network of sentinel clinics, or from passive surveillance using symptomatic case notifications (for instance, [52]). Such studies may provide an incomplete view of the true burden of disease. Obtaining a more comprehensive picture requires synthesizing available information from studies across all severity levels, from serological studies that estimate infection risk, to notification data that provide information on symptomatic cases that could be clinically identified, to hospital records that showcases the risk of more severe expressions such as CNS complications or death. Such estimates need to be constructed carefully to allow uncertainty to be properly reflected and accounted for. This can be achieved through Bayesian methods which provide a flexible framework to integrate data from multiple sources and have been used to estimate infection rates and severity indices for influenza [163, 164] and HIV [165].

To quantify the extent of HFMD across Asia, its mortality risk, and the economic burden of control measures against it [46], we conducted an analysis using data from a previous systematic review of research across East and South East Asia [159] to develop estimates of disease burden at different levels of severity. Specifically, we adopted a modeling approach that combined serological surveys, epidemiological studies, and surveillance data to estimate the risk of children being infected, symptomatic, hospitalized, afflicted with complications and dying. These estimates and the age-weighted method [166] were used to derive estimates of disability adjusted life years (DALY) lost across Asia due to HFMD.

3.3 Methods

The severity of infection was classified into tiers which require different levels of care: i) Clinically apparent, or symptomatic, cases; ii) Hospitalizations; iii) Complications, or severe cases when the child displays neurological or cardiopulmonary complications; and iv) Fatal cases. These manifestations were matched against similar definitions from the Global Burden of Disease 2013 (GBD2013) study [167] that have existing estimates of disability weights, and the overall disease burden is the average of each level of severity, weighted by the proportion of cases entering into each level.

These severity tiers were arranged as a pyramid where lower echelons represent the majority

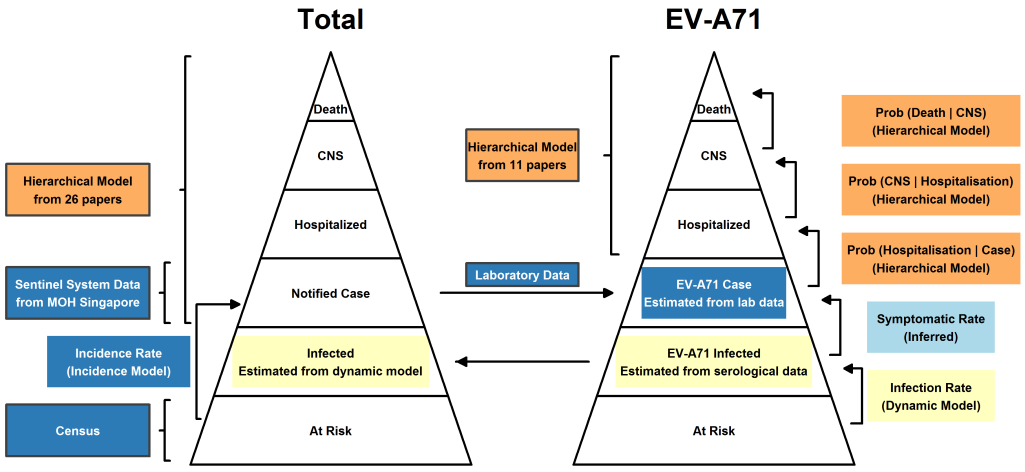


Figure 3.1: The Severity Pyramid Structure for HFMD (Total, All Etiologies) and EV-A71 in Particular. Bordered Boxes are data sources, while boxes within the pyramids are estimated quantities. Pale yellow boxes represent the models and outcomes of submodel 1, the Infection Rate Model. This model uses serological data and differential equations to estimate the rate which infection must occur to match with observed seropositivity level of EV-A71. Blue boxes represent the data, model, and outcomes of submodel 2, the Symptomatic Incidence Rate Model. This model uses surveillance data from Singapore to infer the incidence rate of HFMD and EV-A71 specific HFMD. Together with submodel 1, the estimated symptomatic rate of EV-A71 (cyan box) can be calculated. The orange boxes represent the data and models of submodel 3: the hierarchical model, which is used to estimate indices that quantifies the higher level severities of HFMD.

of (milder) infections with decreasing frequency but increasing severity toward the apex [163, 168] (Figure 3.1). The parameters which describe the severity of HFMD are the proportions of individuals in one state who reach the next higher severity level. This framework implicitly assumes that cases with higher severity are a proportion of cases at the preceding level. For example, those with significant CNS complications are modelled to be a subset of hospitalized cases, and similarly, all hospitalized cases are a subset of notified cases. We built two parallel pyramids for total HFMD (of all etiologies) and specifically for EV-A71, due to the abundance of data on EV-A71 compared to other viruses, as well as this virus being thought to be responsible for more severe symptoms.

Quantifying the disease burden of HFMD requires an estimate of the total number of cases, which was calculated from the infection rate (both symptomatic and asymptomatic) obtained from seroepidemiological studies. From a systematic review previously published

by the authors [159], we extracted 10 studies of EV-A71 serology (Table 3.1) from which the seropositive rate of EV-A71 by age was found using submodel 1 to model the required infection rate to achieve the observed seroprevalence.

Table 3.1: Infection Rate Model Data

Data Origin	Year	Description	Sample Size	Paper
Guangzhou, China	2010	Cross-sectional study of pre and post 2010 outbreak	Pre-epidemic-292 Post-epidemic-330	Li, W. et al. Seroprevalence of human enterovirus 71 and coxsackievirus A16 in Guangdong, China, in pre- and post-2010 HFMD epidemic period. PloS One 8, e80515 (2013).
Jiangsu, China	Sep 2007 to Jul 2009	Women and their neonates were enrolled from 3 hospitals in Jiangsu Province. Blood samples were collected from neonates at 2, 7 months of age. (Not used because data is only for <1 year olds)	266	Mao, Q. et al. Dynamic change of mother-source neutralizing antibodies against enterovirus 71 and coxsackievirus A16 in infants. Chin Med J Engl 123, 1679 (2010).
Lu'an, Auhui, China	Pre-epidemic-2008 Post-epidemic-2010	Community data collected by the CDC	Pre-epidemic-472 Post-epidemic-83	Yu, H. et al. Prevalence of antibodies against enterovirus 71 in children from Lu'an City in Central China. Jpn J Infect Dis 64, 528-532 (2011).
Shanghai, China	Nov 2010 to Apr 2011	Samples collected during health checks at Children's Hospital of Fudan University	614	Zeng, M. et al. Seroepidemiology of Enterovirus 71 infection prior to the 2011 season in children in Shanghai. J. Clin. Virol. 53, 285-289 (2012).
Singapore	July 1996 to Dec 1997	Samples (<=12 years) collected for the seroprevalnce survey in Singapore at the National University Hospital.	856	Ooi, E.-E., Phoon, M.-C., Ishak, B. & Chan, S.-H. Seroepidemiology of human enterovirus 71, Singapore. Emerg. Infect. Dis. 8, 995-997 (2002).

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Data Origin	Year	Description	Sample Size	Paper
Singapore	Aug 2008 to Jul 2010	Residual sera from Singapore residents aged 1-17 in 2 hospitals using microneutralization test for EV-A71.	1200	Ang, L.-W. et al. The changing seroepidemiology of enterovirus 71 infection among children and adolescents in Singapore. BMC Infect Dis 11, 270 (2011).
Taipei, Taiwan	1994, 1997, 1999	Cross-sectional collections of serum	1994:202 1997:245 1999:1258	Lu, C.-Y. et al. Incidence and case-fatality rates resulting from the 1998 enterovirus 71 outbreak in Taiwan. J. Med. Virol. 67, 217-223 (2002).
Taoyuan, Taiwan	Jul 1997 to Dec 1997	Samples obtained from healthy children who participated in trials or health examinations in Chang Gung Children's Hospital	539	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).
Taipei, Taiwan	May to Jun 1999	Serologic tests and surveys done by interview	484	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).
Taoyuan, Taiwan	Jan 1999	Serologic tests and surveys done by interview	1081	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).

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Data Origin	Year	Description	Sample Size	Paper
Taichung, Taiwan	Apr 1999	Serologic tests and surveys done by interview	692	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).
Kaohsiung, Taiwan	Jul 1999	Serologic tests and surveys done by interview	906	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).
Kaohsiung, Taiwan	Jan 1999	Serologic tests and surveys done by interview	851	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).
Ilan, Taiwan	Jan 1999	Serologic tests and surveys done by interview	605	Chang, L.-Y. et al. Risk Factors of Enterovirus 71 Infection and Associated Hand, Foot, and Mouth Disease/Herpangina in Children During an Epidemic in Taiwan. Pediatrics 109, e88-e88 (2002).
Taiwan	Jun 2006 to Jun 2008	Serologic tests done on cord blood and at 6 months at Chang Gung Memorial Hospital	309	Luo, S.-T. Enterovirus 71 Maternal Antibodies in Infants, Taiwan. Emerg. Infect. Dis. 15, 581-584 (2009).

Continued on next page

Data Origin	Year	Description	Sample Size	Paper
Northern Taiwan	2008	Sera obtained from cohort study of 529 children, new-born cord blood, 6 months, 12 months, 24 months	Birth:529 6mth:494 12mth:347 24mth:64	Lee, M.-S. et al. An Investigation of Epidemic Enterovirus 71 Infection in Taiwan, 2008: Clinical, Virologic, and Serologic Features. Pediatr. Infect. Dis. J. 1 (2010).
Khon-Kaen, Bangkok, Thailand	2009 to 2012	Cross-sectionally stored sera collected between 2009 and July 2012 from healthy individuals of different age groups who had participated in previous serosurveillance studies	161	Linsuwanon, P. et al. Epidemiology and seroepidemiology of human enterovirus 71 among Thai populations. J Biomed Sci 21, 16 (2014).
Ho Chi Minh City, Vietnam	Sep 2006 to Sep 2007	Sera obtained from cohort study of 200 children, new-born cord blood, 3 months, 6 months, 9 months, 12 months from Hung Vuong Obstetric Hospital, HCM City	200	Tran, C. B. N. et al. The Seroprevalence and Seroincidence of Enterovirus 71 Infection in Infants and Children in Ho Chi Minh City, Viet Nam. PLoS One 6, e21116 (2011).
Ho Chi Minh City, Vietnam	2006 to 2007	Plasma samples collected between birth and 2 years of age, Hung Vuong Obstetric Hospital, HCM City	263	Tran, C. B. N. et al. The Seroprevalence and Seroincidence of Enterovirus 71 Infection in Infants and Children in Ho Chi Minh City, Viet Nam. PLoS One 6, e21116 (2011).
Ho Chi Minh City, Vietnam	Sep 2005 to Jan 2009	Plasma samples collected for children between 5 to 15 years of age, District 8 Hospital, HCM City	120	Tran, C. B. N. et al. The Seroprevalence and Seroincidence of Enterovirus 71 Infection in Infants and Children in Ho Chi Minh City, Viet Nam. PLoS One 6, e21116 (2011).

To estimate clinically apparent cases, we used data on notified cases from Singapore, where HFMD is a legally notifiable disease and where active surveillance in preschools is conducted via daily temperature taking; this was used to calculate a lower bound for the overall asymptomatic rate of HFMD, in submodel 2. Indices for higher severity levels of HFMD and EV-A71, which include deaths, complications, hospitalizations and symptomatic cases used information from 26 papers [83, 96, 18, 79, 28, 169, 127, 170, 116, 24, 81, 8, 12, 171, 92, 75, 6, 94, 172, 9, 173, 126, 174, 90, 3, 85, 117] and 11 papers [83, 127, 170, 8, 175, 176, 22, 177, 178, 179, 180] respectively from across Asia and were modelled in submodel 3. Finally, the severity indices estimated in the three submodels were fused to collectively estimate the overall disease burden of HFMD in Asia.

3.3.1 Submodel 1: Infection Rate Model

The infection rate model estimates the age-specific proportion of children who must be infected by the EV-A71 virus each year to reach the observed level of seroprevalence. Serological data from China [95, 99, 20], Singapore [64, 65], Taiwan [130, 134, 181], Thailand [39] and Vietnam [182] were extracted from the original papers (Figure 3.2, Left), from which it can be observed that seropositivity to EV-A71 starts low at infancy and reaches a plateau around 40% to 60% by adolescence, a finding which is corroborated by cord blood seropositivity (indicative of maternal antibody levels) of 55% in Vietnam [182], 47% to 50% in Taiwan [183, 181], 38% in Singapore [64] and 26% in China [184].

To accommodate these patterns, we developed and fit models of the age-specific seroprevalence using seven dynamical models which make varying assumptions for the reason behind these phenomena. The basic structure of these models is $\frac{dI_t}{dt} = c(1-I_t)\alpha_t - I_t\beta$ where I_t represents the seropositive rate at time t, α_t is the rate which seropositivity increases (due to new infections) and β is the rate which seropositivity decreases (due to a decay of antibody levels). The assumptions made puts certain restrictions: In models 1 and 2, $\beta > 0$ is assumed to be constant, and in models 3 to 7, $\beta = 0$ is assumed (i.e. models 1 and 2 allow a decay in seropositivity, such that antibodies for EV-A71 in humans will diminish over time like influenza [185, 186], and eventually reach an equilibrium from new infections and loss of immunity). In models 1, 3 and 5, $\frac{d\alpha_t}{dt} = -k\alpha_t$ is used to estimate decreasing infection rate, while infection rate is assumed to be constant for models 2 and 5. In models 4 and 5, $\alpha_t = 0$ after time T to simulate low infection rate during teenage years. Finally, while $c = 1$ for models 1 to 5, this assumption is relaxed in models 6 and 7, allowing it to take on other positive real number values.

The Metropolis Hastings Algorithm (Markov Chain Monte Carlo) with non-informative prior distributions were used to estimate the parameters. The likelihood is evaluated by comparing the observed number of seroconverted persons to the total number at each age to the simulated proportion from the model using a binomial relationship. The model is chosen based on the Deviance Information criterion (DIC), and detailed information can be found in Table 3.2.

Table 3.2: Seven Infection Rate Models.

Mod	Model Spec	Assumptions	Estimated Parameter	DIC
1	$\frac{dI_t}{dt} = (1 - I_t)\alpha_t - I_t\beta$ $\frac{d\alpha_t}{dt} = -k\alpha_t$	Decreasing Infection Rate Immunity to EV-A71 wanes over time	k: 12.8% (12.2% - 13.5%) β: 0.1% (0% - 0.4%)	1860.3
2	$\frac{dI_t}{dt} = (1 - I_t)\alpha - I_t\beta$	Constant Infection Rate Immunity to EV-A71 wanes over time	α: 11.7% (11.0% - 12.5%) β: 7.2% (6.1% - 8.4%)	1855.6
3	$\frac{dI_t}{dt} = (1 - I_t)\alpha_t$ $\frac{d\alpha_t}{dt} = -k\alpha_t$	Decreasing Infection Rate	k: 12.7% (12.1% - 13.4%)	1857.2
4	$\frac{dI_t}{dt} = (1 - I_t)\alpha_t$ $\frac{d\alpha_t}{dt} = -k\alpha_t$ $I_t = 0$, for $t \geq T$	Decreasing Infection Rate Infection ends at time T	k: 12.8% (12.2% - 13.5%) T: 16 (14 - 17) years	1857.2
5	$\frac{dI_t}{dt} = (1 - I_t)\alpha$ $I_t = 0$, for $t \geq T$	Constant Infection Rate Infection ends at time T	α: 9.5% (9.2% - 10.1%) T: 9 (8 - 9) years	1863.3
6	$\frac{dI_t}{dt} = c(1 - I_t)\alpha_t$ $\frac{d\alpha_t}{dt} = -k\alpha_t$	Decreasing Infection Rate Random Initial Infection Rate	k: 11.4% (9.7% - 13.2%) c: 1 (0.98 - 1.21)	1856.8
7	$\frac{dI_t}{dt} = c(1 - I_t)\alpha_t$ $\frac{d\alpha_t}{dt} = -k\alpha_t$ $I_t = 0$, for $t \geq T$	Decreasing Infection Rate Random Initial Infection Rate Infection ends at time T	k: 4.79% (1.92% - 8.07%) c: 2.3 (1.47 - 5.28) T: 9 (8 - 9) years	1849.8

Posterior distributions of parameters were estimated in R [73] using Markov chain Monte Carlo (MCMC) and R package 'deSolve' [187] with non-informative prior distributions. The optimal model to represent infection rate was determined by the fit to the data and the model complexity and was evaluated using the Deviance Information Criteria (DIC), which Model 7 has the lowest DIC.

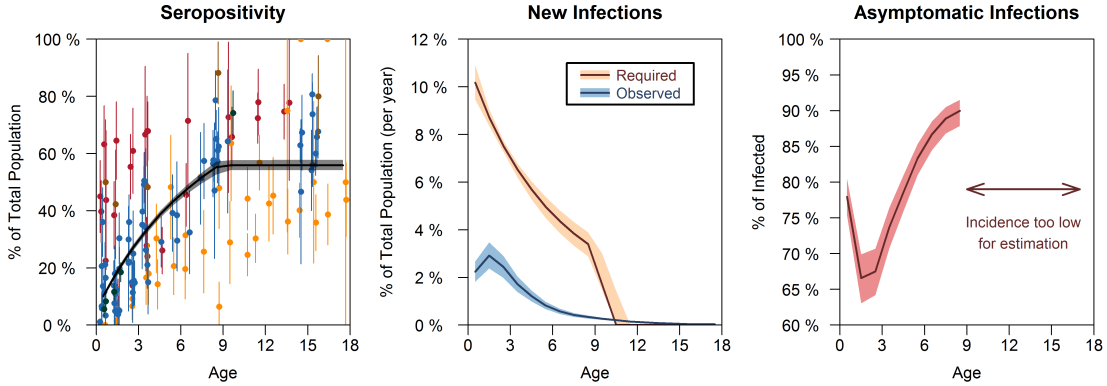


Figure 3.2: Data and Results from the Infection Rate Model (Submodel 1). Left: Seropositivity levels of EV-A71 with fitted curve. Red data points are from China, Blue from Taiwan, and Orange from Singapore, Green from Vietnam and Brown from Thailand. The black line is the fitted model with 95% Bayesian credible intervals. Middle: Required and observed rate of EV-A71 infection. The brown line shows the age-specific infection rate of EV-A71 implied from the seropositivity curve, and is the required rate to get the observed seropositivity levels; the blue line shows the calculated average EV-A71 infection rate based on Singapore's data from submodel 2. The difference between the two lines are the unobserved cases. Right: Asymptomatic infection rate. Calculated from the percentage difference between the Required and Observed rates of infection in the middle.

3.3.2 Submodel 2: Symptomatic Incidence Rate Model

Since October 2000, Singapore has adopted a comprehensive approach for HFMD surveillance including a mandatory case reporting system by medical practitioners and childcare centers [188], with children at childcares being screened once or twice daily for symptoms and then subsequently isolated for 10 days if symptomatic. The clearly defined population catchment in the city-state and the dual layers of doctor- and teacher-driven surveillance make the data on notified symptomatic presentations of HFMD collected by the Singapore's Ministry of Health unusually complete compared to the rest of Asia. In calculating symptomatic incidence rate, we make the assumption that all symptomatic pediatric HFMD were clinically apparent, and captured by the surveillance system in Singapore.

We constructed a matrix containing the incidence m_{ay} of cases in year y (2003 - 2012) among age a (1 - 18), scaled by the number of individuals in the resident population for that age and year derived from the census [189]. To separate the effects of age and time

on incidence rate, singular value decomposition was performed on the incidence matrix to find the first left-singular vector u_a which corresponds to the largest singular value. This vector is normalized to $b_a = u_a/\Sigma u_a$, which is equivalently the normalized first principal component of mm', and represents the systematic variation attributable to age (Table 3.7). The second vector is a time series index, $\gamma_y = (u_a' m)\Sigma u_a$, representing the stochastic variation attributable to HFMD incidence level each year. Values for years 2013 to 2015 were estimated using linear regression using data from Singapore's Weekly Infectious Disease Bulletin [190]. The modelled HFMD incidence rate is estimated from the product of the two vectors, $\hat{m}_{ay} = b_a \gamma_y$.

The EV-A71-specific incidence is a proportion of all HFMD cases. We obtain γ_y^{ev}, the year effect due to EV-A71, by taking it as a binomial sample from the total year effect, $\gamma_y^{ev} \sim Bin(\gamma_y, p_y)$ (Table 3.6). The proportion of cases attributed to EV-A71 in 2008 to 2015, p_y, was estimated from data obtained from the National Surveillance Program in Singapore which collects clinical specimens from patients in 107 clinics and two hospitals, KK Women's and Children's Hospital and National University Hospital (Figure 3.3, right). Throat or vesicle swabs were collected from patients below 12 years of age with clinical diagnosis of HFMD. Specimens were stored at room temperature or 4 degrees (preferred) for no longer than 48 hours, before being sent to the National Public Health Laboratory for testing and typing. Data on p_y prior to 2008 was estimated in a study by Ang et al [3]. The average EV-A71 incidence rate by age is the average of the EV-A71 year effect, weighted by the age effect, or $b_a E(\gamma_y^{ev})$.

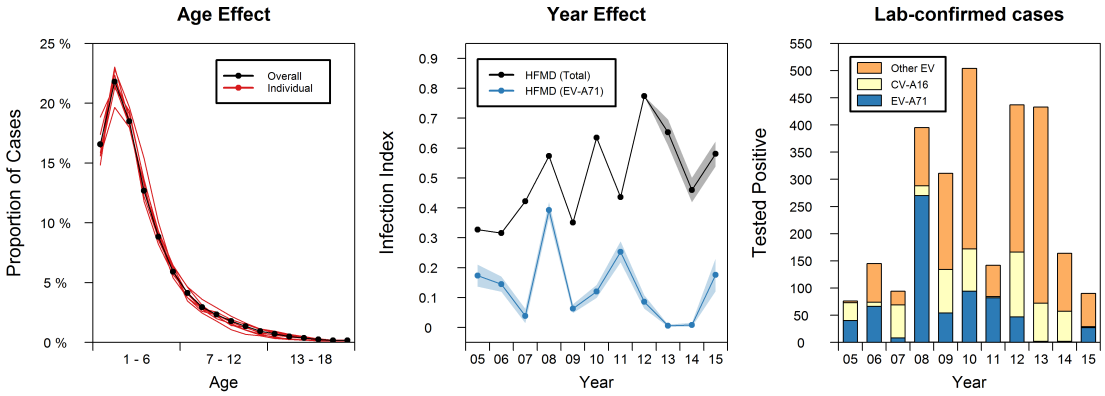


Figure 3.3: Age and Year Trend of HFMD Symptomatic Infection from the Incidence Rate Model (Submodel 2). Left: The Age Effect. The black line reflects the overall proportion of incidence for each year of age, and red lines represents the age effect for individual years from 2005 to 2012. Middle: The Year Effect. The black line is an index which reflects the number of notified cases for 2005 to 2012. The Year Effect for 2013 to 2015 is estimated from data (notified cases) obtained from the Weekly Infectious Disease Bulletin released by the Ministry of Health, Singapore, using linear regression. The year effect of EV-A71 (blue) is mathematically derived from the proportion of cases due to EV-A71 estimated by laboratory data (right). Right: Laboratory data from Singapore's surveillance system. This shows the proportion of cases due to each virus in each year from 2005 to 2015.

3.3.3 Submodel 3: Hierarchical Model of Disease Severity

The upper levels of the severity pyramid from across Asia were modelled as nested binomial distributions, with the probability of reaching state i or higher given the case reaches the lower state j denoted as $\alpha_{i|j}$. The states are D (death), S (severe disease, 'CNS'), H (hospitalized) and C (symptomatic case) so that, for example, $\alpha_{H|C}$ denotes the case-hospitalisation rate. Standard rules of conditional probability mean that the product of these probabilities give the risk of progressing through multiple levels; for instance, the overall case-fatality rate is $\alpha_{D|C} = \alpha_{D|S} \times \alpha_{S|H} \times \alpha_{H|C}$. As a result, the likelihood function for data with missing tiers of the pyramid is specified via combinations of the basic parameters. For example, 4625 hospitalizations and 11 deaths were documented in the Sarawak 1997 outbreak [126], with no mention of severe cases. This dataset will allow us to estimate $\pi_{D|H}$, and thus make inferences on $\alpha_{D|S}$ and $\alpha_{S|H}$ through the multiplicative relationship, where $\pi_{i|j,p}$ is the individual level estimate for paper p. Detailed description of all data sources can be found in Table 3.3 and Table 3.4. The posterior distribution of parameters were estimated using the Metropolis Hastings algorithm (MCMC) in R.

Specifically, in a Bayesian hierarchical framework, the parameters are described in a full probability model where data are expressed as likelihood functions:

$$D_p \sim Bin(S_p, \pi_{D|S,p});\ S_p \sim Bin(H_p, \pi_{S|H,p})\ ;\ H_p \sim Bin(C_p, \pi_{H|C,p})$$

where $\pi_{D|S,p}$, $\pi_{S|H,p}$ and $\pi_{H|C,p}$ are the individual-level severity proportions from paper p, which are related multiplicatively: $\pi_{D|C,p} = \pi_{D|S,p} \times \pi_{S|H,p} \times \pi_{H|C,p}$. This structure allows the handling of missing tiers of data. For instance, if S_p and H_p is missing, D_p and C_p can be modelled as

$$D_p \sim Bin(C_p, \pi_{D|C,p}) = Bin(C_p, \pi_{D|S,p} \times \pi_{S|H,p} \times \pi_{H|C,p})$$

In a hierarchical model, the overall proportions $\alpha_{D|S}$, $\alpha_{S|H}$ and $\alpha_{H|C}$ are known as the hyper parameters, and are related to the individual proportions with the following relationships, with state j being in a lower state of severity than state i:

$$\pi_{i|j,p} \sim N(\alpha_{i|j}, \sigma^2_{i|j})$$

This means that the mean of individual proportions follows the same parameter, providing additional structure to the hierarchical model and lowers the chance of overfitting. This approach will also automatically assign weights to each piece of evidence, through the likelihood functions of each data set. As we do not possess any other information, we chose non-informative priors to 'allow the data to speak for itself'.

$$\alpha \sim N(0, 1000^2); \sigma^2 \sim U(0, 1000)$$

Table 3.3: Data on HFMD Severity (Total)

Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T1_1$	Xu, W. et al. Distribution of enteroviruses in hospitalized children with hand, foot and mouth disease and relationship between pathogens and nervous system complications. Virol J 9, 8 (2012).	Cases are children treated in Shengjing Hospital in Liaoning Province.	2010	China, Liaoning	6027,423,197,9
$T1_2$	Zeng, M. et al. Epidemiology of hand, foot, and mouth disease in children in Shanghai 2007 - 2010. Epidemiol. Infect. 140, 1122 - 1130 (2012).	Reported HFMD cases in the Children's Hospital of Fudan University, Shanghai.	2007 to 2010	China, Shanghai	28058,3948,730,11
$T1_3$	Wang, Y. et al. Epidemiology and clinical characteristics of hand foot, and mouth disease in a Shenzhen sentinel hospital from 2009 to 2011. BMC Infect Dis 13, 539 (2013).	Recorded cases of HFMD in the Affiliated Shenzhen Third Hospital, Guangdong Medical College, Shenzhen.	2009 to 2011.	China, Shenzhen	12132,2944,190,6
$T1_4$	Ma, E., Lam, T., Chan, K. C., Wong, C. & Chuang, S. K. Changing epidemiology of hand, foot, and mouth disease in Hong Kong, 2001 - 2009. Jpn J Infect Dis 63, 422 - 426 (2010).	Surveillance findings of HFMD collected from a general-practitioner-based sentinel surveillance system, and outbreaks reported by institutions.	2001 to 2008	Hong Kong	8082,103,1,0

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T1_5$	Gao, L.-D. et al. Correlation Analysis of EV71 Detection and Case Severity in Hand, Foot, and Mouth Disease in the Hunan Province of China. PLoS One 9, e100003 (2014).	Surveillance system data from Human Provincial Center for Disease Control and Prevention, Hunan China. The paper was not clear on the selection criteria for the 23297 blood samples. We assume these cases were hospitalized cases as the severity rate seems to be very high for non-hospitalised cases.	2010 to 2012	China, Hunan	X,23297,5160,158
$T1_6$	Ooi, M. et al. Identification and validation of clinical predictors for the risk of neurological involvement in children with hand, foot, and mouth disease in Sarawak. BMC Infect Dis 9, 3 (2009).	Prospective study conducted from Jan 2000 to Dec 2006 which included 3 distinct outbreaks in 2000/1, 3003 and 2006 at the pediatric wards and intensive care unit at Sibu Hospital, Sarawak, Malaysia. Severe cases were diagnosed with CNS after cerebrospinal fluid test.	Jan 2000 to Dec 2006	Sarawak, MY	X,1455,352,7
$T1_7$	Lu, H. et al. Prognostic implications of myoclonic jerk in children with enterovirus infection. J Microbiol Immunol Infect 37, 82 (2004).	Hospitalised patients with culture-confirmed enterovirus infection at Chang Gung Children's Hospital, Taoyuan.	2000 to 2001	Taiwan	X,665,54,11

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T1_8$	Fujimoto, T. et al. Outbreak of central nervous system disease associated with hand, foot, and mouth disease in Japan during the summer of 2000: detection and molecular epidemiology of enterovirus 71. Microbiol. Immunol. 46, 621 - 627 (2002).	HFMD inpatients admitted to Shinko Kakogawa Hospital due to severe complications.	Jun to Aug 2000	Japan	X,X,28,1
$T1_9$	Chen, K.-T., Chang, H.-L., Wang, S.-T., Cheng, Y.-T. & Yang, J.-Y. Epidemiologic Features of Hand-Foot-Mouth Disease and Herpangina Caused by Enterovirus 71 in Taiwan, 1998 2005. Pediatrics 120, e244 - e252 (2007).	Surveillance data from Taiwan Center for Disease Control.	1998 to 2005	Taiwan	X,X,1548,245
$T1_10$	Yang, T.-T. et al. Clinical features and factors of unfavorable outcomes for non-polio enterovirus infection of the central nervous system in northern Taiwan, 1994-2003. J Microbiol Immunol Infect 38, 417 - 424 (2005).	Clinical specimens with positive EV isolation and central nervous system disease from a medical center in northern Taiwan.	1994 to 2003	Taiwan	X,X,333,3

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T1_{11}$	Zhu, D. et al. A new factor influencing pathogen detection by molecular assay in children with both mild and severe hand, foot, and mouth disease. Diagn Microbiol Infect Dis 76, 162 - 167 (2013).	Clinically diagnosed samples from Youan Hospital, Beijing.	Jun to Oct 2010	China, Beijing	X,1104,233,X
$T1_{12}$	Liu, M. Y. et al. Characterization of an Outbreak of Hand, Foot, and Mouth Disease in Nanchang, China in 2010. PLoS One 6, e25287 (2011).	Patients with clinical symptoms of HFMD who came/were referred to the outpatient clinic at the 9th People's Hospital of Nanchang.	Apr to May 2010	China, Nanchang	109,3,X,X
$T2_1$	Tian, H. et al. Clinical Features and Management Outcomes of Severe Hand, Foot and Mouth Disease. Med. Princ. Pract. 21, 355 - 359 (2012).	Patients with clinical diagnosis of HFMD who were admitted to the hospital for treatment.	Jan to June 2009	China, Shandong	15823,X,147,3
$T2_2$	Wang, J. et al. Epidemiological Analysis, Detection, and Comparison of Space-Time Patterns of Beijing Hand-Foot-Mouth Disease (2008 - 2012). PLoS One 9, e92745 (2014).	Surveillance data provided by the Beijing Center for Disease Prevention and Control from the National Surveillance System.	2008 to 2012	China, Beijing	157707,X,1465,33

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T2_3$	Yan, L. et al. Distribution and risk factors of hand, foot, and mouth disease in Changchun, northeastern China. Chin Sci Bull 59, 533 - 538 (2014).	Surveillance data collected to the web-based China Information System for Disease Control and Prevention.	2008 to 2011	China, Changchun	17464,X,304,8
$T2_4$	Wang, Y. et al. Hand, Foot, and Mouth Disease in China: Patterns of Spread and Transmissibility. Epidemiology 22, 781 - 792 (2011).	Surveillance data collected to the web-based China Information System for Disease Control and Prevention.	2008 to 2009	China, entire	1645115,X,15045,480
$T2_5$	De, W. et al. A large outbreak of hand, foot, and mouth disease caused by EV71 and CAV16 in Guangdong, China, 2009. Arch. Virol. 156, 945 - 953 (2011).	Surveillance data reported to the Guangdong CDC.	Mar to Dec 2009	China, Guangdong	92749,X,306,23
$T2_6$	Xie, Y., Chongsuvivatwong, V., Tang, Z., McNeil, E. B. & Tan, Y. Spatio-Temporal Clustering of Hand, Foot, and Mouth Disease at the County Level in Guangxi, China. PLoS One 9, e88065 (2014).	Surveillance data collected to the web-based China Information System for Disease Control and Prevention.	2008 to 2013	China, Guangxi	796072,X,5636,350

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T2_7$	Xu, W., Jiang, L., Thammawijaya, P. & Thamthitiwat, S. Hand, Foot and Mouth Disease in Yunnan Province, China, 2008-2010. Asia Pac J Public Health (2011). doi:10.1177/1010539511430523	Surveillance data sent to the national database.	2008 to 2010	China, Yunnan	75109,X,1111,33
$T2_8$	Mou, J. et al. Severe hand, foot and mouth disease in Shenzhen, South China: what matters most? Epidemiol. Infect. 142, 776 - 788 (2014).	HFMD cases were extracted from the former Shenzhen local reporting system.	2008 to 2011	China, Shenzhen	64428,X,286,18
$T3_1$	Shekhar, K. et al. Deaths in children during an outbreak of hand, foot and mouth disease in Peninsular Malaysia - clinical and pathological characteristics. Med. J. Malaysia 60, 297 - 304 (2005).	4625 admissions to hospital.	1997 to 1998	Malaysia, Peninsular	X,4625,X,11
$T3_2$	Chong, C.-Y. et al. Hand, foot and mouth disease in Singapore: a comparison of fatal and non-fatal cases. Acta Paediatr. 92, 1163 - 1169 (2007).	Cases seen at the KK Women's and Children's Hospital Emergency.	Sep to Nov 2000	Singapore	X,129,X,3

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T4_1$	Zhang, Y. et al. An outbreak of hand, foot, and mouth disease associated with subgenotype C4 of human enterovirus 71 in Shandong, China. J. Clin. Virol. 44, 262 - 267 (2009).	An outbreak of HFMD included 1149 people in Linyi Shandong Province, China in 2007.	2007	China, Shandong	1149,X,X,3
$T4_2$	Ang, L. W. et al. Epidemiology and control of hand, foot and mouth disease in Singapore, 2001-2007. Ann Acad Med Singap. 38, 106 - 112 (2009). Wu, Y. et al. The largest outbreak of hand; foot and mouth disease in Singapore in 2008: The role of enterovirus 71 and coxsackievirus A strains. Int. J. Infect. Dis. 14, e1076 - e1081 (2010).	Ang - 7 deaths in 2000-2001; Wu - 1 death in 2008; Cases - MOH data	2001 to 2014	Singapore	279382,X,X,8
$T4_3$	Wang, J.-R. et al. Change of Major Genotype of Enterovirus 71 in Outbreaks of Hand-Foot-and-Mouth Disease in Taiwan between 1998 and 2000. J. Clin. Microbiol. 40, 10 - 15 (2002).	Data recorded from Taiwan CDC.	2000	Taiwan	80677,X,X,41

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
$T4_4$	Nguyen, N. T. et al. Epidemiological and clinical characteristics of children who died from hand, foot and mouth disease in Vietnam, 2011. BMC Infect Dis 14, 341 (2014).	Vietnamese surveillance data.	2011	Vietnam	113121,X,X,170
	Yan, X.-F. et al. Epidemic characteristics of hand, foot, and mouth disease in Shanghai from 2009 to 2010: Enterovirus 71 subgenotype C4 as the primary causative agent and a high incidence of mixed infections with coxsackievirus A16. Scand. J. Infect. Dis. 44, 297 - 305 (2012).	Cases admitted to Children's Hospital of Fudan University, Shanghai. This data is excluded from the analysis as it appears to be a repeat of Zeng, M. et al.	2009 to 2010	China, Shanghai	X,3208,750,7

Table 3.4: Data on EV-A71 Severity

Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
E_1	Xu, W. et al. Distribution of enteroviruses in hospitalized children with hand, foot and mouth disease and relationship between pathogens and nervous system complications. Virol J 9, 8 (2012).	Cases are children treated in Shengjing Hospital in Liaoning Province. Out of 423 hospitalised cases, 177 were tested positive for EV.	2010	Liaoning, CN	89,47,5
E_2	Ooi, M. et al. Identification and validation of clinical predictors for the risk of neurological involvement in children with hand, foot, and mouth disease in Sarawak. BMC Infect Dis 9, 3 (2009). & Ooi, M. H. et al. Human Enterovirus 71 Disease in Sarawak, Malaysia: A Prospective Clinical, Virological, and Molecular Epidemiological Study. Clin Infect Dis 44, 646-656 (2007).	Study conducted at pediatric wards and intensive card unit at Sibu Hospital, Sawawak, Malaysia.	2000 to 2006	Sarawak, MY	277,56,4
E_3	Chang, L.-Y. et al. HLA-A33 Is Associated With Susceptibility to Enterovirus 71 Infection. Pediatrics 122, 1271-1276 (2008).	Virologically confirmed cases at Chang Gung Children's Hospital	May 2001 to Apr 2003	Taiwan	219,125,12

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
E_4	Liu, C.-C., Tseng, H.-W., Wang, S.-M., Wang, J.-R. & Su, I.-J. An outbreak of enterovirus 71 infection in Taiwan, 1998: epidemiologic and clinical manifestations. J. Clin. Virol. 17, 23-30 (2000).	Passive surveillance data from Taiwan Ministry of Health. Clinical and virological investigations carried out at National Cheng Kung University Medical Center.	Apr to Dec 1998	Taiwan	119,53,9
E_5	Lu, H. et al. Prognostic implications of myoclonic jerk in children with enterovirus infection. J Microbiol Immunol Infect 37, 82 (2004).	Hospitalised patients with culture-confirmed enterovirus infection at Chang Gung ChildrenâĂŹs Hospital, Taoyuan	2000 to 2001	Taiwan	140,39,9
E_6	Tu, P. V. et al. Epidemiologic and virologic investigation of hand, foot, and mouth disease, southern Vietnam, 2005. Emerg Infect Dis 13, 1733-1741 (2007).	Children below 15 years who were admitted to a large pediatric hospital in Ho Chi Minh City, Vietnam. EV-A71 found using RT-PCR.	2005	Vietnam	173,51,3
E_7	Ryu, W.-S. et al. Enterovirus 71 Infection with Central Nervous System Involvement, South Korea. Emerg. Infect. Dis. 16, 1764-1766 (2010).	Surveillance data from 62 clinics reported to the Korea Centers for Disease Control and Prevention.	2009	South Korea	168,92,2

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Ref	Authors/Papers	Description	Date	Location	Data (Case, Hospitalization, Severe, Death)
E_8	Zhu, D. et al. A new factor influencing pathogen detection by molecular assay in children with both mild and severe hand, foot, and mouth disease. Diagn Microbiol Infect Dis 76, 162-167 (2013).	Clinically diagnosed samples from Youan Hospital, Beijing. RT-PCR done on nasopharyngeal swabs.	June to Oct 2010	Beijing, CN	512,99,X
E_9	Wang, S.-M. et al. Reemerging of enterovirus 71 in Taiwan: the age impact on disease severity. Eur. J. Clin. Microbiol. Infect. Dis. 31, 1219-1224 (2012).	Children at Cheng Kung University Hospital who were infected by EV-A71.	2008	Taiwan	134,38,X
E_{10}	Wang, S.-M. et al. Clinical spectrum of enterovirus 71 infection in children in southern Taiwan, with an emphasis on neurological complications. Clin Infect Dis 29, 184-190 (1999).	Medical records from National Cheng Kung University Hospital for 1998 outbreak.	Apr to Dec 1998	Taiwan	97,34,X

3.3.4 Disability Adjusted Life Years

To estimate the DALY lost across Asia to HFMD, we conservatively estimated the incidence of HFMD infection to be twice the incidence of EV-A71 using the incidence rate model (submodel 2), to allow for infection by other etiological agents of HFMD which tend to be more common than EV-A71. The number of symptomatic cases was then calculated using the asymptomatic rate from the infection rate model (submodel 1). The numbers of more severe manifestations (hospitalized, CNS and deaths) were then estimated using the parameters from the hierarchical model (submodel 3). Again, posterior distributions were used to characterize uncertainty in these estimates.

We used the age-weighted approach [166] which assigns greater weights to illnesses that occur during periods when the social role is higher, most notably between ages 10 to 55. As disability from HFMD usually occurs among children, this method provides a conservative estimate of DALY. Under this framework, the formula for DALY for an individual is:

$DALY = -[\frac{DCe^{-\beta a}}{(\beta+r)^2}[e^{-(\beta+r)L}(1+((\beta+r)(L+a))-(1+(\beta+r)a)]]$

D is the disability weight (1 for premature mortality)

$r = 0.03$ is the discount rate

$C = 0.166$ is the age weighting correcting coefficient

$\beta = 0.04$ is the parameter from the age-weighting function

a is the age of onset, and

L is the duration of disability or time lost due to premature mortality

Other than inputs which could be calculated by the severity models, the expected disability for each individual are therefore also affected by their disability weights and associated duration for each level of severity. Disability weights from GBD2013 [167] for Severe (0.133, 0.088 - 0.19), Moderate (0.051, 0.032 - 0.074) and Mild (0.006, 0.002 - 0.012) Acute Episodes of Infectious Disease were used as estimates for HFMD with complication, hospitalization, and symptomatic HFMD respectively.

The disability duration for fatalities was determined using the period life expectancy of Japan [191], which represents the expected lifespan of humans as healthcare improves in the current century. This might even be an underestimate given how much human longevity has improve over the last 50 years. The disability duration for morbidity can be estimated from the literature. We extracted 7 key parameters, including sample size, mean duration, median duration, minimum duration, maximum duration, confidence intervals and standard deviation from 30 studies (54 rows of data) and fit these with Weibull distributions, chosen for its versatility in shape.

For studies with either means or medians along with their standard deviations (23 of 54 data rows), the scale parameter, λ and the shape parameter, κ, can be estimated directly as there are closed form solutions. For studies that only reported their mean or median, we assume that the duration is a point estimate (6 of 54 data rows). The studies which reported other combinations of data (24 of 54 data rows) are also modelled with a Weibull distribution with the following steps:

1. As the means and medians have closed form solutions, we find combinations of the shape and scale parameters which matches the means/medians.
2. If the paper contains information about 95% CIs, we will find the shape and scale parameter which produces the CDF that closely matches the values at the 2.5 percentile and 97.5 percentile.
3. If the paper contains information about minimum and maximum, we will find the shape and scale parameter which produces the CDF that closely matches the values at the $100/(n+1)$ percentile and $100n/(n+1)$ percentile. This reflects that a paper with more samples are more likely to hit extreme values.
4. Papers without means and medians will be imputed.
5. Using the estimated scale and shape parameters, the distribution is simulated and used to check whether it matches the original summary statistics from data. The statistics matches very closely (around 2-3% of each other).

After estimating the distribution of disability duration from each data row, a simulation

approach is used to combine these distributions together, weighted by their sample size (Figure 3.4). In the table, Type 1 are the disability duration associated with mild HFMD; Type 2 indicates the duration before the child is sent to the hospital; Type 3 is the actual hospitalization duration for non CNS cases; Type 4 is the actual hospitalization duration for CNS cases. The disability length for mild symptomatic HFMD is estimated at 5.97 (5.82 - 6.15) days (Type 1); hospitalization will require about 8.32 (8.12 - 8.53) days (Type 2 + Type 3), while HFMD with complications is estimated at 10.72 (10.43 - 11.05) days (Type 2 + Type 4).

3.4 Results

3.4.1 Submodel 1: Serological Evidence for EV-A71 Infection Rate

Among the seven contesting models, model 7 with an unconstrained initial rate of infection and a decreasing rate of infection that reaches 0 at time T has the best fit to the data. The model formulation is:

$$\frac{dI_t}{dt} = c(1 - I_t)\alpha_t$$

$$\frac{d\alpha_t}{dt} = -k\alpha_t$$

$$I_t = 0, \text{ for } t \geq T$$

where I_t represents the seropositive rate at time t, c and k are positive real numbers, α_t is the rate which seropositivity increases (due to new infections)

The fitted values are $c = 2.3$ (95% CI: 1.47, 5.28), $k = 4.79\%$ (95% CI: 1.92%, 8.07%), $T =$ 9 years (95% CI: 9, 10), which translates to around 10% required infection rate in the first year of life, decreasing to 5% at ages 5 - 6, and drops to infinitesimally near 0% at around age 9 (Figure 3.2, mid). Details of all models, their estimated parameters and the DIC can be found in Table 3.2 and the results from the chosen model are summarised in Table 3.5.

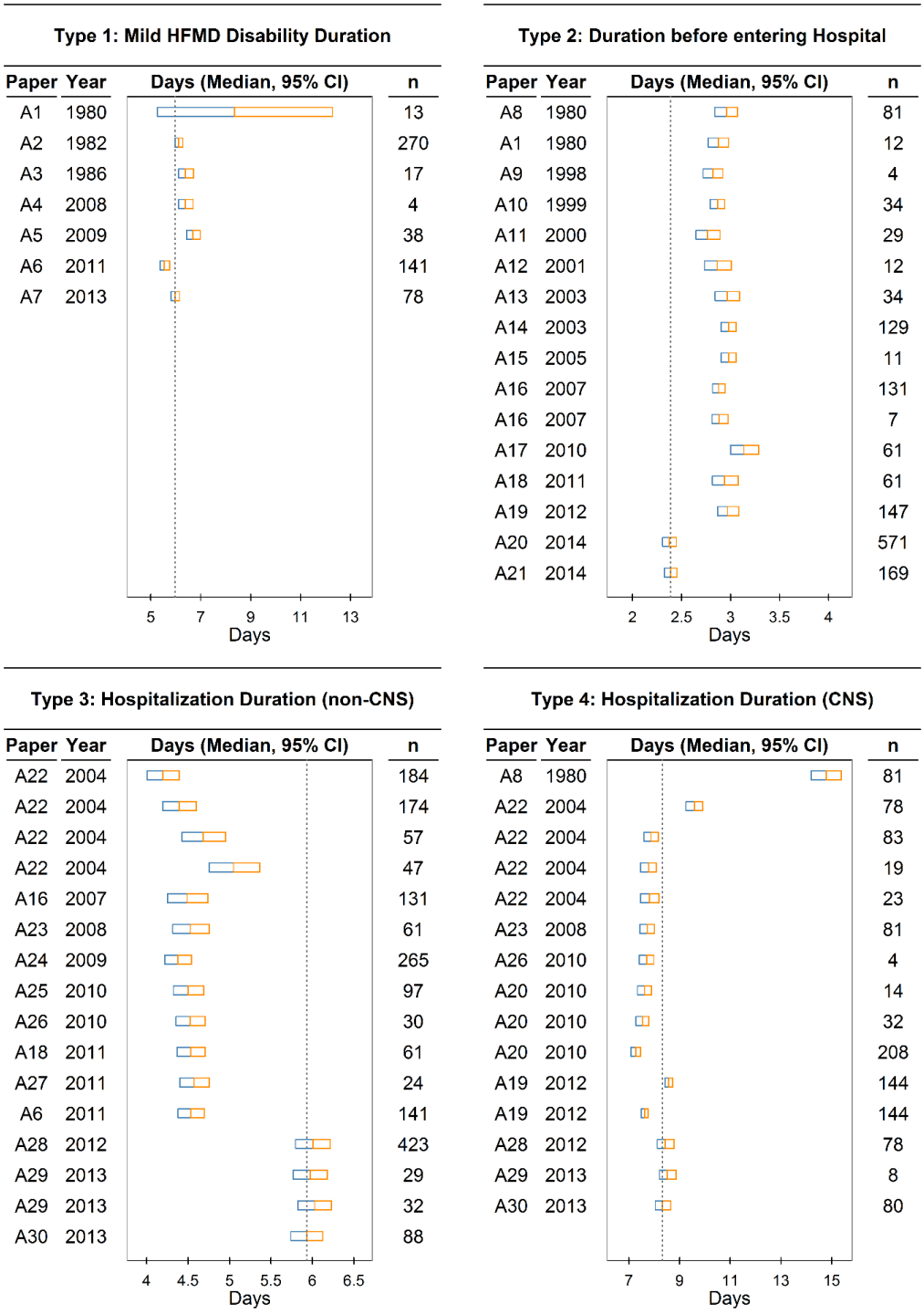


Figure 3.4: Disability Duration at Each Level of Severity. Blue boxes - Below median, Orange boxes - Above median. The information is arranged chronologically and added sequentially - i.e. the box at line 2 consist of data from paper 1 and 2, and box at line 4 consists of data from paper 1 to 4.

Table 3.5: Estimated Asymptomatic Rate of EV-A71 (per 100). Observed EV Infection is calculated from submodel 2 and is calculated by formula: $b_a E(\gamma_y^{ev})$, while Asymptomatic Rate is 1 less the ratio of the observed EV Infection and the Infection Rate: $1 - \frac{b_a E(\gamma_y^{ev})}{\alpha}$

Age	Seropositive Rate (I)	Infection Rate (α)	Observed EV Infection	Asymptomatic Rate
1	10.1 (9.5 - 10.9)	10.2 (9.5 - 10.9)	2.2 (1.8 - 2.7)	77.9 (75.2 - 80.4)
2	18.9 (17.8 - 19.9)	8.7 (8.4 - 9.1)	2.9 (2.4 - 3.5)	66.6 (63.1 - 69.9)
3	26.4 (25.2 - 27.5)	7.5 (7.3 - 7.7)	2.4 (2.0 - 2.9)	67.4 (64.2 - 70.6)
4	32.9 (31.7 - 34.0)	6.5 (6.2 - 6.8)	1.7 (1.4 - 2.0)	73.7 (70.8 - 76.4)
5	38.5 (37.4 - 39.7)	5.7 (5.3 - 6.0)	1.2 (1.0 - 1.4)	78.6 (75.9 - 81.0)
6	43.5 (42.1 - 44.7)	5.0 (4.5 - 5.4)	0.8 (0.7 - 1.0)	83.3 (80.9 - 85.4)
7	47.9 (46.1 - 49.3)	4.4 (3.9 - 4.8)	0.6 (0.5 - 0.7)	86.7 (84.5 - 88.4)
8	51.8 (49.6 - 53.4)	3.8 (3.3 - 4.3)	0.4 (0.3 - 0.5)	88.9 (86.8 - 90.5)
9	55.2 (52.6 - 57.1)	3.4 (2.9 - 3.9)	0.3 (0.3 - 0.4)	90.0 (87.9 - 91.6)
10	55.9 (54.2 - 57.7)	1.7 (1.5 - 3.0)	0.3 (0.2 - 0.3)	NA

3.4.2 Submodel 2: Incidence Analysis within Singapore

The year effect, γ_y, which captures the yearly variation of total HFMD incidence (Figure fig:met3, mid), has a correlation >99% with the overall number of cases of HFMD in Singapore. There is a positive slope of gradient 0.026 (95% CI: 0.0003, 0.052) indicating an increase in number of cases over time. The year effect of EV-A71 is estimated using laboratory data (Figure 3.3, right), and does not have a significant trend with slope -0.0091 (95% CI: -0.033, 0.015).

The age effect b_a which captures the systematic variation of incidence attributable to age, is estimated from data from years 2005 to 2012, and is representative of the proportion of total incidence for each age, showing that more than 50% of disease incidence are due to children below four years of age. Around 2.2% to 2.9% of the population is affected by EV-A71 each year during their younger ages (1 - 3), but quickly reduces to 1.2% by age five (Figure 3.2, mid).

For Singapore, the required rate for new infections is much higher than the actual observed level, and the difference are due to asymptomatic infection. For the most vulnerable age

Table 3.6: Incidence Analysis: Year Effects

Year	HFMD Year Effect	HFMD Cases	Proportion EV-A71	EV-A71 Year Effect
2005	0.308	15232	0.53 (0.41 - 0.64)	0.162 (0.128 - 0.197)
2006	0.297	15278	0.46 (0.37 - 0.54)	0.135 (0.111 - 0.159)
2007	0.399	19984	0.09 (0.03 - 0.14)	0.034 (0.011 - 0.056)
2008	0.542	29443	0.68 (0.64 - 0.73)	0.371 (0.346 - 0.395)
2009	0.331	16867	0.17 (0.13 - 0.22)	0.058 (0.044 - 0.071)
2010	0.601	30749	0.19 (0.15 - 0.22)	0.112 (0.092 - 0.133)
2011	0.412	20610	0.58 (0.50 - 0.66)	0.238 (0.204 - 0.271)
2012	0.732	37035	0.11 (0.08 - 0.14)	0.079 (0.058 - 0.1)
2013	0.617 (0.576 - 0.659)	31779	0 (0 - 0.01)	0.003 (0 - 0.007)
2014	0.435 (0.396 - 0.473)	22187	0.01 (0 - 0.03)	0.005 (0 - 0.013)
2015	0.549 (0.510 - 0.589)	28209	0.3 (0.21 - 0.39)	0.165 (0.113 - 0.217)

group, around 70% of EV-A71 cases do not show sufficient symptoms to be identified either at presentation to their preschool or by their doctor (Figure 3.2, right). Given the high intensity of Singapore's notification system, this is arguably a lower bound on how much unnotified infection there is across other Asian countries.

Table 3.7: Incidence Rate and Age Effect in %

Age	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	b_a
1	4.68	5.02	6.49	9.41	5.88	11.24	7.47	14.27	10.24	7.22	9.11	0.166
2	7.37	6.99	9.03	11.97	7.80	13.32	9.43	18.83	13.45	9.48	11.97	0.218
3	6.73	6.15	8.53	11.11	6.39	11.58	8.08	13.81	11.41	8.05	10.16	0.185
4	5.16	4.64	6.04	8.20	4.69	8.10	5.53	9.02	7.84	5.52	6.97	0.127
5	3.46	2.95	4.22	5.80	3.07	5.76	4.00	6.71	5.43	3.83	4.83	0.088
6	2.05	2.03	2.85	4.17	2.15	3.60	2.93	4.64	3.64	2.57	3.24	0.059
7	1.13	1.37	2.13	2.96	1.42	2.50	1.93	3.44	2.53	1.78	2.25	0.041
8	0.83	0.89	1.42	2.39	1.04	1.83	1.36	2.54	1.85	1.31	1.65	0.03
9	0.61	0.67	0.98	1.95	0.86	1.58	1.29	1.91	1.42	1.00	1.26	0.023
10	0.37	0.55	0.70	1.45	0.56	1.22	0.94	1.72	1.11	0.78	0.99	0.018
11	0.24	0.37	0.49	1.13	0.50	0.91	0.78	1.36	0.86	0.61	0.77	0.014
12	0.20	0.25	0.30	0.79	0.31	0.62	0.61	1.02	0.56	0.39	0.49	0.009
13	0.12	0.15	0.25	0.56	0.26	0.46	0.51	0.87	0.43	0.30	0.38	0.007
14	0.10	0.12	0.12	0.42	0.20	0.38	0.39	0.59	0.31	0.22	0.27	0.005
15	0.08	0.08	0.13	0.30	0.15	0.28	0.29	0.47	0.25	0.17	0.22	0.004
16	0.06	0.08	0.09	0.21	0.11	0.20	0.18	0.34	0.19	0.13	0.16	0.003
17	0.05	0.07	0.08	0.13	0.08	0.17	0.12	0.22	0.12	0.09	0.11	0.002
18	0.05	0.07	0.12	0.11	0.08	0.17	0.09	0.22	0.12	0.09	0.11	0.002

3.4.3 Submodel 3: Hierarchical Model of Disease Severity

For HFMD in general (i.e. without reference to the specific viral agent), the symptomatic case-hospitalization proportion is estimated to be 6% (2.8% - 14.9%). Among hospitalized cases, 18.7% (6.7% - 31.5%) develop CNS complications, of which 5.0% (2.9% - 7.4%) were fatal. Overall, the case-fatality rate for HFMD is estimated to be 52.3 (24.4 - 92.7) per 100,000 symptomatic infections. Among the papers which identified EV-A71 as the pathogen, the majority recruited children seeking treatment in hospitals. Of these, 36.9% (25.9% - 48%) were associated with CNS complications, of which 10.5% (4.9% - 17.8%) died. In contrast, there were no discernable differences in the fatality rate of HFMD patients with CNS complications between cases with EV-A71 and all HFMD infections, possibly because most severe cases were indeed caused by EV-A71. Estimates within studies and overall are presented in Figure 3.5 and Table 3.8, which shows a summary of all estimated severity

quantities of HFMD in Asia with their associated uncertainties. The number of cases, as calculated by the severity pyramid model, is shown in Table 3.9.

There is some heterogeneity observed in the estimates, particularly for the estimates which involves Hospitalization which is dependent on human factors (whether the parents wishes to send the child to the hospital), rather than diagnostic (Severe and Case). Through the Bayesian model, the estimates for Case-Complications and Case-Fatality will remain accurate.

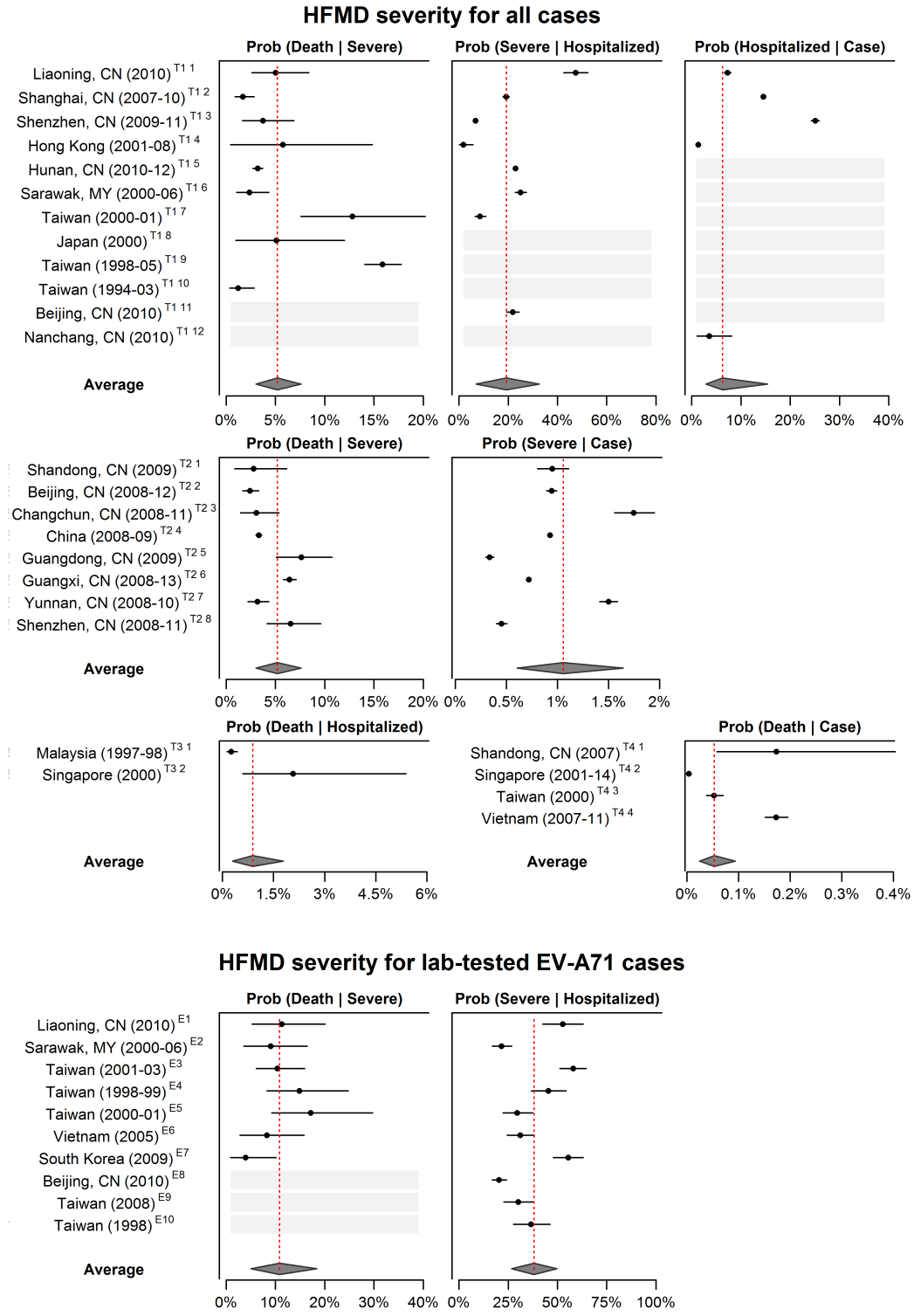


Figure 3.5: Data and Results from the Hierarchical Model (Submodel 3). The median and 95% credible intervals for each individual paper are shown by the black dot and lines, vertices of the diamond are the median and 95% Bayesian CI of the hyperparameter. The estimates are linked together with a multiplicative relationship: $P(D|C) = P(D|S) \times P(S|H) \times P(H|C)$. Case-hospitalization ratio estimate for EV-A71 cannot be calculated as studies on EV-A71 cases are almost performed exclusively from hospital data, and thus the lowest level is 'hospitalized' and not 'cases'. Data sources for HFMD totals can be found in Table 3.3 and EV-A71 in Table 3.4.

Table 3.8: Severity Pyramid of HFMD.

Rate (95% CI)	HFMD	Combined	EV-A71
	Mortality (per 1000 CNS)		
Pr(Death | CNS)	50.2 (29.2, 73.5)	-	104.7 (49.0, 177.8)
	CNS (per 1000 hospitalizations)		
Pr(CNS | Hospital)	186.7 (66.6, 314.9)	-	368.8 (258.5, 480.0)
	Hospitalization (per 1000 cases)		
Pr(Hospital | Case)	-	60.2 (28.2, 148.6)	-
	Symptoms (per 1000 infections)		
Pr(Case | Infection)			
Age 1	-	220.6 (195.8, 247.7)	-
Age 2	-	334.6 (299.7, 369.7)	-
Age 3	-	325.4 (293.8, 357.6)	-
Age 4	-	263.1 (236.0, 291.5)	-
Age 5	-	214.0 (189.8, 240.6)	-
Age 6	-	166.6 (146.6, 191.0)	-
Age 7	-	133.0 (115.3, 154.8)	-
Age 8	-	111.2 (95.1, 131.9)	-
Age 9	-	100.3 (84.8, 121.2)	-
	Infection (per 1000 person years)		
Age 1	44.9 (36.4, 53.4)	-	22.4 (18.2, 26.7)
Age 2	58.4 (47.4, 69.5)	-	29.2 (23.7, 34.7)
Age 3	49.0 (39.8, 58.3)	-	24.5 (19.9, 29.1)
Age 4	34.3 (27.9, 40.8)	-	17.2 (13.9, 20.4)
Age 5	24.3 (19.8, 28.9)	-	12.2 (9.9, 14.5)
Age 6	16.5 (13.4, 19.7)	-	8.3 (6.7, 9.8)
Age 7	11.6 (9.4, 13.8)	-	5.8 (4.7, 6.9)
Age 8	8.5 (6.9, 10.2)	-	4.3 (3.5, 5.1)
Age 9	6.8 (5.5, 8.1)	-	3.4 (2.8, 4.1)

Table 3.9: Estimated Cases at Each Severity Level. The number of 'infections' include asymptomatic cases, and is estimated from serological information all around Asia. 'HFMD' is an estimate of number HFMD symptomatic cases, if each country puts in as much resources as Singapore in case-finding. The number of children at higher levels of severity are estimated as if all these countries have the same level of medical care as the rest of countries in the hierarchical model.

Country	**Infection**	**HFMD**	**Hospitalized**	**CNS**	**Death**
China	18.1M (16.7M - 20.6M)	3.99M (3.65M - 4.58M)	240K (109K - 604K)	42.4K (21.4K - 110.7K)	2,121 (884 - 5,648)
Hong Kong	71.7K (49.7K - 85.5K)	15.7K (10.9K - 18.9K)	899 (391 - 2,348)	159 (76 - 422)	8 (3 - 22)
Japan	1.30M (1.19M - 1.45M)	289K (259K - 323K)	17.3K (7.9K - 43.3K)	3,052 (1,563 - 7,848)	153 (63 - 409)
Malaysia	661K (591K - 695K)	145K (129K - 156K)	8.6K (4.0K - 21.7K)	1,522 (767 - 3,988)	76 (32 - 201)
Singapore	57.5K (48.5K - 100.7K)	12.7K (10.6K - 22.3K)	825 (353 - 2,288)	144 (71 - 399)	7 (3 - 21)
Taiwan	243K (202K - 373K)	53.9K (44.1K - 82.5K)	3,441 (1,489 - 9,170)	605 (297 - 1,651)	30 (12- 86)
Thailand	915K (850K - 1.18M)	203K (184K - 260K)	12.5K (5.6K - 31.7K)	2,198 (1,127 - 5,764)	110 (46 - 300)
Vietnam	1.74M (1.66M - 1.82M)	385K (360K - 409K)	23.0K (10.6K - 57.4K)	4.1K (2.1K - 10.5K)	203 (84 - 536)

3.4.4 Disease Burden

The disease burden estimated using our approach is 96,900 (40,600 - 259,000) DALYs per annum in eight high endemic countries in East and South East Asia. Country level estimates for DALY are provided in Table 3.10. Our approach is conservative and will represent the minimum expected burden of disease. Despite the low case-fatality rate, the majority of disability comes from mortality ($\sim 99\%$) because deaths from 1 - 3 year olds contribute very long disability lengths, as shown in the disability length models, which estimates disability length for mild symptomatic HFMD at 6.0 (5.8 - 6.2) days; hospitalization about 8.3 (8.1 - 8.5) days, while HFMD with complications at (10.4 - 11.1) days.

Table 3.10: Annual Disability Adjusted Life Year (DALY) Losses in Eight Asian Countries.

Country	DALY	95% CI
China	75,881	(31,835 - 202,591)
Hong Kong	285	(115 - 767)
Japan	5,456	(2,290 - 14,589)
Malaysia	2,723	(1,138 - 7,281)
Singapore	259	(104 - 748)
Taiwan	1,084	(435 - 3,052)
Thailand	3,928	(1,644 - 10,536)
Vietnam	7,248	(3,042 - 19,414)

3.5 Discussion

In this study, we used a statistical modelling approach to synthesize information on HFMD from countries in East and South East Asia where the disease is highly prevalent. To estimate a conservative lower bound for the asymptomatic rate of EV-A71, we used case notification data from Singapore. The small, clearly demarcated jurisdiction with policies that lead to comprehensive case finding of symptomatic HFMD cases-notification being currently mandatory for physicians and childcare centers - may permit a higher fraction of symptomatic cases to be notified than other settings. Supporting evidence comes from the higher reported incidence rate of HFMD in Singapore (around 6% to 14% at age 1) than in

other areas (e.g. about 3 to 5 times higher than China where it is $\sim 3\%$ at age 1), despite Singapore having lower average EV-A71 seropositivity rates (Figure 3.2, left). Under the assumption that conditions in Singapore permit all symptomatic HFMD to be detected, then the difference between the infection rate of EV-A71 and the incidence rate of EV-A71 gives the asymptomatic rate which, at 71.4% (68.3% - 74.3%) for ages 1 - 4, is comparable to previous estimates [159].

Such a high asymptomatic rate for HFMD has implications on estimates of R_0, the basic reproduction number. A large proportion of unobserved infections imparts a downward bias on estimated infection rate parameters in compartmental models (such as SIR variants), and accounting for the high asymptomatic rate could be the key to reconciling the highly heterogeneous R_0 estimates in the literature [159, 192, 193].

When performing data synthesis, there is a particular challenge in the allocation of collected information into the severity tiers defined in this study. For instance, we were concerned that the identified EV-A71 cases in the literature was subjected to selection bias, as it is more likely for physicians to take samples from sicker children. We therefore excluded studies that reported clinically diagnosed cases of EV-A71 because of this potential for bias, and as a result, the lowest level on the severity pyramid we could estimate for EV-A71 was 'hospitalizations' rather than 'symptomatic cases'. To obtain the case-fatality rate for EV-A71, therefore, required assuming the case-hospitalization rate is the same for EV-A71 and other etiologies (i.e. 6%, 95% CI: 2.8% - 14.9%). This is a lower bound estimate, since EV-A71 is usually more severe than others. Under that assumption, the CFR of EV-A71 infection is at least 0.23% (0.10% - 0.42%). This estimate is lower by a factor of 7 than the 2015 systematic review of EV-A71 case-fatality rate of 1.7% (1.2% - 2.4%) [157] which we attribute to a difference in denominator, with the Zhao review more in-line with our estimated hospitalization-fatality rate of 3.8% (1.7% - 7.0%). Based on symptomatic case-fatality rates, EV-A71 is less severe than the relatively mild influenza A(H1N1-2009pdm) strain, which had a symptomatic CFR ranged between 0% - 1.2%[194] in 50 published studies. Our other estimates, of case-hospitalization, case-severity, and case-fatality rates of symptomatic HFMD align with those from an epidemiological analysis of Chinese data [50], giving credibility to both sets of estimates.

Despite the high incidence rate of HFMD, the relative scarcity of severe symptoms and the generally short disability durations led to estimated annual DALY losses in eight high incidence Asian countries of only 96,900 (40,500 - 258,400), considerably below the disability caused by URTI 603k (343k - 1m) but comparable to dengue (100,000 in the same countries) which has low incidence in China [195]. However, the disability caused by HFMD is mostly associated with Years of Life Lost (YLL) (>99%), driven by long expected remaining lifespan of children in comparison to the mild, temporary disability from acute HFMD symptoms. Overall, the YLL of HFMD 96,322 (39,911 - 257,613) is higher than dengue 43,762 (25,444 - 76,488) and URTI 10,767 (8,459 - 13,636) because the majority of the DALY were caused by Years Lived with Disability (YLD) for the latter two diseases. Since EV-A71 causes the vast majority of deaths in HFMD, the adoption of EV-A71 vaccine will eliminate most of the disease burden.

The challenge in calculating the disease burden of HFMD lies in integrating and synthesizing available data from decades of research into useful quantities for public health actions. In doing so, many assumptions were made. In particular, we integrated data from serological and epidemiological studies from a mixture of country- and city-level studies across Asia, to develop an overall 'Asian' average which summarizes the severity across affected parts of the continent, which differ in economic development and healthcare systems. These differences cannot be easily adjusted using statistical methods due to low amounts of information from each individual country. Differences between and within countries, however, mean that these overall estimates may not apply to local decision making. Secondly, the age-specific asymptomatic rate estimates were based on surveillance data from Singapore, which we believe has a notification system for symptomatic infections presenting at primary care or in education that is close to complete, but with room for misdiagnosis. Thirdly, societal impacts - such as the indirect costs that result from caregivers taking time off work or other impacts of control policies [196] - are not measured in our disease burden estimates, and future research should seek to quantify such indirect impacts of hand foot and mouth disease so that the full cost effectiveness of intervention strategies can be evaluated.

3.6 Footnote

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3.7 Additional Materials

Table 3.11: Summary Statistics of Disability Duration

Paper	Year	Type	n	Mean	Median	Min	Max	CI	Std Dev	Special	Shape	Scale
A1	1980	1	13		7	1	18				1.328	9.226
A2	1982	1	270		6					Median only		
A3	1986	1	17	11		2	29				1.16	11.587
A4	2008	1	4			5	7			Imputed median = 6	5.468	6.416
A5	2009	1	38	9.13				5.34 - 12.9			5.65	9.874
A6	2011	1	141	2.78				1.78 - 3.78			6.62	2.981
A7	2013	1	78	8.6				7.1 - 10.1			14.14	8.923
A8	1980	2	81			2	4			Imputed median = 3	6.429	3.176
A1	1980	2	12							0 days, n=1; 1 days, n=4; 3 days, n=5; 4 days, n=1; 5 days, n=1		
A9	1998	2	4			1	2			Imputed median = 1.5	2.931	1.7

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Paper	Year	Type	n	Mean	Median	Min	Max	CI	Std Dev	Special	Shape	Scale
A10	1999	2	34		3					Median only		
A11	2000	2	29		2	1	10				1.474	2.565
A12	2001	2	12			1	7			Imputed median = 4	2.227	4.716
A13	2003	2	34			2	5			Imputed median = 3.5	4.582	3.791
A14	2003	2	129		3					Median only		
A15	2005	2	11		3	2	5				2.756	3.427
A16	2007	2	131	2.61					0.65		4.56	2.86
A16	2007	2	7	3.43					4.88		0.72	2.77
A17	2010	2	61		4	1	42				1.188	5.445
A18	2011	2	61		1	0	10			Imputed min = 1	1.474	1.282
A19	2012	2	147	3.1		1	4		0.5		7.31	3.31
A20	2014	2	571	1.59					1.05		1.55	1.77

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Paper	Year	Type	n	Mean	Median	Min	Max	CI	Std Dev	Special	Shape	Scale
A21	2014	2	169							77% of cases sought treatment in hospital within 3 days (130/169). Imputed median to be 2.5.		
A22	2004	3	184	4.2		0	34		1.35		3.44	4.67
A22	2004	3	174	4.6		0	66		2.4		2	5.19
A22	2004	3	57	6.5		1	90		5.4		1.21	6.92
A22	2004	3	47	8.3		0	90		6.5		1.29	8.97
A16	2007	3	131	2.51					1.46		1.78	2.82
A23	2008	3	61			3	7			Imputed median = 5	5.3	5.358
A24	2009	3	265	4					1.05		4.3	4.39
A25	2010	3	97		4	1	63				1.021	5.728
A26	2010	3	30	5.4					0.6		10.88	5.66
A18	2011	3	61		4.1	1	14				1.606	5.151

Continued on next page

Paper	Year	Type	n	Mean	Median	Min	Max	CI	Std Dev	Special	Shape	Scale
A27	2011	3	24		4	1	75				0.882	6.062
A6	2011	3	141	4.21				4.1 - 4.32			95.31	4.235
A28	2012	3	423		9	1	101				1.474	11.541
A29	2013	3	29	4						Mean only		
A29	2013	3	32	9						Mean only		
A30	2013	3	88		4				1.52		2.9	4.54
A8	1980	4	81			10	20			Imputed median = 15	6.429	15.88
A22	2004	4	78	4.2		0	34		1.35		3.44	4.67
A22	2004	4	83	4.6		0	66		2.4		2	5.19
A22	2004	4	19	6.5		1	90		5.4		1.21	6.92
A22	2004	4	23	8.3		0	90		6.5		1.29	8.97
A23	2008	4	81			5	10			Imputed median = 7.5	6.429	7.94
A26	2010	4	4	5.5					1.2		5.27	5.97

Continued on next page

Paper	Year	Type	n	Mean	Median	Min	Max	CI	Std Dev	Special	Shape	Scale
A20	2010	4	14	5.46					0.83		7.79	5.81
A20	2010	4	32	6.55					1.29		5.92	7.07
A20	2010	4	208	6.75					1.67		4.59	7.39
A19	2012	4	144	14.2		12	17		0.8		22.07	14.55
A19	2012	4	144	2.55		1.67	4		0.27		11.44	2.67
A28	2012	4	78	18		1	101				1.01	18.075
A29	2013	4	8	18						Mean only		
A30	2013	4	80		6				2.27		2.9	6.81

Shape and Scale are estimated parameters. Data imputation is performed for papers which reported only min/max only such that the median is between the min and max.

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Chapter 4

Eliciting Parental Preferences for the Control of Hand, Foot and Mouth Disease: A Discrete Choice Experiment.

4.1 Abstract

Background: Hand, foot, and mouth disease (HFMD) is a high incidence viral illness in Asia that mostly affects young children. In Singapore, these children often spend much of the day at childcare centers as most families have two working parents. As such, control measures for HFMD, such as school closures, can be disruptive and inconvenient to the parents, but weaker measures could result in higher risk of disease. This paper describes how parents with young children in childcare centers trade off inconvenience and cost with HFMD risk as measured through a discrete choice experiment (DCE), to assist policymakers with control measure optimization.

Methods: A qualitative study and pilot study was conducted to determine the attributes for the DCE. Childcare centers (CCC) were stratified by location and randomly selected from a sampling frame of all CCC that experienced at least one outbreak in the past year. The parents involved with each CCC were contacted through the school principals, and invited to participate in the DCE which consists of 10 pairs of hypothetical scenarios with varying levels of policy strength, cost and disease outcome. A total of 575 parents responded completely to the survey, and the data were analyzed with Bayesian random utility models.

Results: Overall, when faced with a tradeoff between reducing the risk of HFMD versus

the inconvenience caused by such measures, parent deem isolation sufficiently effective for implementation, although they do not show a preference towards different lengths of isolation. Closure policies were deemed to be not sufficiently effective to justify the burden. Daily temperature screening and mandatory sanitization after outbreaks were also seen to be important.

4.2 Introduction

Hand, foot and mouth disease (HFMD) is a childhood disease endemic in East and Southeast Asia, caused predominantly by Enterovirus A71 (EV-A71) and Coxsackievirus A16 and A6. Of these, EV-A71 infections have the greatest potential to cause debilitating neurological and pulmonary complications, leading to childhood mortality [140, 52, 198, 50]. In Asia, the infection rate of EV-A71 can reach 3 - 12% during preschool, with around 80% of notified cases occurring before age 6 [159, 150, 199]. This results in millions of symptomatic cases annually in the region.

As HFMD mostly affects young children, Singapore's Ministry of Health (MOH) has implemented a number of control policies targeted at preschools and childcare centres (CCC) which include i) mandatory participation in the surveillance program, with institutions being required to notify MOH when two or more cases of HFMD occur within a 10-day period; ii) home isolation of diagnosed cases (up to 10 days); and iii) school closure whenever the attack rate reaches 23% or >16 cases over 24 days [150].

These measures are similar to those commonly considered for outbreak response in governmental pandemic preparedness plans: isolation of confirmed cases, mandatory quarantine, school closure and temperature screening. Their effectiveness depends in part on how readily they were accepted by the community.

The choice of which control measures to implement, and when, involves trade-offs between the anticipated effectiveness of such measures in reducing disease burden and cost to the community. The effectiveness of an intervention can be evaluated through observational or modelling studies, which have suggested for example that school closures can be most effica-

cious when initiated early in an epidemic [199], although uncertainties in the epidemiology of novel (and indeed some extant) threats make the effectiveness of even common control strategies ambiguous. For instance, with HFMD, it is unclear whether schools are the main environment in which transmission occurs [159], while triggers based on outbreak size may fail in the face of a high asymptomatic rate which can reach close to 70% [23].

At the same time, effective interventions may not translate into successful policies: policies that do not have the support of the actors involved in implementing may become ineffective as a result. During the influenza A-H1N1 pandemic in 2009 in Victoria, Australia, parents whose children were nominally prevented from school attending due to quarantine showed generally poor compliance with these quarantine restrictions [200], while in Singapore, similar actions that undermine HFMD control, such as attempts to circumvent isolation, have been reported [46]. Even under the most restrictive of policies, effectiveness ultimately often depends on parental choice. A pragmatic approach to determining optimal control strategies therefore requires an understanding of subjective preferences over these strategies.

This paper presents findings from a discrete choice experiment (DCE) designed to elicit the preferences of the Singaporean public over control measures for HFMD to allow public preferences to be accounted for in policy decision making [201]. It explores how individuals for whom the interventions are directly salient weigh up competing preferences in a priority-setting context.

4.3 Methods

4.3.1 Pilot and Questionnaire Development

The perceptions of preschool staff and parents toward existing control policies were investigated in a qualitative study which included 44 in-depth interviews with teachers, principals and parents, as previously reported [46], the findings of which were used to inform the choice of attributes and levels in this study. In addition, a pilot study was conducted on 59 parents of children at three purposefully selected childcare centers from 30 Sep 2014 to 20 Oct 2014.

4.3.2 Design of Discrete Choice Experiment

An example of the choice set presented to participants may be found in (Figure 4.1). The DCE was designed to elicit preferences over four existing control policies (isolation, closure, temperature screening and mandatory sanitization), two outcomes (expected numbers of cases and hospitalizations) and a cost factor, represented as an increase in school fees (Table 4.1). The inclusion of the cost attribute allows the conversion from pure utility measure to a measure of willingness-to-pay (WTP) for modifications to extant policies. The outcome attributes were included to quantify how respondents trade-off risk of disease and inconvenience to parents when measures were implemented. The number of attributes were restricted to seven to ensure that task complexity did not cause excessive cognitive burden. Levels were selected based on the epidemiology of HFMD [166] and existing policies: isolation durations were based on postulated serial intervals [166], school closure and sanitization based on a simplified version of the current school closure policy, temperature screening were based on systems reported by teachers [46], changes in costs were selected based on typical current monthly fees (∼$600 - 800), while the number of cases and hospitalization were based on typical annual numbers (with an option of a substantial reduction in both added). The survey was fielded on-line (see later for details) and the DCE was framed as being a choice between two scenarios and the associated expected outcomes.

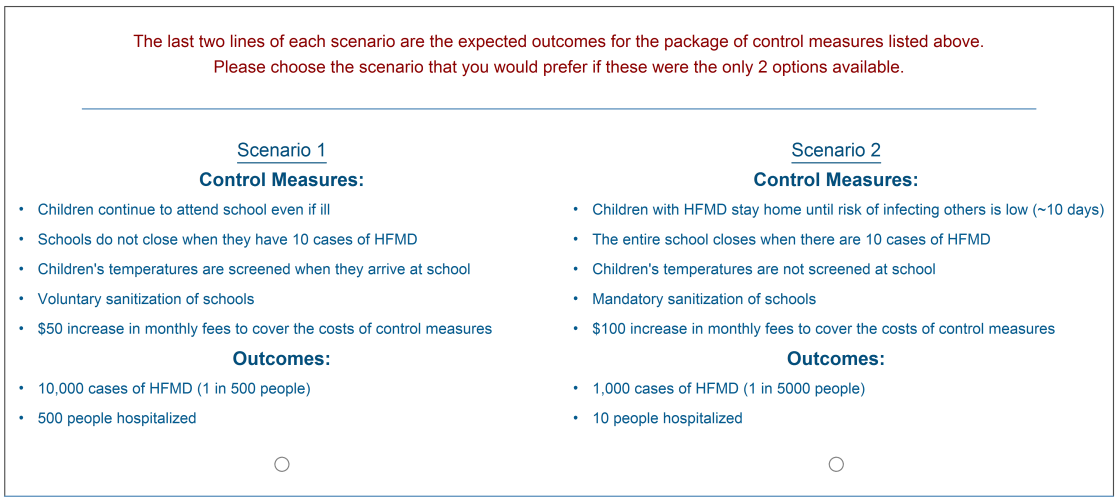


Figure 4.1: Example Choice Set from the Discrete Choice Experiment. Participants viewed ten of these in succession, and for each were requested to select one of the two sets as reflecting their preference.

With a total of 1,944 combinations (arrays) of levels or $1,944 \times 1,943$ possible choice sets, an algorithm was applied to reduce the number of choice sets to 105. Specifically, each array is coded as a vector of integers that represents the levels of the seven attributes. These numbers were scaled and the Euclidean distance between every possible choice set calculated. The array with the shortest average distance from the rest was removed sequentially until 50 arrays were left. Choice sets were selected by choosing pairs with the greatest distance from each other, until all arrays were selected at least once and the levels appear uniformly. Each respondent was provided 10 randomly selected choice sets and requested to indicate a preference in each pair.

This method satisfies three principles: Firstly, to prevent choice sets with dominant options, more intrusive measures were accompanied with better outcomes, reflecting better results if the respondents were willing to accept more inconvenience (Figure 4.2 Top). Secondly, to be able to better estimate the main effects of each variable, each attribute level should appear with the same frequency (Figure 4.2 Bottom). Finally, the two arrays in a choice set should be dissimilar to one another to increase the amount of information carried in each set.

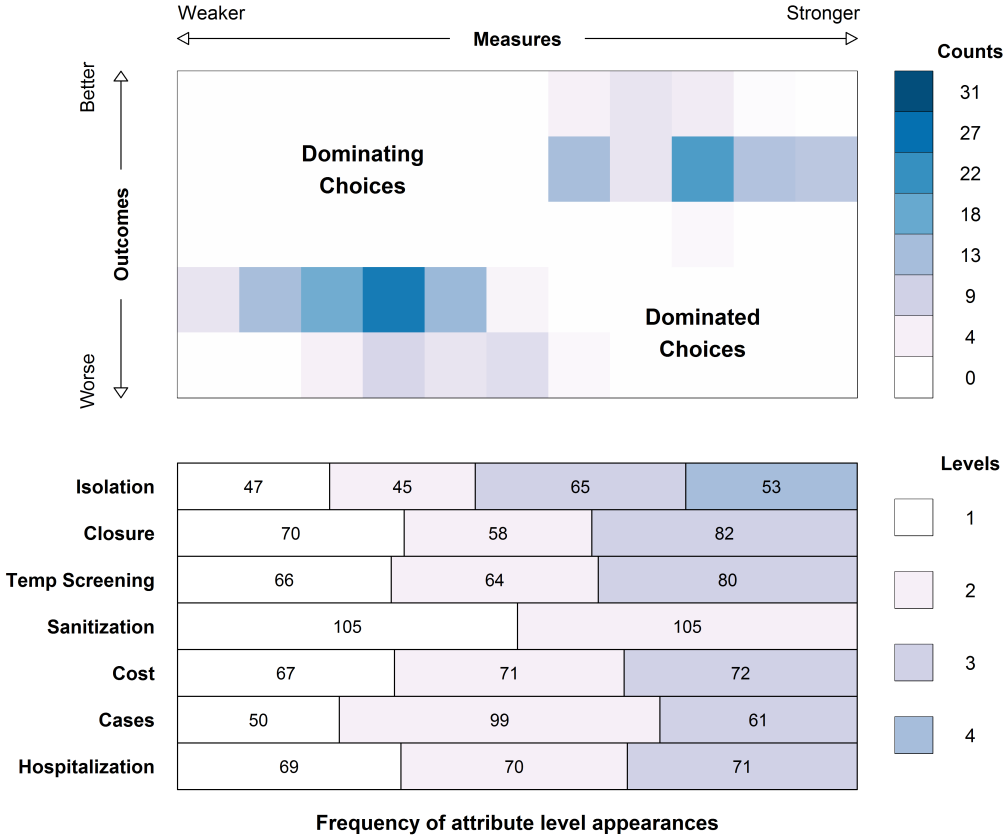


Figure 4.2: Characteristics of Attribute Levels in Choice Sets. Top: Heat map of the frequency of attribute level appearances in the final 105 selected choice sets, sorted by the strength of outcome and measures. The instrument was designed such that stronger measures were associated with better outcomes, and vice versa. Arrays with stronger measures and worse outcomes are always dominated by other arrays, while arrays with weak measures and better outcomes are always dominating. Bottom: Frequency of attribute level appearances. Level 1 are weaker measures/better outcomes. Each level within each attribute appears approximately the same number of times.

4.3.3 Fielding of Questionnaire

We used a sampling frame of 799 institutions derived from a complete list of all CCC registered under the Ministry of Community Development, Youth and Sports (MCYS) and preschools registered under the Ministry of Education (MOE) which had at least one outbreak in the past 12 months, defined as at least two cases within ten days. 39 CCC and 10 kindergartens were selected by random sampling with stratification by geographical region in Singapore. The survey consisted of two parts: a face-to-face interview with the principals to understand more about their CCC, and an online questionnaire on Qualtrics [202] for the parents. Principals gave written informed consent and were asked to invite parents to the online questionnaire using emails drafted by the researchers. Having an intermediary reach the parents was likely to reduce survey response rates, but was considered necessary to conform to Singapore's personal data protection laws. The survey was conducted between 18 December 2014 and 16 February 2016.

Parents were given two weeks to respond to the questionnaire which consisted of the DCE and other demographic and knowledge based questions. Measures to boost response rate included reminder emails, which the principals were requested to forward to parents; and monetary incentives in the form of SGD$10 (∼USD8) supermarket coupons for completion of the survey that was mailed to participants who provided their (delinked) address.

No data were available on the target population (parents of children in child care in Singapore) against which we could compare the sample to assess representativeness. Instead, we used (i) the whole population, (ii) Singapore residents aged 30 to 44, and (iii) ever-married females aged 30 to 44 as comparator groups, with demographic data taken from the national Department of Statistics [203, 204].

4.3.4 Statistical Analysis

Under the Random Utility Framework [205], which assumes that individuals choose the options which maximizes their utility, there are a number of ways to model the results of a discrete choice experiment. A common approach is the logistic regression model which has the

advantage of simplicity but usually assumes homogeneous preferences. Taste heterogeneity can be accommodated with more flexible models such as hierarchical models, generalized linear mixed models and latent class models, which have the freedom to explain differences in preferences between participants, improving the fit at the cost of introducing more parameters. We used both fixed effect logistic regression and hierarchical variants to quantify preferences.

The logistic regression model assumes that the utility of individual i (among 575 individuals), for choice j (2 choices) in task t (10 questions each) is $U_{itj} = x'_{itj}\beta + \epsilon_{itj}$ where x_{itj} is the set of attributes of the j choices in the choice set for individual i, task t. By assumption, individual factors do not interact with features of the DCE in this formulation. The random error terms, ϵ_{itj}, allow for deviations both from the linear model and from the assumption that participants are perfectly rational. The probability that an individual i picks choice $j = 1$ in task T is equal to $Pr(U_{it1} - U_{it2} > 0)$, which is a difference between two utilities of the same person. The likelihood function is thus derived from:

$$Pr(U_{it1} > U_{it2}) = \frac{\exp(x'_{it1}\beta - x'_{it2}\beta)}{1+\exp(x'_{it1}\beta - x'_{it2}\beta)}$$

The hierarchical variant of this model allows groups of individuals, differing in demographics, to differ by having their own set of coefficients δ_G distributed about an overall mean β, with a covariance matrix Σ quantifying the correlations of the coefficients, i.e. $U_{itj} = x'_{itj}\delta_{G_i} + \epsilon_{itj}$ where $\delta_G \sim MN(\beta, \Sigma)$.

Parameters under this framework can be estimated using a Bayesian approach where the posterior distribution was sampled using Markov chain Monte Carlo (MCMC) simulation using JAGS and R [206, 73]. Non-informative uniform distribution priors were used for the hyperparameters, where individual coefficients of $\beta \sim$ iid $N(0, 1000)$ and $\Sigma \sim W^{-1}(I, \nu)$, i.e. an inverse Wishart.

We made three modelling choices in the hierarchical model. Firstly, we elected to set all parameters (excluding cost) to be multilevel, rather than keeping some as fixed effects, following arguments presented elsewhere [207]. Secondly, we explored three models for the cost variable, which determines willingness to pay for the interventions: (i) setting it as a fixed effect, (ii) setting it as a random effect with Normal distribution and (iii) setting it as a random effect with (negatively signed) log-Normal distribution. For all models, we used

the deviance information criterion (DIC) [208] to measure goodness of fit and select between models.

The third modelling choice was the selection of groups in the hierarchy. We determined the groups by identifying demographic factors which improve model fit when stratified over, based on the likelihood ratio test, and found five such variables: Age (older vs younger than median), House size (Large vs small), Working Parents (both working or not), Domestic Helpers (presence vs absence) and Infants in the household (presence vs absence). The five dichotomous factors split the respondents into $2^5 = 32$ groups, which are the groupings used for the hierarchical model, forming 32 sets of coefficients and 1 overall set (Table 4.2).

4.4 Results

4.4.1 Characteristics of the Respondents

A survey completion rate of 60.7% (585/964) and response rate of 27.1% (964/3559) was attained, which was lower than most online surveys [209], due to our reliance on an intermediary to reach the target group. Compared to Singapore residents aged 30 to 44, our respondents had a slightly higher average number of children per household (1.76 vs 1.58 per ever married female), which is to be expected because our respondents have a minimum of 1 child, while the national data includes all married females regardless of parity (Table 2). Since 11.6% of married women have no children, a crude estimate would be 1.79 children per married women with at least 1 child, very close to 1.76 in our sample. The sample had a higher socio-economic status than the general population, as evinced by the larger proportion of respondents staying in larger apartments or private properties (the common proxy for socioeconomic status in Singapore). It also underrepresented people from the Malay ethnic group. Finally, women were more represented in the survey but this may reflect mothers being more likely to be the CCC's contact person in the family.

4.4.2 Willingness to Pay for Interventions

The willingness to pay measures for the fixed effects model demonstrate that the community favoured i) mandatory sanitization of childcare centers (WTP: $291, 95% CI $153, $671); ii) any duration of isolation (WTP for 10d: $178, 95%CI $92, $455, and similar for 5d and 15d isolation); and iii) temperature screening once daily (WTP: $68, 95% CI $18 , $177); and iv) closure of affected classes during outbreaks (WTP: $49, 95% CI $-18, $137). All other levels were valued similarly to the reference level for other attributes.

4.4.3 Group Heterogeneity in Willingness to Pay

Table 4.4 presents DIC scores for the hierarchical models together with the fixed effect model. There is overwhelming evidence that the hierarchical models outperform the fixed effects model in describing preferences. However, there is a close correspondence between the coefficients of the fixed effect logit model and the average coefficients of the hierarchical model (Figure 4.3), except for slighter higher variance which makes the preference for class closure not significantly different from no closure, indicating that the fixed effects model adequately describes the overall preferences of participants. In addition, the low variances of the hierarchical models' hyperparameters suggest that differences between groups, while present, were relatively unimportant. (The individual estimates for each of the 32 groups is shown in Figure 4.3 (orange circles) and summary statistics and their correlations are shown in Figure 4.4.) The interventions with most divergence of opinion on their value are isolation and sanitization.

4.4.4 Policy Comparison

Table 4.5 shows the willingness to pay for three policy options. One is the current policy (at the time of writing) in which children are screened a minimum of once daily for symptoms while at childcare, those with HFMD are isolated for 10 days, schools with an outbreak are closed for 10 days, and schools are advised to undertake regular sanitization of the school environment. The second is a policy currently being considered by Singapore's Ministry of

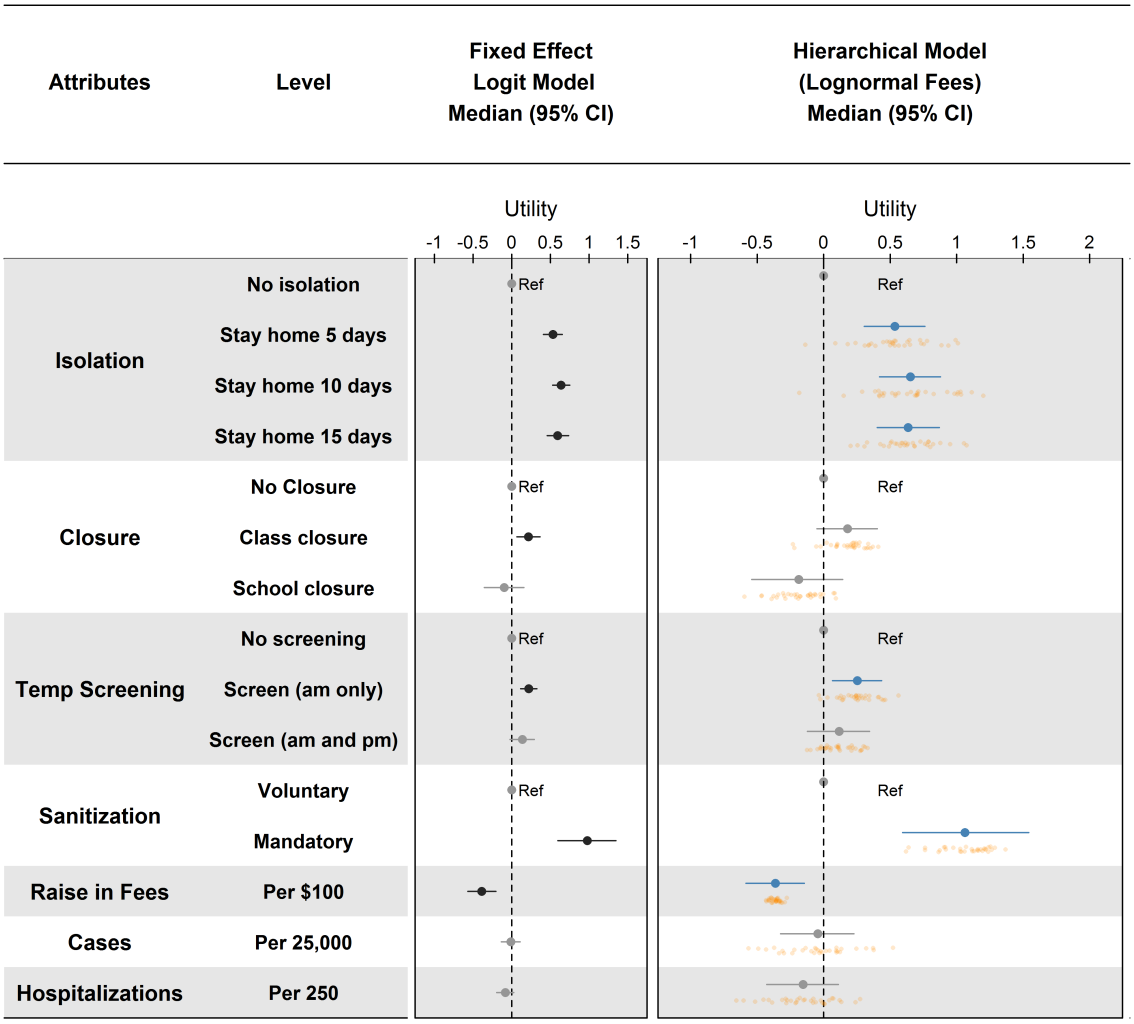


Figure 4.3: Coefficients of the Utility Models. Left: Estimated utility coefficients from the fixed effect logistic regression model. Right: Estimated utility coefficients of the hierarchical model. Blue dots indicate the median and 95% Bayesian credible intervals of the hyperparameters (the overall estimates). The orange circles are the median estimates at the group level.

Predictor Correlation	Isolation: Stay home 5 days	Isolation: Stay home 10 days	Isolation: Stay home 15 days	Closures: Affected Class	Closures: Whole School	Temperature Screening: Morning	Temperature Screening: Morning & Afternoon	Sanitization: Mandatory	Cost: Per $50	Incidence: Per 50,000 Cases	Hospitalizations: Per 500
Isolation: Stay home 5 days	1	0.83	0.69	0.28	0.65	0.04	0.58	0.59	-0.02	-0.24	0.45
Isolation: Stay home 10 days		1	0.77	0.47	0.58	-0.01	0.46	0.61	0	-0.31	0.41
Isolation: Stay home 15 days			1	0.2	0.51	0.11	0.12	0.49	-0.04	-0.25	0.53
Closures: Affected Class				1	0.3	-0.37	0.28	0.51	0.04	-0.44	-0.02
Closures: Whole School					1	-0.02	0.66	0.82	-0.1	0.05	0.46
Temperature Screening: Morning						1	0	-0.3	0.11	0.25	-0.12
Temperature Screening: Morning & Afternoon							1	0.54	-0.24	-0.07	0.23
Sanitization: Mandatory								1	-0.16	-0.26	0.36
Cost: Per $50									1	-0.09	0.01
Incidence: Per 50,000 Cases										1	0.14
Hospitalizations: Per 500											1
Mean (Utility)	0.54	0.65	0.63	0.18	-0.19	0.25	0.12	1.06	-0.04	-0.02	-0.06
Standard Deviation Between Groups	0.254	0.301	0.211	0.154	0.169	0.139	0.126	0.191	0.004	0.104	0.092
Mean Standard Deviation Within Groups	0.007	0.007	0.007	0.006	0.008	0.006	0.007	0.01	0.001	0.003	0.003

Figure 4.4: Correlation and Summary Statistics of the Group Level Coefficients. Deeper red ellipses represent high positive correlation, and gray ellipses represent high negative correlation. Mean (Utility) is the average of the 32 coefficients for the utility scale. Standard Deviation Between Groups represent the heterogeneity of the 32 groups, with higher standard deviations reflecting more heterogeneity. Mean Standard Deviation Within Groups is the average standard deviation of the posterior estimate of each group level coefficient. Low values mean that the estimate is precise.

Health, in which school closure is not routinely implemented. The third policy was developed based on the most favored implementations elicited via the discrete choice experiment, i.e. in which only classes affected by an outbreak are closed, and schools are mandated to undertake frequent sanitization. The overall value of the proposed policy is similar to that of the current one ($197 vs $248 respectively, MWU p-val <0.001, statistically significant but the dollar difference is not that large), and lower than the one created from the preferences expressed by participants in this study ($590.59, MWU p-val <0.001).

4.5 Discussion

This paper studied the preferences of 575 parents of children who are in childcare centres or preschools which experienced an outbreak in the past 12 months. They were asked to choose between multiple hypothetical combinations of HFMD-control policies involving different extents of isolation, school closure, temperature screening and sanitization, which were designed following a qualitative study [46] and a pilot study.

Under Singapore's regulations, since a fatal outbreak of HFMD in the year 2000 [37, 64], children diagnosed with HFMD have needed to obtain a doctor's letter to return to school. Despite parents being generally negative about isolation of sick children due to the logistical complications of keeping a child at home [46], the general consensus from the DCE shows that parents see value in isolation measures, although they did not express any difference in preferences toward different durations of isolation. This could reflect asymmetric preferences towards isolation measures, where parents want other children to be isolated if they are sick but think it is a burden when it happens to their children and they are unable to determine how long isolation should be. This phenomenon might also be attributable to lack of knowledge about the infectious period of HFMD (the evidence for 10 day isolation is weak [159]), or because of their trust in the ability of what has been called Singapore government's 'paternalistic and authoritarian rule' [210] to make the correct decisions.

It was hypothesized in a previous, qualitative study that closure measures were perceived to be effective but socially and economically burdensome [46]. This study, which quantifies the

tradeoff between preferences of prevention of disease and of avoiding burdensome interventions, found that, interestingly, parents preferred class closure more than either school closure or no closure at all in response to an outbreak. This may reflect their perception that transmission mostly occurs at a class level, and so closure of the entire school may be perceived as unduly inconveniencing.

Temperature screening and mandatory sanitization measures differ from isolation and school closure as the social burden lies with the childcare centre and not the parents. Parents exhibited no strong preferences about screening, but very strong preferences for mandatory sanitization. The government provides guidelines on sanitization of childcare centers but this is not currently enforced in Singapore. The high willingness to pay for mandatory sanitization ($200-$400 per month) is in spite of the paucity of evidence to support it reducing transmission.

Finally, the experiment revealed that parents' preferences did not depend on the number of cases and hospitalizations. We hypothesize that this indifference could be due to the reduced salience of societal level impacts compared to factors such as increasing school fees or isolation of sick children that parents may have experienced first-hand. Also, by phrasing these options as an 'incidence rate' in our questionnaires, parents might not have interpreted it as health risk to their child, but rather to other children, or to the general population, and thus deemed to be less important. Also, as the number of deaths due to HFMD in Singapore is lower than in other countries in the region (chapter 3) even after adjusting for population size, perceived risk of severe illness and death might be lower compared to other countries in the region.

There are a number of limitations to this study. Despite the DCE only containing seven attributes, there was some item non-response that may reflect difficulty in answering the DCE questions: out of the 964 responses, 790 attempted at least 1 DCE question, but only 608 (77.0%) completed the full set of ten questions (and only 575 had full responses to demographics). The lengthy survey time (about 30 mins) might also have contributed to lower completion rates [211, 212]. However, the characteristics of the 215 who did not complete the study were not statistically different from those who did, in their gender

($\chi^2 : p = 0.09$), age (Mann-Whitney U [MWU]: $p = 0.77$), citizenship ($\chi^2 : p = 0.38$), type of housing ($\chi^2 : p = 0.50$), household size ($MWU : p = 0.53$) and race ($\chi^2 : p = 0.60$), which makes selection bias unlikely.

Although CCCs were approached using a formal sampling frame, their principals could decline to participate, introducing potential selection biases. In particular, a number of institutions under the PAP Community Foundation – a charitable foundation with lower school fees – did not consent to participate, and there was some evidence that our surveyed population was of higher social economic status than average for the population as a whole. This was shown by comparing the average fees of the surveyed CCCs: the fees charged by participating kindergartens (median $278) were higher than the general community (median $161), while the fees charged by the CCCs surveyed (median $1070) were also higher than the general community (median $850) [213]. As a result, the sample is likely to over-represent higher-income households, who might have higher willingness-to-pay.

To conclude, a better understanding of the preferences of the general community to control of HFMD and other diseases will allow affected jurisdictions to optimize existing guidelines. Previous research in Singapore suggested that parents frequently sought to 'beat' isolation policies, by consulting multiple doctors in the hope that one would certify their child well for school, or negotiating with schools to allow their children back early [46]. The impact on parents of isolation or temporary school closure was in many cases inconvenient [46], or even costly, similar to findings from studies on influenza [213, 214]. The current study shows, however, that even with such an impact–and even in a population that has, generally, experienced isolation and closure (given the high incidence of HFMD and school closure)– support for isolation to prevent transmission remains high. School closure itself, however, received less support, echoing the results of in-depth interviews [46]. Compounded with the lack of evidence for its effectiveness HFMD outbreak control, this policy may need to be re-evaluated for routine outbreaks of HFMD.

4.6 Appendix

4.6.1 Alternative Approach: Latent Class Model

With the focus on modeling the heterogeneity of preferences, another common approach is to use latent class models (LCM). Such models can accommodate taste heterogeneity by classifying respondents into unobserved groups which display similar characteristics.

The LCM for discrete choice among $j = 2$ alternatives, by individual i for 575 respondents, observed in $t = 10$ tasks (each respondent answers 10 questions) is similar to the logistic regression model, except each individual in class q is allowed to have their own coefficients. We model the utility as:

$$U_{itj} = x'_{itj}\beta_q + \epsilon_{itj}$$

where $\boldsymbol{x_{itj}}$is the set of attributes of the j choices in the choice set for individual i, task t, and there is a set of $\boldsymbol{\beta}$ for each class, q.

$$\begin{aligned}&\Pr(\text{Choice } j = 1 \text{ by individual } i \text{ in task } T \mid \text{Class } q)\\ &= \frac{\exp((x'_{it1} - x'_{it2})\beta_q)}{1+\exp((x'_{it1} - x'_{it2})\beta_q)}\\ &= \Pr(U_{it1} > U_{it2} \mid \text{Class } q)\end{aligned}$$

If there are M latent classes, class membership probabilities follows the Dirichlet distribution with a vector of M non-negative parameters $\boldsymbol{\alpha}$. $\boldsymbol{z'_{it,j}}$ is the set of attributes of the j choices in the choice set for individual i, task t. $\boldsymbol{\beta_q}$ is the vector of parameters to be estimated using Bayesian inference. Non-informative uniform distribution priors were used for all coefficients, while non-informative Dirichlet priors were used for membership probabilities of latent classes. To determine the number of latent classes, we adopted two criteria for model selection. Firstly, the chosen model should have an advantage in terms of their goodness-of-fit, measured by Akaike information criteria (AIC). Secondly, as over-fitted latent class models are known to suffer from large uncertainties in the coefficients [215] and low membership probabilities in redundant latent classes, models which displayed such features were identified were removed post-hoc.

4.6.2 Results

Based on 10,000 samples of the posterior distributions of the coefficients, the average AIC $= 2k - 2\times$ logL is 6745.4 for the logistic regression model (close to the DIC of the fixed effect model of the actual analysis) and 5139.9 for the 2 latent class logit model. The standard logistic model has $k = 11$ parameters, while the 2 latent class logit model has $k = 23$ parameters from 22 coefs and 2 membership probabilities.) The membership probabilities associated with the 2 latent class logit model were 53.1% (47.8% - 58.4%) and 46.9% (41.6% - 52.2%). However, for models that had more than 2 latent classes, converge for the parameters beyond the first 2 latent classes was poor, with large standard deviations for the posterior distributions of parameters and with low membership probabilities, suggesting there was no evidence in the survey of more than two latent classes. Furthermore, in the models with three or more classes, the two more populous classes resembled the two from the 2-class model. Thus, the 2 latent class model was selected for further analysis. The modeling results using the standard logistic regression and 2 latent class logistic regression are presented in Figure 4.5.

The standard logistic regression coefficients represent the preferences of the community as a whole. Overall, the community favored isolating sick children, but did not display any preference towards the duration of isolation. Secondly, closure of affected classes only during outbreaks was favored over closing the entire school. Parents also prefer temperature screening once a day (in the morning) over doing nothing and screening twice a day. Lastly, mandatory sanitization was the most important policy to parents in terms of willingness-to-pay (WTP). The WTP was calculated by comparing all the utility coefficients (numerator) to the cost coefficient (denominator). For the general public, compared to doing nothing, the median WTP estimates derived for 10 day isolation was \$163 (95%CI \$105 - \$317); \$50 (\$12 - \$100) for closure of affected classes; \$55 (\$28 - \$107) for temperature screening (once daily); and \$276 (\$166 - \$505) for mandatory sanitization. Relaxing the assumption of taste homogeneity with the latent class logit model found that 53.1% of surveyed parents (LC1) had particularly strong views and high willingness-to-pay compared to the 46.9% (LC2) whose primary revealed aim was isolation of sick children from their peers. For LC1,

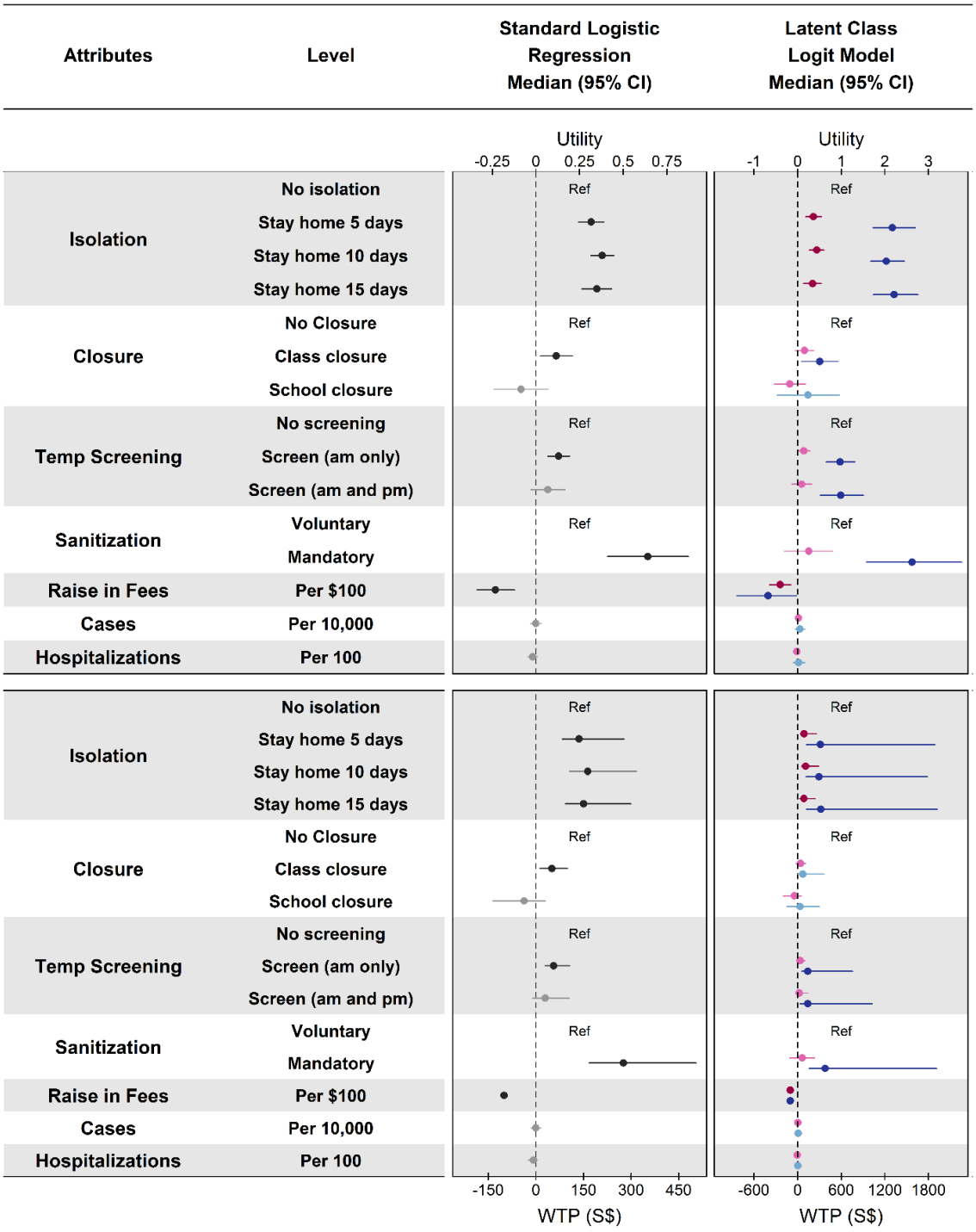


Figure 4.5: Results of Latent Class Model. Coefficients of the standard logistic regression model and the 2 latent class model in red (Class 2) and blue (Class 1). Darker coloured points are statistically significant. The willingness to pay metric is obtained by dividing the individual coefficients with the cost coefficient. The blue group prefers stronger policies and is more willing to pay to see them enforced. However, the distributions of the latent class model coefficients appear to have very high skewness, as seen by the length of the 95% CIs (particularly for the WTP).

mandatory sanitization 2.63 (1.58, 3.77) and isolation of any duration 2.14 (1.71, 2.63) were rated as the most important policies relative to the rest. LC1 also prefers class closure 0.51 (0.08, 0.93) to school closure 0.23 (-0.95, 0.47), and does not differentiate between screening temperature once or twice a day 0.98 (0.59, 1.41). This translates to a median WTP of $380 for mandatory sanitization, $308 for isolation, $72 for class closure and $140 for temperature screening. In comparison, LC2 was only concerned about isolation of any duration 0.38 (0.19, 0.56) which translates to WTP of $96.

Respondents preferring stronger interventions (LC1) were more likely to be older, $p = 0.0510$. The latent classes also differ in the ethnic group, with fewer Chinese and more 'others', a loose category for everyone who is not Chinese, Malay or Indian ($p = 0.019$) These households also tended to have dual working parents ($p = 0.009$) and to have the option of working from home ($p < 0.001$), less likely to flexible working hours ($p = 0.055$) and less like to be able to take unpaid leave ($p = 0.033$). They are less likely to have grandparents who are available to help care for the child ($p = 0.062$), but far more likely to have hired a domestic worker ($p < 0.0001$). The main results from the analysis of the demographics of the 2 latent classes are summarized in Table 4.6.

4.6.3 Predictive Analytics

The mean coefficient for each measure at the group level are first converted to ordinal variables: average (utility score between 30th and 70th percentile), high (>70th percentile), and low (<30th percentile), and cross-validation classification tree analysis was performed on isolation, school closure, screening (twice) and mandatory sanitization coefficients by breaking the dataset into two-thirds training data (for model building) and one-third test data (for cross-validation). The classification tree analysis is performed with the âĂŸtreeâĂŹ package in R [216], which fits data based on the Gini Impurity. Results from 10,000 simulations indicate training data misclassification rate of 32.8% (95% CI: 26.6%, 46.1%) and test data misclassification rate of 46.9% (95% CI: 34.4%, 60.9%). This can be improved to 30.8% (95% CI: 24.3%, 41.1%) and 45.3% (95% CI: 32.1%, 60.4%) respectively by excluding temperature screening, which has lower heterogeneity and correlation compared to the

rest. The classification tree outperforms random selection misclassification rate of 66% significantly.

While classification tree models have a clear advantage in its interpretability, they were known to suffer from large variances, for example, our sampleâĂŹs misclassification rate ranges widely between 30% to 60%. Under this algorithm, data is fitted based on best outcome at each split, and not by the best overall fit, and thus could be sensitive to the chosen training data. This weakness can be mitigated with Random Forests, which improves the model with a combination of bootstrap aggregation (bagging) and random sampling of variables at each split to increase the independence of individual trees. Using the same cross-validation strategy with the random forest, performed with the 'randomForest' package in R [217], yielded training set misclassification error of 35.5% (27.1% - 45.7%) and testing misclassification error of 35.8% (26.4% - 49%) for the selected measures, a significant 10% improvement over the classification tree.

Leave-one-out cross validation was performed for all measures and outcomes (Figure 4.6). There are three possible outcomes from the model: Correct, partially incorrect and totally incorrect, all of which occurs with equal chance in a random allocation (i.e. 33% each). The model beats random allocation completely with 68.7% correct, 28.1% partially incorrect and 3.1% totally incorrect, showing that the random forest performs very well in predicting the perception towards isolation measures given demographic data, even with only five predictors. The relative importance of the demographic variables can be accessed by the average change in misclassification rates or Gini Impurity when the variable was permuted. The order of importance is Infants, Housing, Age, Working parents, Domestic helpers (13.2%, 12.8%, 11.4%, 10.8%, 7.3%).

The heterogeneity amidst the parents who displayed a consensus of favouring isolation measures was modelled with the random forest algorithm which divided them into three groups with mean willingness-to-pay of $104.3, $161.7 and $243.3. Since the algorithm taps uses bootstrap aggregation to improve prediction results, it lacks a clear and intuitive decision rule that explains how demographics can predict WTP. For instance, having infants and staying in larger homes was identified to be the greatest predictors, where parents

without infants but stays in larger homes have higher WTP (rows 17 to 24), while parents with infants and smaller homes have the lower WTP (rows 9 to 16). The parents with lowest WTP are with infants and large homes, and those with neither (rows 4 to 8, 28 to 32).

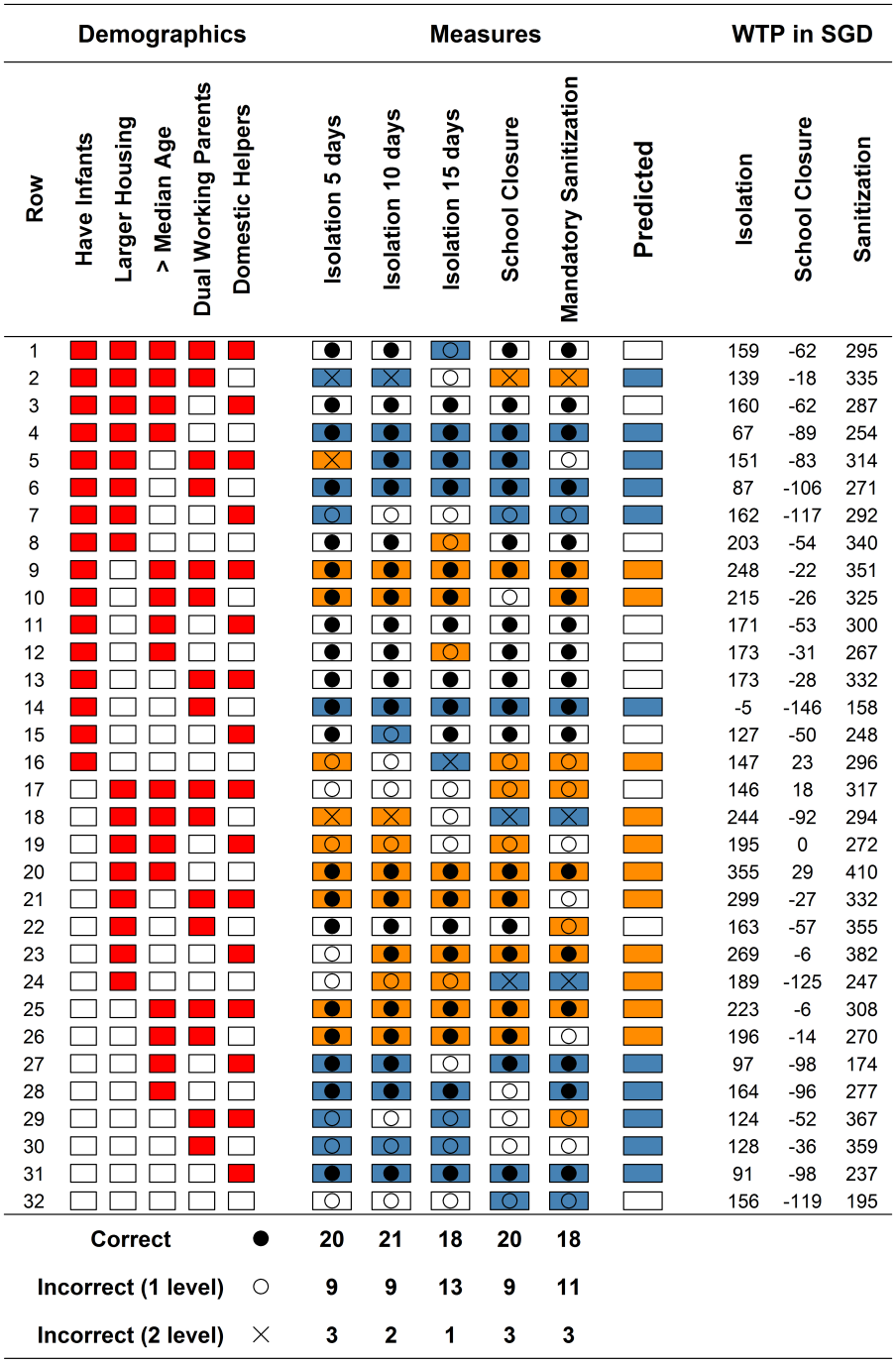


Figure 4.6: Results of Random Forest Modeling. Left: Demographics arranged in order of importance, based on increased Misclassification Rates when that variable is excluded. Red cells means 'Yes' while white cells means 'No'. Mid left: Results of leave-1-out cross analysis. Colours indicate the estimated utility categorized into ordinal variables: Orange means above 70th percentile, blue means below 30th percentile and white is in the middle. Symbols indicate the success of prediction by Random Forest. Black dots: Correct prediction; Empty circle: High/Low was predicted as average; an average predicted as either low or high. Crosses are completely wrong - High was predicted as a low, and vice versa. Mid right: Predicted categories based on entire data and the average WTP. Right: The mean willingness-to-pay for that measure for each group.

Table 4.1: DCE Levels and Attributes.

	Attributes & Levels
1	**Isolation of actual cases**
	Children continue to attend school even if ill
	Children with HFMD stay home until symptoms go away (∼5 days)
	Children with HFMD stay home until risk of infecting others is low (∼10 days)
	Children with HFMD stay home until no risk of infecting others (∼15 days)
2	**School closure**
	Schools do not close when they have 10 cases of HFMD
	Only affected class(es) close when there are 10 cases of HFMD at a school
	The entire school closes when there are 10 cases of HFMD
3	**Temperature screening at school**
	Children's temperatures are not screened at school
	Children's temperatures are screened when they arrive at school
	Children's temperatures are screened in the morning and afternoon
4	**Sanitization of School**
	Voluntary sanitization
	Mandatory sanitization
5	**Raise in monthly fee to cover extra control efforts**
	S$25
	S$50
	S$100
6	**Expected number of infections**
	1,000 cases (1 in 5,000 people are infected)
	10,000 cases (1 in 500 people are infected)
	50,000 cases (1 in 100 people are infected)
7	**Expected number of hospitalized cases**
	10 hospitalized
	100 hospitalized
	500 hospitalized

Table 4.2: Results of Stratification Analysis of Demographics (Logistic Regression). The purpose of the stratification is to find demographic variables which might cause heterogeneity in the perceptions of parents toward the policies. Each demographic is stratified into 2 or more groups and a logistic regression is fitted for each group. The overall likelihood is then compared with the likelihoods of the non-stratified model, to obtain the p-values of the likelihood ratio test. The demographic variables which improves the model significantly are kept for further analysis.

Variable	Group	Size	p-value
Age	$<$ Median	264	0.005
	$\geq$ Median	311	
Housetype	Smaller (1-4 rm)	231	0.080
	Larger (5+, private)	344	
Dual working parents	Both working	397	0.002
	$\geq$ Not working	178	
Domestic Helper	No	3811	0.0002
	Yes	194	
Infants	0	468	0.001
	≥ 1	107	
Household size	Small	275	0.531
	Big	300	
Race	Chinese	456	0.55
	Non-Chinese	119	
Citizenship	Citizen	449	0.15
	Non-Citizen	126	
Sex	Male	145	0.31
	Female	430	
Work from home	No	288	0.16
	Yes	109	
Unpaid leave or Flexihour	No	174	0.66
	Yes	223	
Grandparents	No	449	0.73
	Yes	126	
Young Children	<1	501	0.75
	≥ 1	74	

Table 4.3: Demographics of 575 Respondents and the General Population in Singapore.

Demographics	**Number N=575**	**Proportion**	**Singapore**
Male	145	25%	49%
Age as of 2015	$36(27-45)^b$		-
Singaporean	449	78%	61%
Singaporean PR	87	15%	9%
Non-citizen	23	7%	30%
Housing			**Aged 30 - 44**
HDB 2 Rm	7	1%	3%
HDB 3 Rm	51	9%	14%
HDB 4 Rm	173	30%	36%
HDB 5 Rm	163	28%	26%
Private	158	27%	21%
Others	23	4%	0.4%
Household			**Aged 30 - 44**
Size	$5(3-8)^b$		$4(1-6)^b$
Chinese	456	79%	73%
Indian	57	10%	11%
Malay	14	2%	11%
Others	48	8%	5%
Children	**Total**	**Avg/HH**	**Aged 30 - 44** [c]
Total	1014	1.76	1.5
Infants (0 - 1)	109	0.19	-
Young (2 - 5)	510	0.89	-
Older (6+)	290	0.5	-
Not stated	105	0.18	-
Caregivers			
Dual working parents	407	71%	73.8%[a]
i) With flexible working hours	142/407	35%	-
ii) Who can work from home	117/407	29%	-
iii) Who can take unpaid leave	177/407	43%	-
Non-working Grandparents	126	0.29	-
Non-working Family Members	27	0.06	-
Domestic helper	194	0.35	-

a) Weighted avg of age 30-34, 35-39, 40-44 by population size

b) Median and 2.5 to 97.5 percentiles

c) Per ever married female

Table 4.4: Goodness-of-fit of Contesting Models. DIC = Deviance + Penalty, where the former represents the model fit, and the latter represents the effective number of variables. Models with fixed effect has the least 'freedom' and thus has the least penalty. Log-normal models has lower penalty than normal models as it is constrained to only positive values.

Non-cost variables	Cost variable	Deviance	Penalty	DIC	Selected
Fixed Effect	Fixed effect	6528	8.5	6536	No
Hierarchical Normal	Normal	6319	170.6	6490	No
Hierarchical Normal	Fixed	6328	142.7	6471	No
Hierarchical Normal	Log-Normal	6339	156.7	6495	Yes

Table 4.5: Policy Recommendations.

Policy	Willingness to pay	95% Credible Interval
Current policy		
Isolation of cases (10 days) Screening once daily School closure Voluntary sanitization	$197.08	$62.34 - $497.57
Proposed policy		
Isolation of cases (10 days) Screening once daily No School closure Voluntary sanitization	$248.30	$135.18 - $609.74
Optimally preferred policy		
Optimally preferred policy Isolation of cases (10 days) Screening once daily Class closure	$590.59	$364.02 - $1314.63

Table 4.6: Demographics of the Latent Classes. Latent class 1 is the group which is more willing to pay (blue group in Figure 4.5). Italic p-values calculated from the Mann Whitney U test, while the others are based on the chi-square test.

Demographics	**Latent Class 1 N=307**	**Proportion**	**Latent Class 2 N=268**	**Proportion**	**P-val of Difference**
Male	81	26.4%	64	23.9%	0.55
Age as of 2015	36 (27.7 - 45)		36 (25.7 - 45.3)		*0.051*
Citizenship					
Singaporean	236	76.9%	213	79.5%	
Singapore PR	50	16.3%	37	13.8%	0.70
Non-citizens	21	6.8%	18	6.7%	
Housing					
HDB 2 Rm	3	1.0%	4	1.5%	
HDB 3 Rm	27	8.8%	24	9.0%	
HDB 4 Rm	85	27.7%	88	32.8%	0.29
HDB 5 Rm	92	30.0%	71	26.5%	
Private	91	29.6%	67	25.0%	
Others	9	2.9%	14	5.2%	
Household					
Size	5 (3 - 9)		5 (3 - 8)		*0.93*
Chinese	236	76.9%	220	82.1%	
Indian	28	9.1%	29	10.8%	0.019
Malay	7	2.3%	7	2.6%	
Others	36	11.7%	12	4.5%	
Children					
Total	542	1.763	472	1.761	*0.92*
Infants (0 - 1)	60	0.190	49	0.183	
Young (2 - 5)	256	0.887	254	0.948	0.15
Older (6+)	162	0.504	128	0.478	
Not stated	64	0.183	41	0.153	
Caregivers					
Dual working parents	232	75.6%	175	65.3%	0.009
i) With flexible working hours	76	32.8%	66	37.7%	0.055
ii) Who can work from home	72	31.0%	45	25.7%	<0.001
iii) Who can take unpaid leave	94	40.5%	83	47.4%	0.033
Non-working Grandparents	68	29.3%	58	33.1%	0.062
Non-working Family Members	12	5.2%	15	8.6%	0.99
Domestic helper	117	50.4%	77	44.0%	<0.0001

Chapter 5

Conclusion

One of our original hypotheses was that policies made by affected jurisdictions were based on quick decisions spurred by the mortality driven by EV-A71 outbreaks, and thus are likely to be stronger than required, such that there is some room for optimization. The systematic review made it apparent that the current knowledge on the epidemiology of HFMD is not sufficiently comprehensive to permit economic studies such as cost-effectiveness analyses. Instead, there exist gaps in the literature that were filled up by assumptions. For example, the incubation period of HFMD is written to be 3-5 days almost everywhere [135, 38, 136, 29, 88, 26, 137, 34, 94, 74, 138, 12, 36, 133] but are actually assumptions based purely on expert opinion and without sufficient quantitative evidence. Estimates from the literature for the R_0 of EV71 has been reported as 1.1-1.2 [6], 4.2-6.5 [138] and 1.1 [144], which have been discussed in detail in Chapter 2, and more recently 23.1-30.4 [193] (Uses Time Series SIR model that factors in cross-reaction) and 2.36-4.53 [192] (same method as [138]). These widely varying estimates can clearly not all be correct. The R_0 estimates range from being barely epidemic potential ($\sim$1) to something that requires close to the entire population to be seropositive to have sufficient herd immunity ($\sim$25). The common issue with all five estimates is that the modeling is only performed on the observed data, which is likely to be the minority of infections as the asymptomatic rate estimated to be between $\sim$30% [141] to almost $\sim$90% [20].

The estimates of a number of HFMD epidemiological parameters, which include R_0, asymptomatic rate, incubation period, infectious period and severity, are heterogeneous or poorly defined. Many of these features cannot be investigated directly or experimented upon. For example, a study which challenges seronegative people with EV-A71 to find the asymptomatic

rate is likely to get very informative results, but inappropriate given the risk for severe neurological complications. Filling this gap in the literature, which is driven by overlapping and partial information, will require careful analysis and data synthesis by the statistician for reconciliation, which is the overall theme of this thesis - finding parameters which cannot be observed directly.

5.1 Research Findings

5.1.1 The Gaps in the Literature

The first step in this project was to take a deep look at the underlying assumptions behind all the possible control strategies, and what are the required information for policymakers to make evidence-driven decisions. Chapter 2 addresses the problem with a rigourous systematic review on many questions about HFMD. For example, we found that the case for closing schools (socio-distancing) is actually weaker than expected because the literature is unclear on the relative importance of school versus community transmission [82, 38, 134, 23]. There is a real possibility that a closed school will just lead to more cases outside in the community. Also addressed is the asymptomatic rate of HFMD which is known to be substantial, and thus school closures based on attack rates or notifications will likely be too slow to stop the outbreak, unless the threshold is set very aggressively, which will in turn create a lot of false positives. The closure of too many schools unnecessarily is a burden to parents who are unable to work from home or take leave from work, causing some to leave their jobs to care for their child [46]. This highlights the importance of understanding control policies both from the epidemiological point of view and the social point of view, and to find ways to reduce the burden for all stakeholders. There are a number of hypotheses which require deeper analysis, for instance, will easier access to alternative caregivers or flexible working hours make parents more likely to support isolation or closure policies? Are there any factors which will change how people perceive different intervention and disease outcomes and their adherence to recommended practices?

Given that HFMD incidence usually peaks during summer in temperate/sub-tropical coun-

tries, many researchers have tried to model incidence with meteorological parameters. The causal relationship is however, suspect, because generic reasons such as 'decreased social contact' during cold weather fails for some droplet borne diseases such as influenza. Humidity was also mentioned as a factor for virus decline (dry weather has faster decline) [48] and would perhaps call for an experimental approach to see if drier environments in childcare facilities could result in lower disease incidence. We also found that research on risk factors are scarce and confined to generic themes such as 'hand-washing' and 'contact' which are factors for all droplet borne diseases [16, 34, 134]. Further studies should also include more specific factors that can also be influenced, such as parents' and teachers' knowledge about disease control, their method of keeping the environment clean, ambient temperature/humidity in the childcare to find more ways of preventing outbreaks. Lastly, we also showed that reproduction numbers are frequently not calculated with the assumption that there is a high rate of unobserved, asymptomatic HFMD, and sometimes modeled with the assumption that the population is immunonaive[138, 111], leading to heterogeneity among estimates of R_0, which is important to consider as it will determine the cost effectiveness of using vaccines to achieve herd immunity in the population.

5.1.2 Disease Burden and Severity

In Chapter 3, we explored the topic on asymptomatic infections further with a model for the entire spectrum of possible disease severity of HFMD, from 'infected but asymptomatic' to 'fatal', and found that asymptomatic rates can be as high as 71.4% (68.3% - 74.3%) for ages 1 to 4, the ages of greatest incidence, showing that we are only observing about one-third of all infections. The direct implication of this is that any modeling study of R_0 using notification case data and does not factor in the rate of asymptomatic infections is likely to greatly underestimate the parameter. Also apparent is that since it might be possible for asymptomatic children to infect others, this finding informs us that effective socio-distancing measures is like to be far more difficult than isolating symptomatic children, as we are only removing one-third of the source of infection this way.

Using observed serological data, we have estimated the infection rate to be between 6.5%

to 10.2% of the entire population per year for ages 1-4 in Asia (with older children less likely to get infections as the number of immune children increases over time). Symptomatic case fatality ratio of HFMD is about 52.3 (24.4 - 92.7) per 100,000 symptomatic infections. Although this number appears low, Years of Life Lost is the major contributor to the overall DALY of 96,900 (95% CI 40,600 - 259,000) age weighted DALYs per annum in eight high burden countries in East and South East Asia (China, HK, Japan, Malaysia, Singapore, Taiwan, Thailand, Vietnam), because children with fatal infections usually are between 1 to 3 years old, with almost 80 years of expected life. This is also an opportunity as vaccination to EV-A71 will then reduce the vast majority of disability.

5.1.3 Policy Preferences of the Public

Chapter 4 explored the preferences of parents toward four measures (isolation, closures, temperature screening and mandatory sanitization of childcare) and two outcomes (incidence and hospitalization incidence) where the childcare centre which they placed their child in experienced at least one outbreak in the past year. Support for isolation and closure measures is a struggle between the inconvenience of keeping a child at home when they have to work, and the perceived effect of keeping their child safe. We found that parents are generally favourable towards isolation of sick children but had little opinion towards how long the isolation should be, reflecting either lack of knowledge towards a suitable timeframe, or asymmetric preferences where they want other sick children to be isolated but not theirs. It was also discovered that parents prefer class closures (where the sick child was found), instead of school closures. This is an area which could potentially be effective.

Temperature screening and mandatory sanitization are burdensome on the childcare facility, not on the parents, and thus there is no struggle between inconvenience and effectiveness like isolation and closure. Temperature screening multiple times a day was not preferred by parents over screening just once a day, showing that parents perceive it to be not effective. Mandatory sanitization is strongly favoured (in Singapore, it is voluntary) as a measure, showing that perhaps more specific guidelines should be worked on to determine a set of instructions for childcare facilities to follow for sanitization during outbreaks.

5.2 Recommendations for Singapore

5.2.1 Regulations on Childcare Facilities and Schools

Despite the lack of direct studies on the effects of school closures on HFMD incidence so far, indicators found in this thesis calls the effectiveness of this policy into suspect. Firstly, school closures based on attack rates (23% currently for Singapore) are likely to be too slow to contain any outbreaks. The high asymptomatic rate of EV-A71, which is shown to hit up to 70% (Chapter 3) in children, will mean that the vast majority of children is already infected (symptomatic or not) by the time the desired attack rate is observed. Secondly, the duration of school closures seem to be arbitrary (10 days) and much longer than the estimates for the incubation period of EV-A71 (3 to 5 days). Thirdly, despite the belief by parents that school closures could be effective [46], the DCE revealed that they are neutral to school closures. This means that their belief that closures are effective was countered by the negative utility from the inconvenience.

The alternative to school closures will be either 1) No closures; or 2) Class closures, of which class closures should be considered and investigated based on the findings from the DCE, where parents have indicated their preference for class closures over other closure measures. This is perhaps because they see it as a more targeted and relevant approach.

As our research has only looked into the preferences of parents in closure policies, and not the actual policy effectiveness, there are a number of questions that should be answered before implementation. Firstly, a prospective experiment on CCCs could be attempted to find an class closure attack rate threshold which is the most effective. Intuitively, classes are small and therefore closures can be performed at much lower attack rates compared to school closures, such as at the first secondary infection. This might make it more effective in halting future transmissions. It is also more targeted, as children of the same class spends the majority of their time together, and thus much more likely to spread within classes rather than between classes. Secondary, the infectious period of HFMD viruses should be investigated with modeling studies to better inform policymakers of the ideal duration of closure.

Class closures are more likely to be effective if the childcare facility practices greater separation of students between classes, such as between meal times and during school-wide events. In actual practice however, classes could be encouraged to mix for reasons related to child social development and learning, such as improving social skills and teaching older children responsibility over younger ones. The benefits to mixing are thus likely to outweigh the risk of transmitting HFMD-viruses, where severe illness is rare.

Risk of HFMD can be reduced by other means, as revealed in the systematic review. Personal hygiene in the form of handwashing of caregivers and children helps to reduce risk. Given that EV-A71 in faeces can continue to shed many weeks after leaving the body, there is a need to wash and dry buttocks to prevent children from getting viruses onto their hands. Mandatory sanitization after class closures could be enforced by the authorities, and is a measure generally supported by the parents. However, the ministry should issue guidelines on how to perform sanitization and provide advice to the childcare centres on how to tender for vendors and the expected cost of service to avoid overpayment from unnecessary cleaning services.

5.2.2 Surveillance

As the mild conditions of HFMD contributes little to the overall disease burden, and following our recommendations to remove school closures from the enforced control policies for HFMD, some desired outcomes of HFMD surveillance would be to:

1. Quickly recognise risk of EV-A71 so that hospitals can be on high alert for potential CNS disease
2. Observe the overall trends of HFMD
3. Collect sufficient information to enforce class closures, if desired
4. Collect enough information to allow optimisation of current policies

Given that the majority of DALY is due to YLL, and that EV-A71 causes the majority of deaths, HFMD surveillance could include random swab samples of children, both healthy and sick, to check for EV-A71. This is a more proactive approach that addresses the high

asymptomatic rates associated with HFMD-causing viruses. Alert systems for EV-A71 could be set up, using modeling methods such as Bayesian Networks, to combine laboratory data and case data for immediate insights. The shift to real-time analytics will require information to be collected in a standardized manner.

Thus, we recommend that HFMD remain on the notification list of government hospitals and polyclinics while preschools and other childcare facilities should no longer be required to report individual HFMD cases. The private healthcare sector, which submits their data in their own formats, can continue to contribute by reporting severe cases of HFMD. Also, there is little reason to publish the names of childcare centres with HFMD outbreaks on the government website as this information does not seem to serve any purpose for other stakeholders (such as parents), and is potentially depressing for the staff of childcare centres which experienced an outbreak despite best efforts in maintaining cleanliness and hygiene [46].

Converting HFMD into non-notifiable disease serves a number of purposes. Firstly, data from surveillance must be analyzed and acted upon for surveillance to be effective as a measure. Data systems of private healthcare sector are made of different and disparate software and usually require manual effort in submitting reports on HFMD to the ministry. Data systems from the public sector clinics, on the other hand, are synchronized and can be automated to send data directly to the ministry, leading to cleaner data with fewer data quality issues. Such data can be processed quickly or automated with programs to produce timely reports to the authorities, allowing them to observe real time trends and thus react to outbreaks faster. Thus, while the approach is counterintuitive, the collection of less overall data in favour of cleaner and more standardized data makes it easier for the authorities to execute timely interventions.

Secondly, from our analysis in Chapter 3, we showed that despite best efforts to collect data, the observed incidence rate is not remotely close to being able to explain the level of seropositivity in the population. About 70% of all infections are not observed in Singapore from combination of asymptomatic and non-notified cases, but the proportion between the two is unknown. Other countries without mandatory reporting, such as volunteer sentinel

clinics, will likely see a greater proportion of non-notified cases. In this context, the 'observed incidence rate' of HFMD does not appear to be an informative statistic, as we know it is incomplete, but do not know the extent, and the statistic is not comparable to other countries as it depends on notification rate of each place. Taking a subset of this data (i.e. data reported from public healthcare) should still reveal the same trends as the 'full' data that is collected today, and thus still able to provide us early warning signals of outbreaks, guide public health policy, and for evaluation of the effectiveness of interventions. The main weakness of this approach is that the use of public healthcare is associated with lower socioeconomic status, and thus there is a need to check for any systematic differences in disease severity between public- and private-reported cases.

5.2.3 Vaccination

We have shown that the majority ($\sim$ 99%) of HFMD DALY is due to deaths of young children, and not from the disability which is generally a short illness of low severity. As severe manifestations and death due to HFMD is usually caused by EV-A71 [2, 3, 4, 5], vaccination from this virus could be an effective measure in reducing disease burden.

A cost effectiveness analysis (CEA), which lies outside the scope of this thesis, will be required before any recommendations could be made support vaccinations. A potential method to perform the CEA is to do it in 2 parts. First, an SIR model could be fitted in an individual level simulation environment for each childcare. At the very least, the model should factor in weekends, holidays, trigger points for school closures, infection rate in school, infection rate out of school (home or community), and asymptomatic infections, which was previously estimated in Chapter 3 of the thesis. Factoring in asymptomatic infections is critical, as misinterpretation of symptomatic incidence as notified-incidence rates in the current situation where up to 70% of cases were not captured will most definitely give the modeler a warped sense of the transmission dynamics.

The outcomes of this simulation could be the R_0, incubation periods, infectious periods, community vs school infection rates. The second part, the CEA proper, can be performed by stress testing policeis using the constructed SIR model. A policy such as mandatory

vaccinations, could be simulated by increasing the initial seropositivity rate at time 0, and the simulation will tell how many fewer cases are there due to the policy. The delta of the incidence could be then be compared with the estimated cost of policy implementation.

5.3 Outstanding Questions

The discussion above revealed that there are a number of outstanding questions which will assist in policy optimization. The individual level simulation can potentially be used to reproduce the results of other studies for R_0, by changing the underlying assumptions to resemble those studies as much as possible. Another way to estimate infectious period could be performed by measuring the quantity / concentration of pathogens which causes HFMD from the surfaces of furniture which the child interacted with and the hands of the child for each day of the infection. The weakness of using this method to estimate the infectious period is the same as papers which describe virus shedding - we do not know how does the shedding of virus affect transmission rate, although it would be intuitive to expect higher viral load to result in higher transmission risk.

The role of school and class closures in mitigating disease transmission can be studied from outbreak data where the intervention was actually implemented. Cohort studies of childcare centres, carefully chosen such that class size, the floor area of the school, density of students, etc, are comparable can be conducted to observe the effect of class closures on overall disease transmission. Swabs should be taken from children in these studies to understand more about asymptomatic infection and their role in transmission. The drawing up of more specific guidelines for sanitizing, isolation, health checks, separation between classes, etc should be performed with such cohort studies to ensure maximum effectiveness.

5.4 Finale

This thesis is the result of a deep collaboration between researchers from the National University of Singapore and Singapore Ministry of Health to enhance strategic efforts against infectious diseases. It investigates many key features of HFMD around Asia and uses

statistical modeling methods to combine the fragmented evidences together to form a more complete story. Through the systematic review, we see that although many unknowns remain, research on HFMD did make large strides of progress in the last two decades. By identifying these unknowns, we embarked on a quest to find them using a mix of serological data, surveillance data, and meta analysis to form estimates of the disease severity and disease burden, which are then used to derive our policy recommendations to the Singapore government. While these ideas can also be translated to other countries, policy should be evaluated on a case-by-case basis with deeper considerations to the individual country's context.

Lastly, we should note that communication with the public is as important as coming up with the best evidence-backed policies, as the population needs to understand and buy-in to the new ideas. Authorities must speak to and educate the public to some basic ideas about epidemiology to help them understand the evidence in the literature and why there is a need to update the control measures for HFMD in light of new findings.

Chapter 6

Appendix

This chapter is meant to give a brief overview of the different methods which was used in the modeling of the data we collected. Throughout the thesis, Bayesian modeling is commonly used due to it's flexibility in handling heterogeneous and hierarchical data. In particular, we used the Bayesian framework in Chapter 3 to model the change in seropositivity towards EV-A71 as age increases, and also to model the severe states of EV-A71 which has both missing and hierarchical data. The Bayesian framework is also used to model the heterogeneity of the parent's responses in our questionnaires, where we tried both hierarchical approaches and dirichlet latent class approach. In the next section, we provide some information about Bayesian modeling, and the two Markov Chain Monte Carlo algorithms which we used for the estimation of the parameter posterior distributions. Other statistical methodologies will be covered in their respective chapters.

6.1 Bayesian Modeling

Bayesian modeling is used extensively in the research work presented in the thesis. In this section, we will provide some basics of Bayesian statistics which is omitted from the actual publications. As the literature on Bayesian methods is extensive, we will only cover the key essentials and algorithms actually employed in the course of the research.

6.1.1 Bayes' Theorem

The use of Bayes' theorem is critical for us to make probability statements about the parameters which we wish to estimate, θ, given the data collected, y, allowing us to use probability to quantify uncertainty. The theorem is based on the concept of conditional probabilities. The joint probability density of θ and y, $p(\theta, y)$, can be written as a product of the *prior distribution* $p(\theta)$, and the *data distribution* $p(y|\theta)$, also called the *likelihood*.

$$p(\theta, y) = p(\theta)p(y|\theta)$$

Conversely, we can also flip the θ and y to get an alternative form for $p(\theta, y)$.

$$p(\theta, y) = p(y)p(\theta|y)$$

Equating them together gets us the Bayes' theorem.

$$p(\theta)p(y|\theta) = p(y)p(\theta|y)$$

$$p(\theta|y) = \frac{p(\theta)p(y|\theta)}{p(y)}$$

$p(\theta|y)$ is known as the *posterior distribution* and represents the probability distribution of the parameters given the data. Other than in a few special cases, the equation of the posterior probability distribution is elusive because the denominator, $p(y)$, also known as the *prior predictive distribution*, is an integral $p(y) = \int_\theta p(y, \theta)\, \mathrm{d}\theta$ that is usually very difficult to work out.

Another way of viewing the posterior distribution is to see $p(y)$ as a proportionality constant independent of θ. We will be able to compute the posterior distribution if we normalize the numerator.

6.1.2 The Prior, Likelihood and Posterior

The prior distribution describes the "prior information" that you have for the parameters, θ. This distribution should include all plausible values of θ, but need not be realistically concentrated around the true value. Priors can be broadly classified into three variants: Non-informative, weakly informative and informative, which represents the amount of 'prior information' that you possess before the research. The use of any priors are always stated clearly in each of the chapters.

The posterior distribution, $p(\theta|y)$, is the probability distribution of the parameters given the data and is proportional to the product of the *prior distribution* and the *likelihood function*, which is formed from a combination of the model specification and the actual data. Closed form solutions of the posterior is impossible to derive for our purposes and is always estimated using *Markov Chain Monte Carlo* (MCMC).

$$p(\theta|y) = k\ p(\theta)p(y|\theta) \propto p(\theta)p(y|\theta)$$

A MCMC sets up a markov chain simulation that generates a series of values which are only dependent on the current value that approaches the posterior as the values are produced. In this thesis, the *Metropolis Hastings Algorithm* [218, 219] are used in Chapter 2 and the *Gibbs Sampler* [220, 221] are used extensively in Chapter 3.

We will describe the Metropolis Algorithm here:

Our goal is to draw samples from the posterior distribution $p(\theta|y)$. Let there be a function $f(\theta)$, which is proportional to $p(\theta|y)$. That is, $p(\theta|y) \propto f(\theta)$, or $p(\theta|y) = k\ f(\theta)$, where k is some normalizing constant. The Metropolis Algorithm is as follows:

1. Start with initial value θ_0 satisfying $f(\theta_0) > 0$.
2. Using current value of the parameters, θ_{t-1}, propose a candidate point θ^* from some proposal distribution $q(\theta_t|\theta_{t-1})$.
3. Metropolis Algorithm: Given the candidate point θ^*, calculate the ratio of the density at the candidate point and the previous point θ_{t-1}.

$$\alpha = \frac{p(\theta^*|y)}{p(\theta_{t-1}|y)} = \frac{f(\theta^*)}{f(\theta_{t-1})}$$

4. Accept the new point θ^* with probability α. If $\alpha > 1$, accept the new point.

The Metropolis Hastings Algorithm generalizes the algorithm with a different acceptance probability which allows for asymmetric proposal distributions.

$$\alpha = min(\frac{f(\theta^*)q(\theta_{t-1}|\theta^*)}{f(\theta_{t-1})q(\theta^*|\theta_{t-1})}, 1)$$

This acceptance probability is the same as the Metropolis algorithm if the proposal distribution is symmetric such as the normal or uniform distribution. Symmetric distributions fulfill this criteria: $q(\theta_{t-1}|\theta^*) = q(\theta^*|\theta_{t-1})$.

The Gibbs Sampling is a special case of the Metropolis Hastings algorithm which the proposal distribution is chosen specifically to make the acceptance probability 100%. This works by having a proposal distribution which consists of d univariate **conditional distributions** if there are d variables in θ, instead of proposing all d variables in a single joint distribution, allowing calculation of high dimension models. For this thesis, Gibbs sampling is performed exclusively with the JAGS software [206].

6.1.3 Model Diagnostics - Checking and Comparisons

A good statistical analysis will involve some check of the adequecy of the model to the data the plausibility of the model for the purposes which the model will the used. To perform this, we should check the sensitivity of the results to the prior distribution and likelihood. A sensitivity analysis is to access how much does the posterior inferences change when other reasonable probability models are used in place of the present model. For all models, we always perform sensitivity analysis on the prior distributions by rerunning the same model with other distributions, for example, $N(0, 1000^2)$ and $U(-200, 200)$ are both candidates for non-informative priors and the posteriors should not be sensitive to the choice of the priors. Other checks which are adopted include posterior predictive checking, which involves

simulating data from the model to check if the simulated data resembles actual data.

6.1.4 Examples

Linear Regression with Metropolis Hastings Algorithm in R

This section provides a simple illustration of the Metropolis Hasting Algorithm through a linear regression, which only has 3 unknown parameters in a simple linear regression - β_0, β_1 and sd.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

$$\epsilon_i \sim N(0, sd^2) = y_i - (\beta_0 + \beta_1 x_i)$$

1. Create new function `logposterior` which computes the log-likelihood using sum of the log-likelihoods based on the normal distribution of ϵ, and the priors using non-informative priors.

```
logposterior <- function(theta){

  #likelihood
  LL = sum(dnorm(data$y-theta$b0-theta$b1*data$x, mean = 0,
sd = theta$sd, log = T))

  #prior (non-informative)
  LP_b0 = dnorm(theta$b0,mean=0,sd=1000,log=T)
  LP_b1 = dnorm(theta$b1,mean=0,sd=1000,log=T)
  LP_sd = dunif(theta$sd,min=0,max=1000,log=T)
  theta$logposterior = LL + LP_b0 + LP_b1 + LP_sd

  return(theta)
}
```

2. Create function which performs 1 iteration of the MH-Algorithm. It requires 2 inputs - the old values θ_{t-1}, and the new proposed values θ^*. It starts by rejecting

invalid proposals (*sd* cannot be negative), and proceeds to calculate the f value using the `logposterior` function which we just defined. Next, we determine whether the proposed values should be accepted or not by comparing $\frac{f(\theta^*)}{f(\theta_{t-1})}$ to the uniform distribution. If it is accepted, the new values will replace the old.

```
mh=function(oldtheta,newtheta)
{
  reject=FALSE
  if(min(newtheta$sd)<0)reject=TRUE
  if(!reject)
  {
    newtheta=logposterior(newtheta)
    logaccprob=newtheta$logposterior-oldtheta$logposterior
    lu=-rexp(1)
    if(lu>logaccprob)reject=TRUE
  }

  if(reject)return(oldtheta)
  return(newtheta)
}
```

3. Load in the data and create variables to store the posterior samples

```
data<-read.csv("simdata.csv",header=TRUE)
n_T=nrow(data)
theta=list(N=n_T,b0=0,b1=0,sd=1)
theta=logposterior(theta)    #calculate the first logposterior

storage=list(b0=matrix(0,MCMCits,1),
             b1=matrix(0,MCMCits,1),
             sd=matrix(0,MCMCits,1),
             logP=matrix(0,MCMCits,1))
```

4. Run the simulation

```
MCMCits=(As many iterations as you wish)
thinning=(As much as you wish)


for(iteration in 1:MCMCits)
{
  if(iteration\%\%100==0)print(paste(iteration,round(theta$logposterior)))
  for(thin in 1:thinning)
  {
      oldtheta=theta;
      theta$b0=rnorm(1,theta$b0,1);     ##the proposal distribution
      theta$b1=rnorm(1,theta$b1,1);     ##the proposal distribution
      theta$sd=rnorm(1,theta$sd,1);     ##the proposal distribution
      theta=mh(oldtheta,theta)
  }
  storage$b0[iteration,]=theta$b0
  storage$b1[iteration,]=theta$b1
  storage$sd[iteration,]=theta$sd
  storage$logP[iteration,]=theta$logposterior
}
```

Logistic Regression

With a logistic regression, the main difference is the likelihood function. Let us write the regression equation in this simplified manner: $logit(\theta_i) = \beta_0 + \beta X_i$, where βX_i represents the rest of the coefficients and covariates. The probability of success is $\theta_i = logit^{-1}(\beta_0 + \beta X_i)$. Since we know the outcome and the probability of success, we can model the probability of each subject i as a *Bernoulli distribution.*

The likelihood function:

$$p(y_i|\beta_0, \beta, X_i, n_i) \propto (\theta_i)^{y_i}(1-\theta_i)^{n_i - y_i}$$

And the posterior is the product of the joint prior distribution and the likelihood:

$$p(\beta_0, \beta | y_i, X_i, n_i) \propto p(\beta_0, \beta)(\theta_i)^{y_i}(1-\theta_i)^{n_i - y_i}$$

The Metropolis Hastings Algorithm was used in Chapter 2 for the hierarchical model for the meta analysis data of high severity state HFMD.

6.1.5 Simulation Diagnostics

After obtaining posterior samples from the MCMC simulation, diagnostics were always performed to ensure convergence and goodness-of-fit. Convergence can be observed by studying the **Trace Plots**, a plot of the value of the simulated samples versus the number of iterations. Simulations which have achieved convergence will show trace plots which fluctuates around a value. Other factors, such as the bandwidth (related to the average distance of the proposed θ^* from θ_{t-1} during each iteration), burn-in (removal of the starting values of the simulation which are not converged yet), thinning (for hard-disk space considerations and posterior sample autocorrelation), and simulating multiple chains with different initial values (to ensure all chains approach the same posterior) are all considered.

6.1.6 Model Comparison

Model comparisons were mostly performed using the Deviance Information Criterion (DIC), which is the sum of the average deviance and a penalty term for models with large number of variables, p_d. We can obtain the deviance for each posterior sample:

$$D(y, \theta) = -2log\ p(y|\theta)$$

The statistic of interest would be the average deviance, $D_{avg}(y)$ can be estimated by storing and averaging the log-likelihoods, multiplied by -2. p_d is the effective (unconstrained) number of parameters in the model, and can be calculated as half the posterior variance of the deviance, which can be estimated from the posterior simulations:

$$p_d = \frac{1}{2}\hat{var}(D(y,\theta|y)) = \frac{1}{2}\frac{1}{L-1}\sum_{l=1}^{L}(D(y,\theta^l) - \hat{D}_{avg}(y))^2$$

The Deviance Information Criterion (DIC) is the sum of the average deviance and p_d. The smaller the DIC, the better the model. The DIC is a versatile measure which can be used for comparing non-nested models.

$$DIC = D_{avg} + p_d$$

.

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